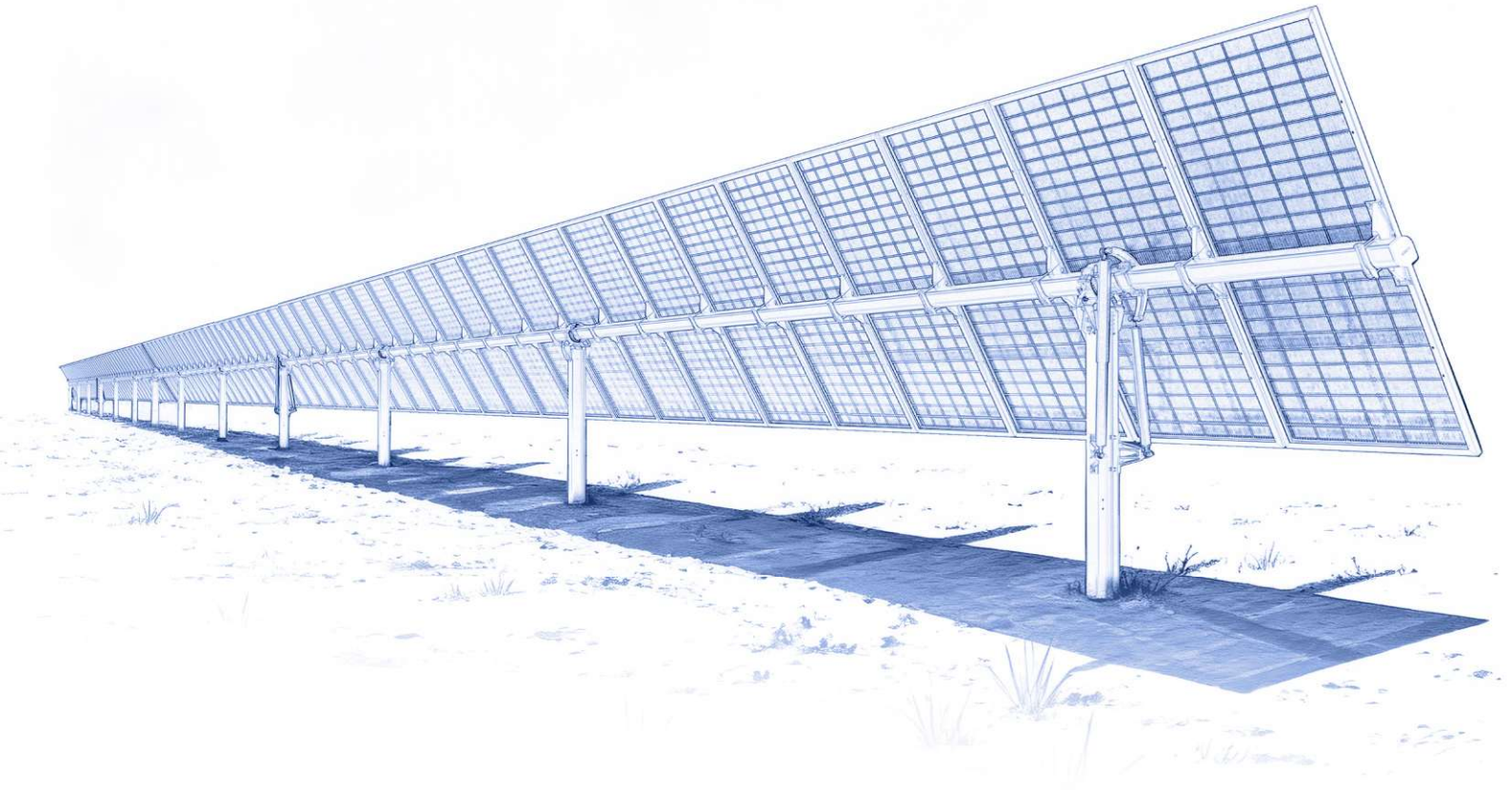
TECHNICAL INSTRUCTIONS AND INSTALLATION MANUAL

Expert in Smart Green Manufacturing

and All-Scenario Mounting Solutions



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1. Foreword

1.1 About this manual

This manual outlines the correct installation procedures and establishes the minimum standards necessary for product reliability and warranty fulfillment. A thorough understanding of this manual is essential for proper installation; failure to adhere to the guidelines provided may lead to property damage, personal injury, or even fatalities. Any alterations to the installation procedures must be promptly communicated to ESET for validation.

- **Professional Installation**: Ensure that the installation is carried out by certified solar installers. All tasks must be supervised by technical personnel designated for this project.
- **Regulatory Compliance**: Adhere to all applicable local or national building codes, including those that may take precedence over this manual.
- **Product Compatibility**: Verify that all products are suitable for the specific installation
- environment and array configuration under the site's loading conditions.
- **Original Parts Usage**: Utilize only "ESEEK-1P" original components or parts that are supplied or recommended by ESET. The use of parts from other sources may invalidate any applicable warranty.
- **Site Validation**: Assess site parameters prior to installation, as a local geotechnical report may be necessary to evaluate soil conditions. It is advisable to consult with a local engineer who is well-versed in local regulations and site requirements, including soil conditions, terrain, and load criteria. All these factors can influence foundation requirements. The lengths of foundation posts mentioned herein are for illustrative purposes only.
- **Fire Safety Compliance**: Comply with all relevant fire codes, including but not limited to, maintaining clear pathways and avoiding obstructions.
- **Accurate Information**: Ensure that the information provided is accurate. Any issues arising from inaccurate information are the responsibility of the installer.
- **Compatibility Verification**: Confirm with the module manufacturer that the ESEEK-1P Ground Mount Tracker is compatible with the selected photovoltaic modules.
- **Electrical Safety**: Ensure the safe installation of all electrical components of the PV system. Electrical installation and procedures should be performed by a licensed and

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bonded electrician or solar contractor. All work must comply with national, state, and local installation procedures, product, and safety standards.

- **Maintenance Guidance**: Refer to the ESEEK-1P Operations and Maintenance manual for detailed information on periodic and preventive maintenance tasks for the trackers.

1.2 Safety instructions

A. Before the construction, please read this installation manual carefully.
B. Installation training and skills assessment must be carried out for operators, who can only participate in the construction after passing the assessment.
C. Workers must wear a helmet and fasten the hat strap after entering the site, and must correctly use personal protection products during work.
D. Strictly implement the operating regulations, no illegal command and illegal operation, workers have the right to refuse the instruction of illegal operation and have the responsibility to stop the illegal operation of others, the installation process must comply with the relevant local safety requirements.
E. Site electricity must be managed by a specified person, while setting up a dedicated distribution box, prohibit indiscriminate pulling of wires, especially to eliminate unauthorized operation, to prevent personal, line, equipment accidents.
F. All tools, electrical equipment, and power supplies must be carefully inspected before installation work begins. Faulty tools or equipment must not be used, and unqualified tools and equipment must be repaired before use.
G. Workers are expected to be alert, focused, and use their tools wisely at all times while on the job. Do not use tools when fatigued, under the influence of alcohol or after consuming drugs, as this may lead to serious personal injury.
H. When using the power tool, other irrelevant personnel must be far away from the working scope of the power tool.
I. Never use tools with damaged or unguarded!
J. Ensure that the construction site has a proper lighting environment, ensure that the construction site is clean and tidy, and the materials and tools related to the installation work should be placed in a reasonable manner, to prevent bumping and tripping over the site workers.
K. During installation work, there may be impact or other potential hazards, so be sure to

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use personal protective equipment: helmet, goggles, earplugs, protective gloves, and work shoes, etc;

L. No unauthorized removal of the site's protective facilities, safety signs, warning signs, if they need to be removed during installation, it must be approved by the person in charge, and other effective ways for the safety warning work should be carried out in this area.

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1.3 Regulation for Material Receiving Management and Implementation

1.3.1 Purpose

This procedure is formulated in order to standardize the tracking process, methods and time limit of documents, correspondence and related matters in the process of receiving materials.

1.3.2 Scope

This procedure applies to the tracking and processing of documents, correspondence and related matters in the process of receiving materials for ESET　Carbon steel project.

1.3.3 Reference Document

1. The relevant laws, regulations, rules, regulations and standards applicable to the equipment within the scope of inspection and acceptance.
2. Procedures, methods and systems for the management of relevant materials acceptance and receiving.

1.3.4 Responsibility

1．Project manager: responsible for internal and external organization coordination, confirming project progress and status

2．Project support engineer: responsible for leading the project site material receiving and handover work, and issuing relevant certificates.

1.3.5 Implementation procedure

1．Material inspection and receiving is in accordance with the delivery schedule.

2．Project manager requests the construction schedule of the projects, including but not limited: ramming, installation, energization, commissioning test and COD, to organize resources at least one (1) month before the goods arrive to project site to organize resources.

3．Project support engineer will share with the Client two (2) documents 《Declaration about goods receiving》and 《Guidance for On-site Handling and Storage -ESET Tracker System》, in which is specified how to properly storage the goods. Both documents must be signed by the Client to guarantee the proper storage of the goods. Otherwise, ESET's warranty would not be applicable.

4．Client shall check the quantity of the Product within two (2) business days upon delivery of each shipment of Product at the Delivery Location ("Period of Check") and shall sign the

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Material Delivery and Acknowledgement Certificate (MDAC) within the “Period of Check” after receiving the goods. In case of any shortage of quantity of the Product that is attributable to ESET, Client shall provide a notice to ESET sale and project manager by e-mail with written evidence or other support documentation. ESET shall be entitled to request any additional support documentation of the claim as it may deem appropriate, within five (5) business days after the receipt of such notice from the Client. If such evidence or documentation indicates that such shortage of quantity exists at the delivery of the product, ESET will make up the deficiency of quantity within a reasonable time, and without prejudice Construction Schedule, until the actual new delivery date of the missing products. Except for those Products that are supplied in bulk by ESET, if client fails to check the quantity of the Product or uses the product without checking the quantity within the period of check, it shall be deemed as that the product is correct in quantity and is delivered in accordance with packing lists or any other shipping documents or relevant documents.

5. Client shall also be entitled to conduct inspection of the product within the period of check set forth in subparagraph c. Client shall inspect whether there is any damage to the packaging or the product that is obvious enough to be identified and whether there is any patent or obvious defect of the product that is not in accordance with these terms and conditions from the inspection. In case any damage to the packaging or the product or any defects are found during such inspection, it shall be notified by client to ESET sale and project manager in writing with evidencing documents or other documentation deemed appropriate within five (5) business days after the period of check. If such evidence or documentation indicates that such damage or defect is attributable to ESET, either ESET will replace such non-conforming product within a reasonable time or remediation works on-site shall be done to not prejudice the construction schedule. Should Client fail to conduct any inspection of the product within the aforesaid period or to notify ESET of any damage or defect within the above specified period, the product shall be deemed to have been accepted by client. Following any such acceptance, any claims with respect to the product must be made as warranty claims and the sole remedies of client with respect to damage to or defect in the Product will be those as provided in “Warranty of goods”.

6. Client is entirely responsible for the correct unloading and storage of the materials; Client

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shall follow the guidance described on “Guidance for On-site Handling and Storage -ESET Tracker System”. Such document must be signed by the Client before receiving the goods.

1.3.6 Record

1. Material Delivery and Acknowledgement Certificate (MDAC) .
2. RF-DS-016 Equipment (Material) Acceptance Report.

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1.4 Requirements for on-site material storage and transportation

1. The environment in which materials are stored should be dry, free of rain, dust and open flames.
2. The materials should be stored away from direct rain and should not be placed in humid places for a long time. The package (box) should be placed on a higher ground, and should be raised with square wood or blocks with a thickness of not less than 10cm. It is prohibited to have direct contact with the ground, and the ground should have good waterproof and drainage measures. H-shaped steel, pipes and other materials with surfaces or internal surfaces that are prone to water accumulation should be placed diagonally, with one end higher and the other lower, to ensure free water dispersion. The support material should be prevented from direct contact with other objects to avoid contamination by dust and other dirt. The support material needs to be checked regularly. If it is found to be damp, deformed, slightly corroded, etc., it should be handled in time.
3. Solar mounting structure materials can be stored according to shape, material, specification and brand. Different types of materials should be placed in different areas to avoid confusion and cross contamination, especially for steel components, bolts, washers and other
4. For unloading or secondary transfer, avoid dragging when using manual or forklift truck unloading, and use the loading tool correctly. For long pieces with a length of more than 3m, the loading and unloading forklift must be equipped with long forks (toothed sleeves). The length of the specific sleeves is customized according to requirements, Hanging forks are forked and forked. Do not drag.
5. In the specific storage, the photovoltaic support should pay attention to the dryness of the storage environment, and should be covered during storage. It should effectively avoid the corrosion of corrosive substances such as acid, salt and alkali. secondly, the components of the photovoltaic support should be classified and placed in the code. The layers of the parts should be properly separated by cushions, and the number of stacked layers in the field should not exceed 3 layers. The cargo storage area is best to be flat and not suitable for low-lying areas, so as to prevent the goods from soaking in water and ensuring that the goods are dry after rain.
6. During the installation of the bracket, it is forbidden to violent construction, or stand directly on the brackets, accessories and other parts to avoid product deformation. For the long

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pieces of the purlins, it is necessary to handle them gently. if the surface of the standard parts is scratched due to the mismatch of the installation tools, it should be repaired in time.

7. For small pieces and standard parts, they should be kept safely to prevent moisture and loss. Do not soak in water, and dry it in time after immersion.
8. In the corrosive environment (acid alkaline or high salt soil, seawater or substances that accelerate corrosion), the bracket product should be avoided from contact with the corrosion source, If the bracket material is accidentally contaminated by corrosive substances, it should be cleaned as soon as possible to avoid long time, Attached to the surface of the product and treated with anti-corrosion treatment.
9. If the product surface treatment layer is damaged by collision during transportation or installation, the dirt should be removed as soon as possible and anti-corrosion treatment should be done.
10. The storage, loading and unloading and construction shall comply with the above provisions, and the company shall not be responsible for any damage caused by improper storage, Improper loading or unloading or improper construction.
11. If corrosion or rust is detected on the parts, the area must be cleaned and sprayed with zinc-rich paint. It is recommended that zinc-rich paint conforming to ISO 3549 be used and that at least two coats of paint be applied, with an interval of approximately one hour between coats.

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2. Bill of installation tools

NO.	NAME	SPECIFICATION	PATTERN
1	Electric wrench	M5/M8/M10/M12/M14/M16/M18	
2	Screwdriver	samll size	
3	Leveling instrument	1m	
4	Allen wrench	6mm (for M8)	
5	Open-end wrench	8/13/16/18/21/24/27mm	
6	Rubber hammer	medium size	
7	Tape measure	10m	
8	Construction line	100m	
9	Angle gauge	±90°	
10	Torque wrench	Torque Range: 0–450 N·m	
11	Long nose pliers	4in	

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3. Schematic diagram of the finished installation product

The installation effect diagram is shown in Figure 3.1:

1. *The following model diagram is for reference only.

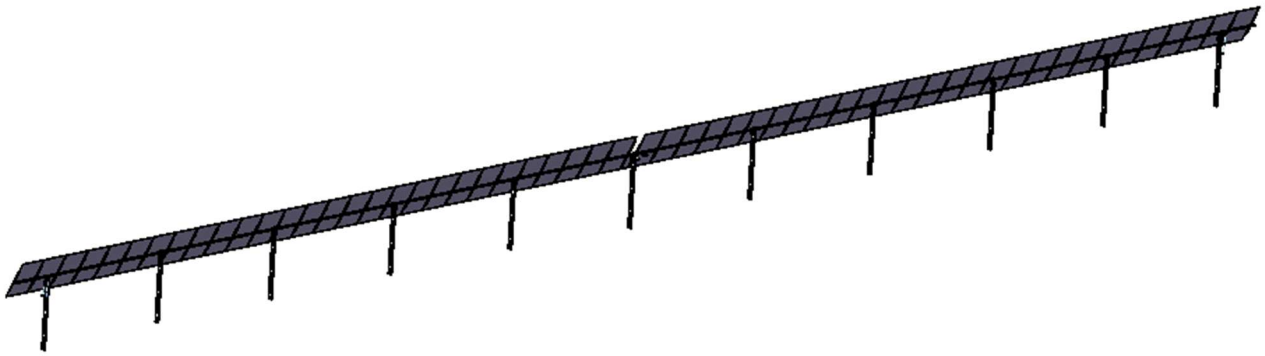


Figure 3-1 Overall effect diagram of the solar tracker for reference.

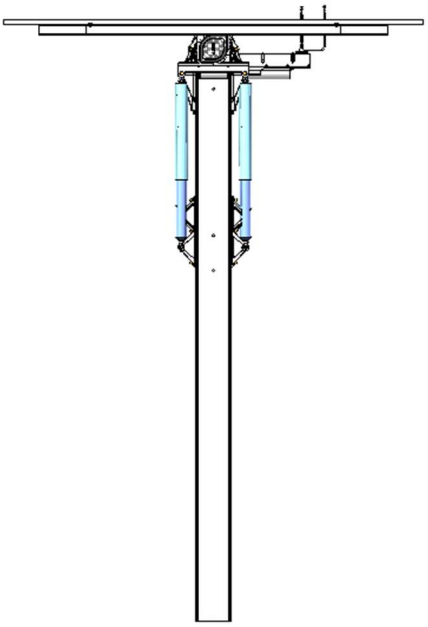
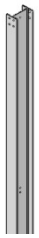
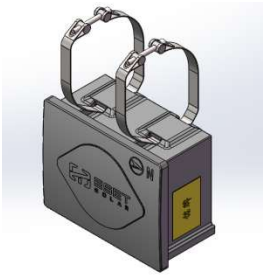
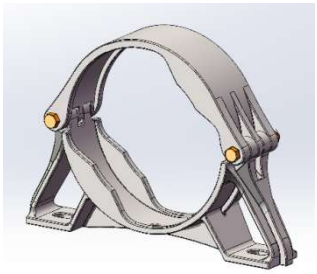
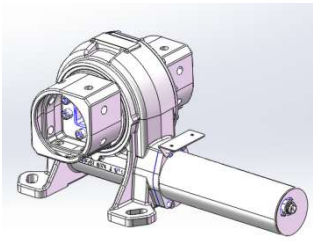
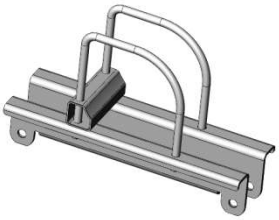
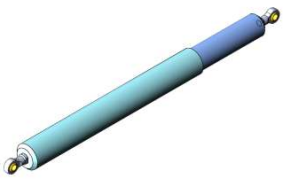
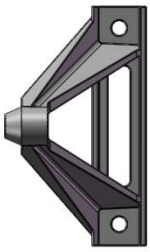


Figure 3-2 Side view effect diagram of the solar tracker for reference.

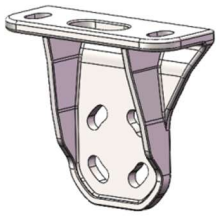
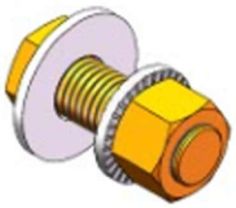
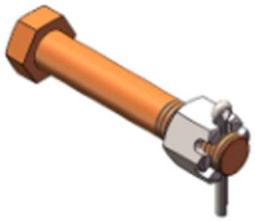
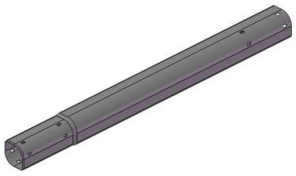
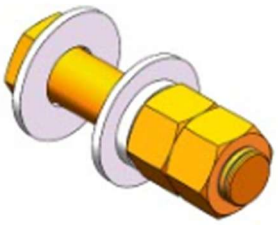
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4. List of major components

Table 4-1 List of major components for reference.

NO	Description	Schematic	NO	Description	Schematic
1	Post		10	TCU BOX	
2	Bearing housing		11	Slewing reducer	
3	Bearing base		12	Polymer bearing wear parts	
4	Damper hoop		13	Damper	
5	Damper lower fixture		14	wireless receiver	

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6	Reducer base		15	M12/M14 Bolts with hole	
7	Torque tube connecting the reducer		16	M14 Assembly for Bearing housing	
8	Connecting torque tube		17	M16 Assembly for Reducer base & Bearing base	
9	Purlin		18	M18 Assembly for Reducer	

For detail tightening torque of the bolts, please check Appendix 1.

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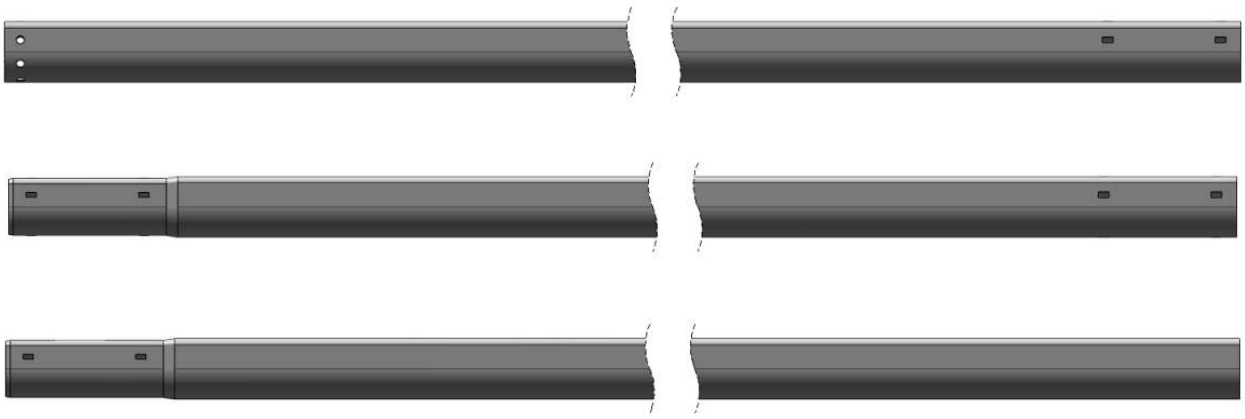
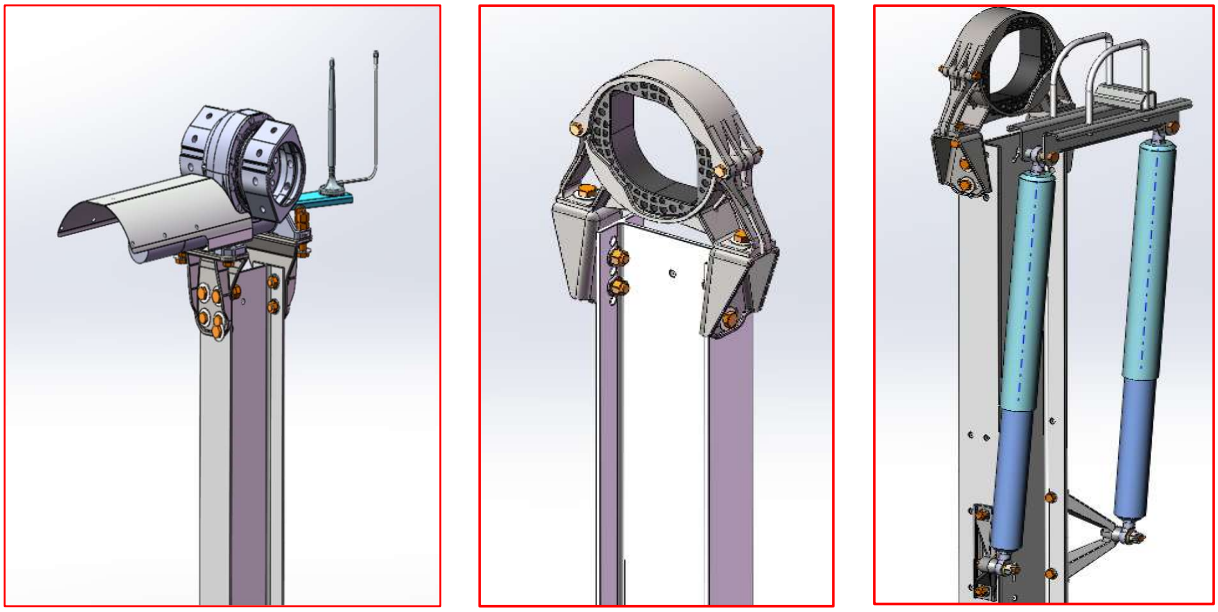


Figure 4-2 Drawing of 3 types of torque tube



A. Driving post B. Non-driving post C. Damper post

Figure 4-3 Schematic effect drawing of the three types of post

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5. Structure and installation

5.1 Foundation procedure (Step 1)

5.1.1 Foundation preparation

1. Follow the foundation positioning coordinates from layout drawings and install positioning nail by using TPI (Total Station with Positioning Intelligence) as figure 5.1-1;
2. Prepare the position and depth mark line of posts as figure 5.1-2 & 5.1-3;
3. Use Pile driver with a minimum of 1200 Joules impact energy for Foundation Construction as figure 5.1-3;
4. Highly recommended to use the pile caps to protect end of post that come with the device or provided by ESET as figure 5.1-6.
5. ☆Some posts adopt cast-in-place pile foundations. For specific locations and requirements, refer to the Layout Drawing.



Figure 5.1-1: Foundation positioning and positioning nail installation

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Figure 5.1-2: Foundation depth reference line

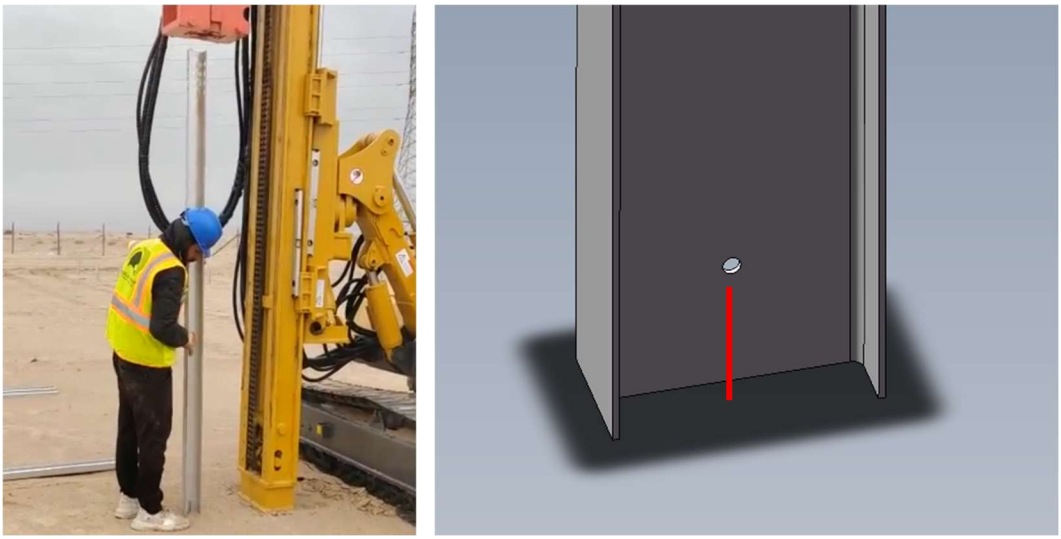


Figure 5.1-3: Key Points of Foundation Construction (positioning nail reference position)



Figure 5.1-4: Key Points of Foundation Construction (post depth mark line reference)

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Figure 5.1-5: Foundation post central reference line

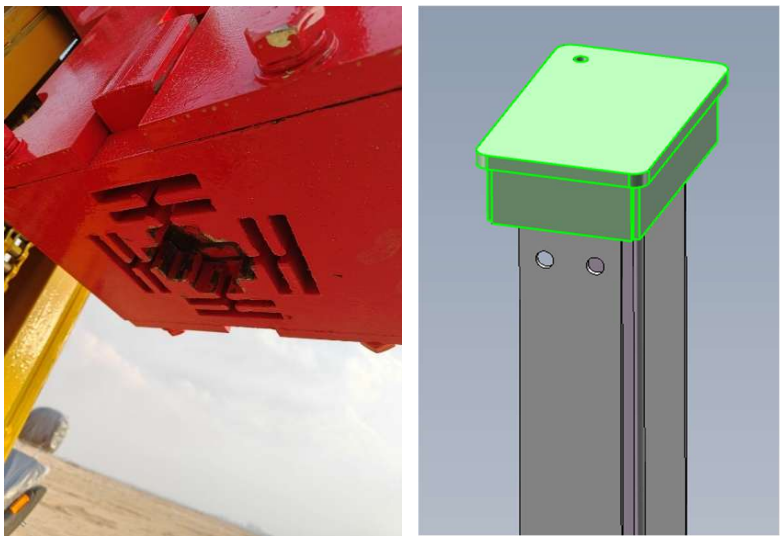


Figure 5.1-6: Foundation pile cap image

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5.1.2 Foundation layout

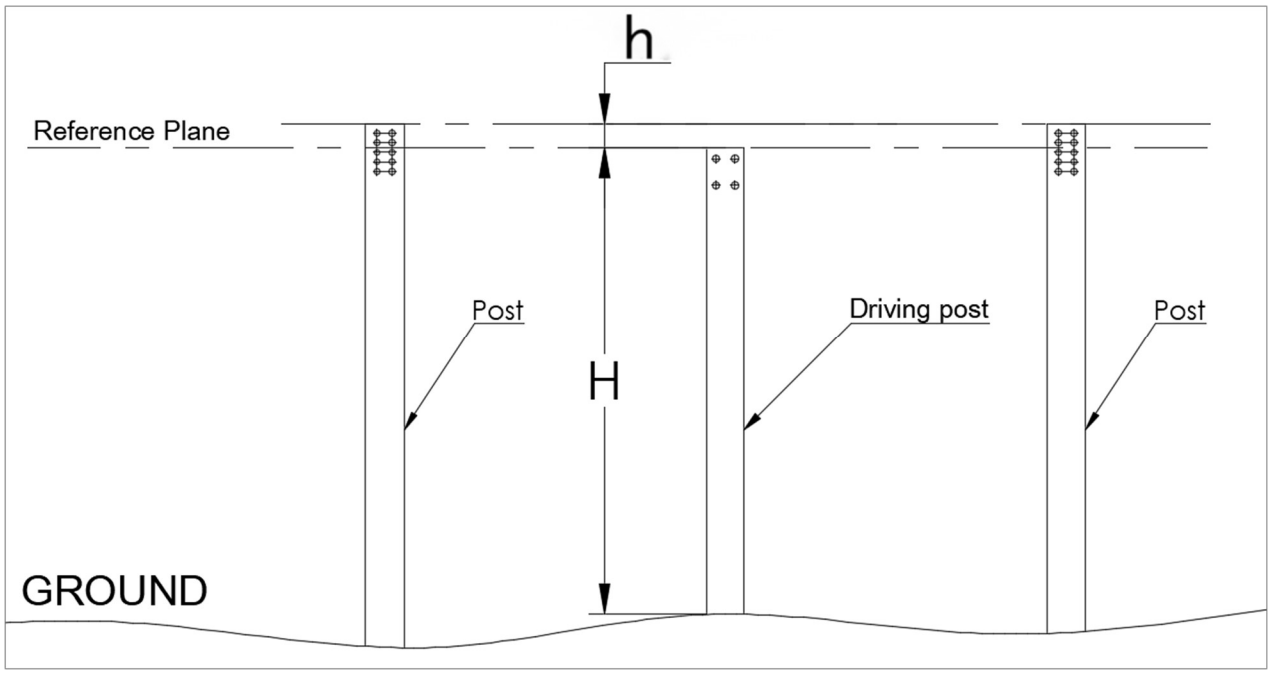


Figure 5.1.2-1: Schematic diagram of foundation depth and height from the ground

Remark:

① All above dimensions identified as “H”, “h”,and numbers of general posts are project specific.Please refer to **Appendix 2.1:Installation of posts**.

② For details about the post distribution refer to the foundation drawings to obtain specific foundation depths and distances to ram.

③ This installation manual takes the top of the posts as reference. The top of each Non-Driving Post must be h refer to **Appendix 2.1:Installation of posts**. higher than that of the Driving Posts. The post is always vertically installed no matter if there is slope or not.

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5.1.3 Terrain requirement

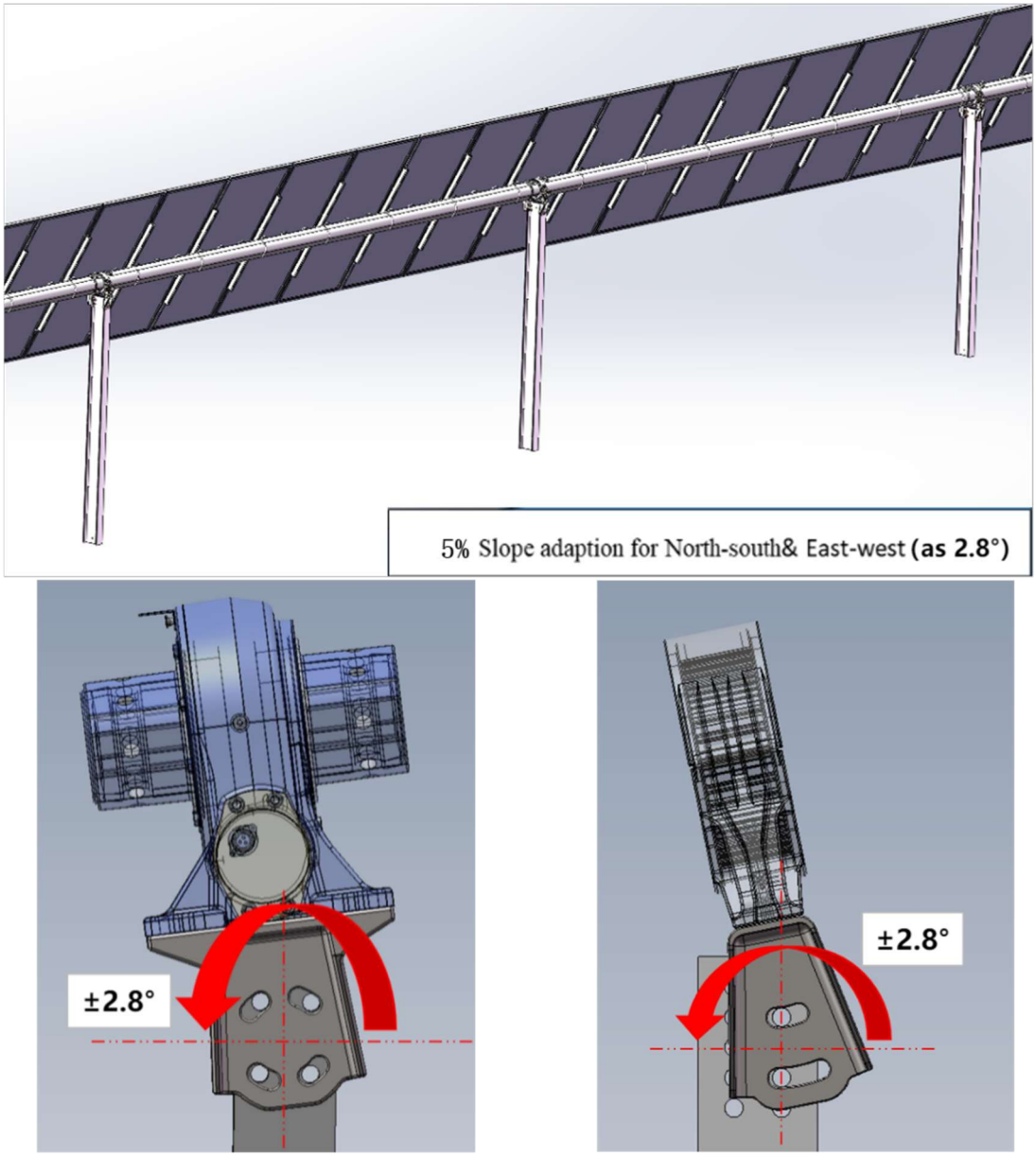
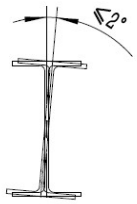


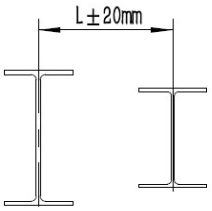
Figure 5.1.3-1: Schematic diagram of pile foundation slope adaption

(≤5% for North-south & East-west slope)

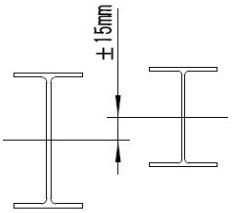
5.1.4 Allowable tolerance of foundation



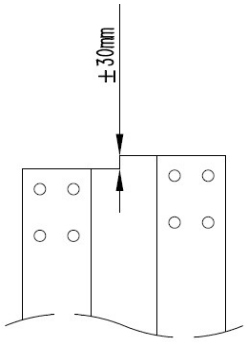
Torsional angle



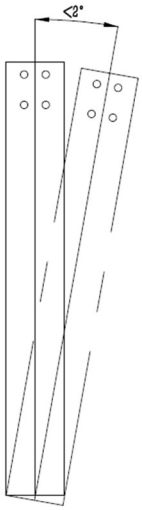
North-south direction span



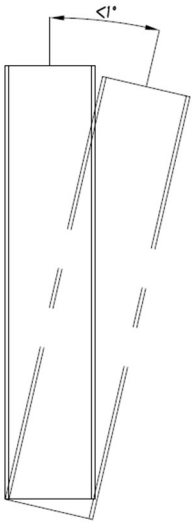
East-west direction span



Height difference



N-S Verticality



W-E Verticality

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5.1.5 Installation of Piles

5.1.5.1 Ramming piles

All procedures of ramming pile installation should also follow the user manual of ramming machine.



Figure 5.1.5-1 Direct Ramming Pile

Generally, the following procedure is required during the pile ramming:

1. A stake is nailed to the location of the pile.

2. The drive is straightened and kept upright in the marked place.

3. The plumb line of the pile is checked.

4. The pile driving hammer is lowered to the top of the pile.

5. Some light blows are given, and the plumb line of the pile is checked.

6. Full driving begins.

8. The pile is driven to the intended depth. In some cases, pile ramming stops.

9. The Engineer must monitor the penetration speed of the pile. Unusually high times may indicate rocks or bedrock. If the driving time is very long (greater than 4 min) and/or the head of the abutment is damaged, remove the abutment and apply the Pre-drilling procedure.

5.1.5.2 Pre-drilling with ramming piles

In case the need arises, the following procedure will be followed. The method is: the diameter of the hole would be less than the maximum sectional dimension of the profile to force the corners of the pile (shaded areas of the following figure) to deform, adapting to the circumference drilled.

The steps to follow in the use of predrilling are as follows:

1. Identify the location where pre-drilling will be performed.
2. Align the machine and drill bit, and mark the target depth.
3. Drill until the target depth is reached.
4. The pre-drilling diameter is around 60% of the outermost diagonal diameter of the steel pile, the final diameter should be determined by the test on site.
5. Position and align both the pile and the pile driving machine.
6. Drive the pile until the target depth is reached.
7. Pre-drill holes should be backfilled after ramming process.

NOTES:

Upon refusal, extract the steel pile and pre-drill a hole to enable smooth pile installation.

The pre-drilling hole shall reach the designed depth.

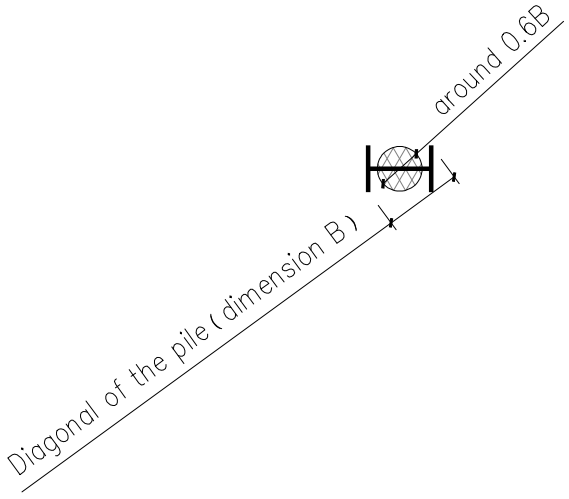


Figure 5.1.5-2 Pre-drilling Diameter Illustration

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5.1.5.3 Pre-drilling with concrete piles

In case the pre-drill diameter increases up to larger than diagonal dimension of the pier cross section, concrete encased pile solution should be used.

The steps to follow in the use of predrilling are as follows:

1. Locate the point where predrilling will be carried out.
2. Align the machine as well as the drill bit and mark the target length.
3. Drill until the target length is reached.
4. Remove the existing soil completely from the hole.
5. Place and Align the Steel pile with the setting template.
6. Piles to be checked with Plumb line.
7. Leave it for at least a minimum of 14 days after casting.

Along with that, 15 Nos. of cylindrical concrete specimens need to be cast and do the same curing process minimum for 14 days, specimens need to be tested in laboratory[3].

Please ensure that the compressive strength of each cylindrical concrete specimen is documented in comparison with the POT results.

The final pile solution should be determined basing on the soil condition and POT test and approved by project owner and manager.

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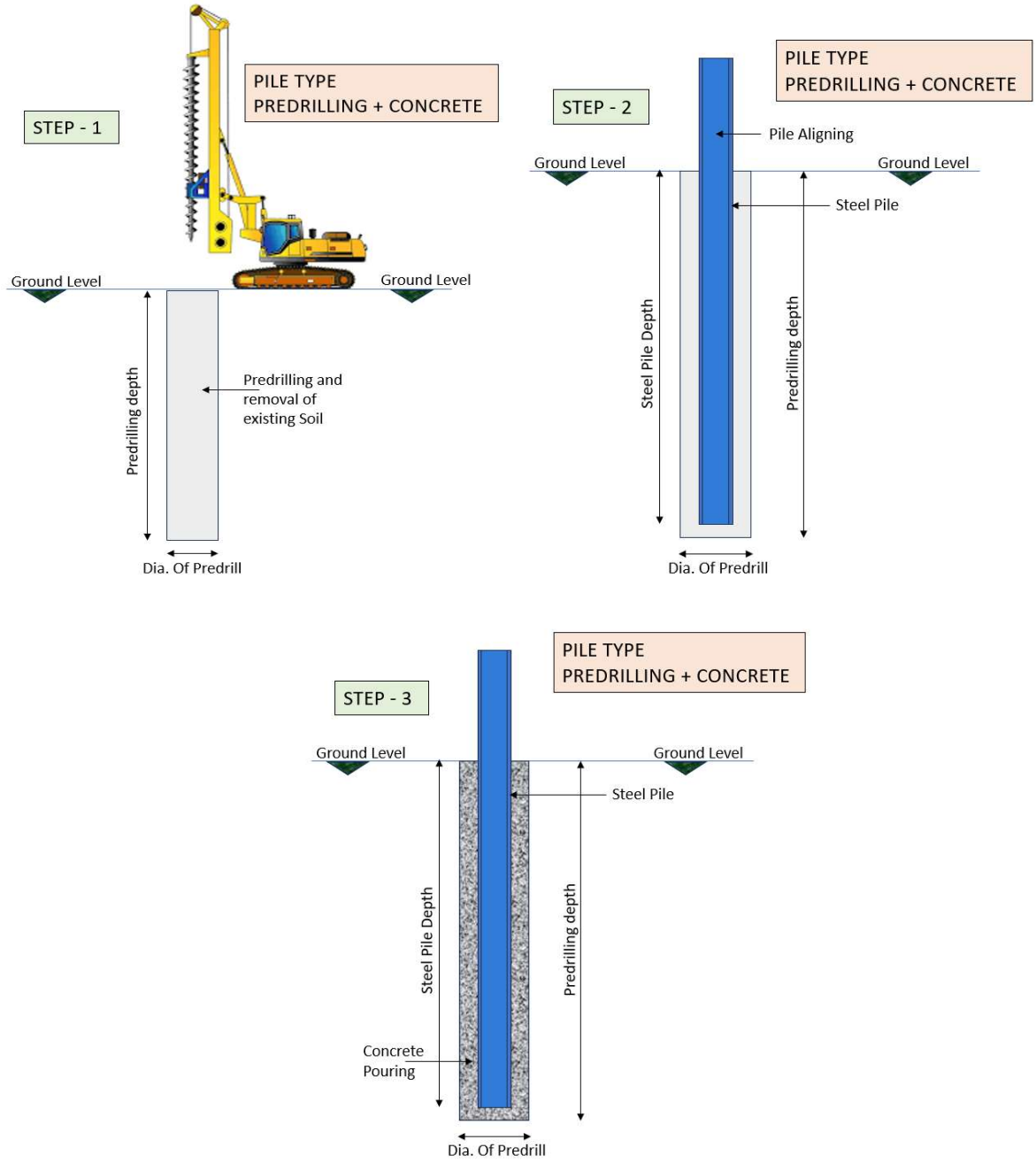


Figure 5.1.5-3 Pre-drill with Concrete Pile

NOTES:

For concrete-encased steel pile foundation, tests should be conducted after the concrete has completely cured to specified strength: the average compressive strength of the concrete shall equal or exceed 85% f'c (empirically, the curing time is a minimum of 14 dry days).

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5.2 Installation of posts (Step 2)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: leveling instrument, and related equipment
3. Embed the posts into the ground according to the designed embedment depth, and measure the levelness and verticality with a leveling instrument;

Remarks:

① The installation direction of posts is the north and south;
② Refer to **Appendix 2.1:Installation of posts** for details on foundation spacing and height requirements.;
③ Refer to **Appendix 2.1:Installation of posts** for post installation location and spacing.
④ *Take the following model as an example for reference.

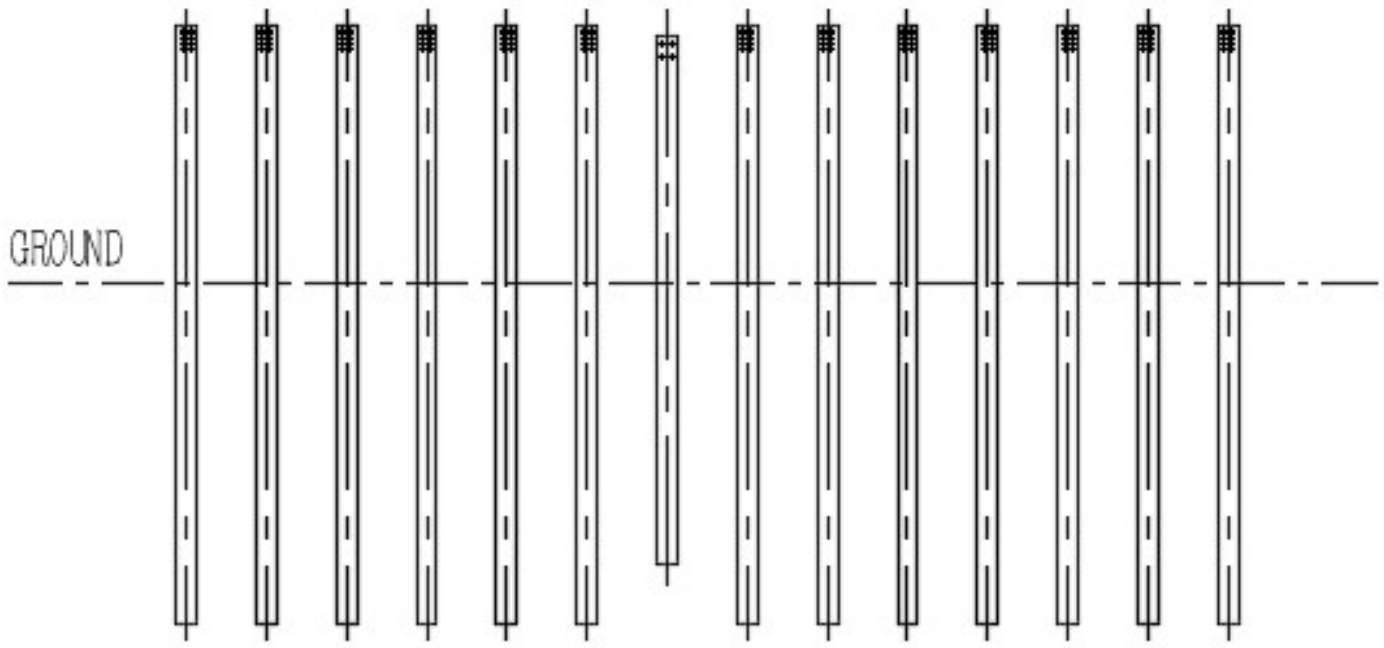


Figure 5.2-1: Schematic diagram of pile foundation spacing.

For detail information, please check Pile foundation construction guidance.

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5.3 Installation of reducer base and bearing base (Step 3)

1. Prepare the reducer base mounting parts and bearing base mounting parts, and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 21mm open wrench * 2, 24mm open wrench * 2, leveling instrument, and related equipment.

Remarks:

① The plane formed by the upper surface of the reducer base shall be parallel to the elevation plane of the non-drive post.

② ☆ After installation, measure the level of the both ends of the reducer driving shaft, the tolerance of torsion angle is ≤ 0.5°.If the angle cannot be adjusted within the tolerance range, report promptly.

③ The 5 holes near the reducer side is preferred when installing the bearing base, if there is interference, can be replaced to the other side of the 5 holes, the bearing base is preferred to be installed in the second and fourth holes, if there is a height difference during installation, the adjustable range is ± 25mm, as shown in Figure 5.3-6.

④ ☆ All double-layer self-locking washers shall be installed on the nut side.

Table 5.3-1: Table of materials required for the installation of the reducer base

NO.	Description	Specification	Remarks
1	Slewing reducer base	BP064	
2	Hex bolt	M16 * 50 * 30 (1 flat washer, 1 double stack self-lock gasket, 1 nut)	Torque 145~193 N.m

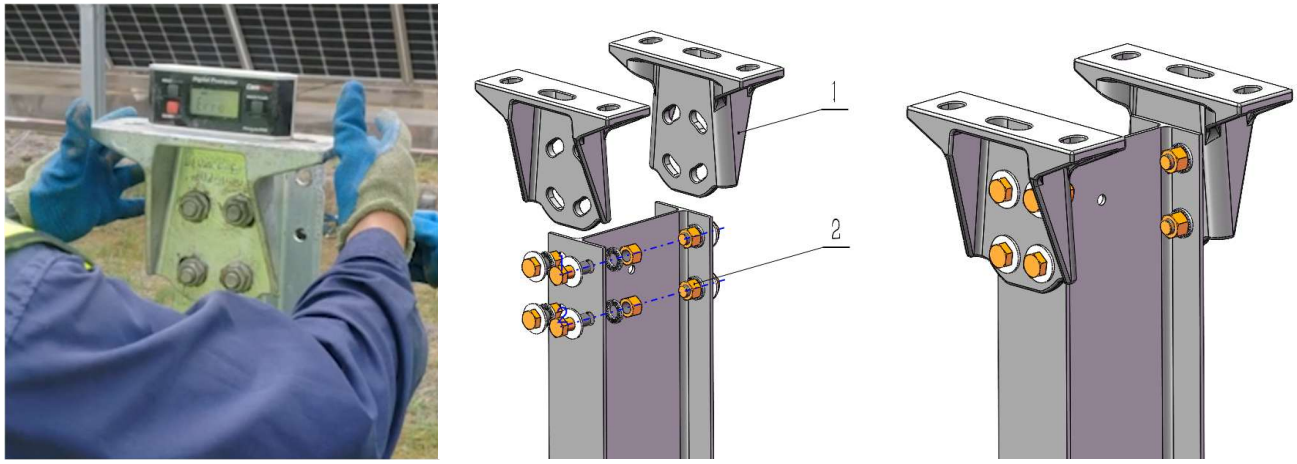


Figure 5.3-2: Installation diagram of reducer base

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Table 5.3-3: Table of materials required for the installation of the reducer base

NO.	Description	Specification	Remarks
1	Slewing reducer base	BP088	
2	Hex bolt	M14 * 40 * 30 (1 flat washer, 1 double stack self-lock gasket, 1 nut)	Torque 93~124 N.m

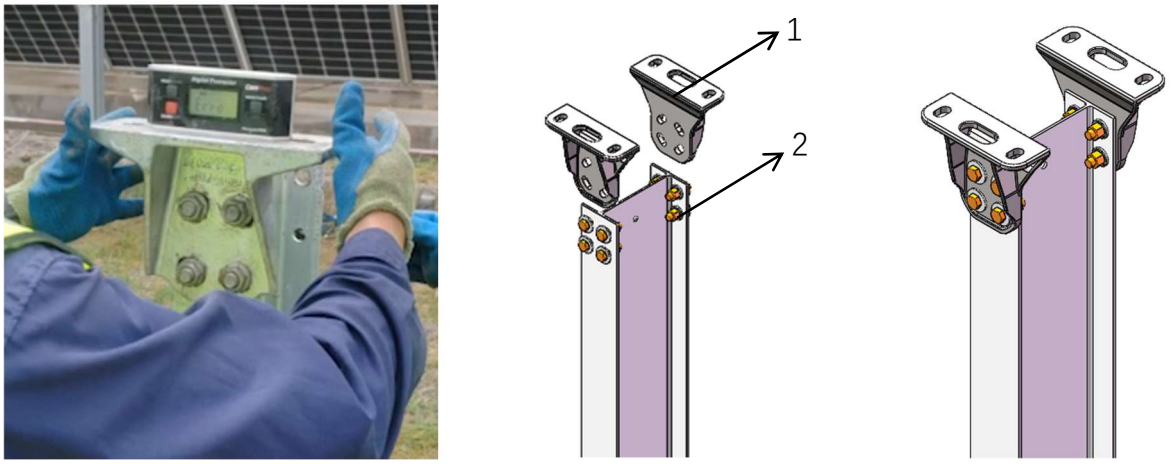
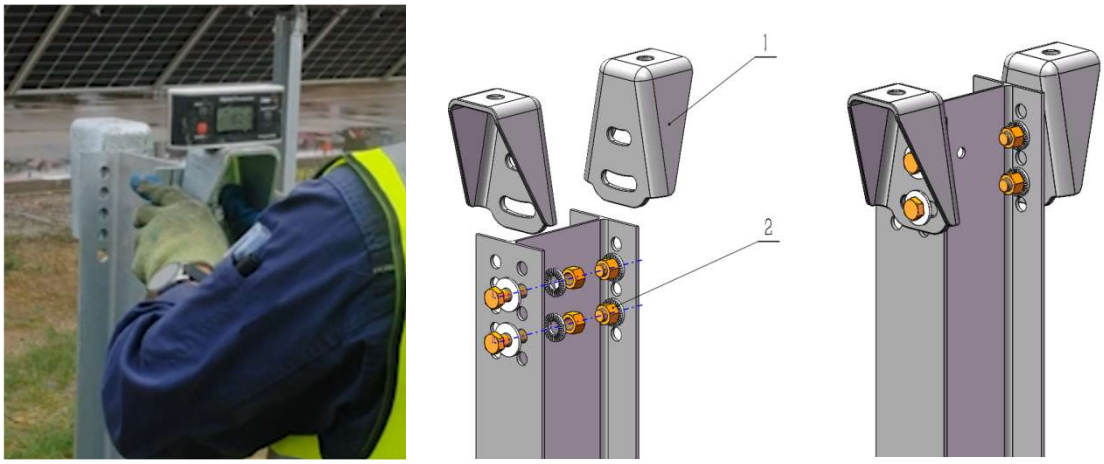


Figure 5.3-4: Installation diagram of reducer base

Table 5.3-5 Materials required for bearing base installation

NO.	Description	Specifications	Remarks
1	Bearing base	BP036H	
2	Hex bolt	M14 * 40 * 30 (1 flat washer, 1 double stack self-lock gasket, 1 nut)	Torque 93~124 N.m



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Figure 5.3-6: Schematic diagram of the double stack self-lock gasket installation

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5.4 Installation of slewing reducer (Step 4)

1. Prepare reducer installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 27mm open wrench * 2, leveling instrument, and related equipment
3. Place the reducer on the reducer base (☆ motor facing east), install bolts, tighten by hand; measure and adjust the level in both directions with a leveling instrument, and tighten all bolts to the specified torque.

Table 5.4-1: Materials required for reducer installation

NO.	Description	Specifications	Remarks
1	Driving post	Refer to **Appendix 2.1: Installation of posts**	
2	Slewing reducer base	BP064 / BP088	
3	Hex bolt	M18 * 80 * 50 / M16 * 75 * 50 (2 big flat washer, 1 spring washer 2 nut)	
4	Slewing reducer		

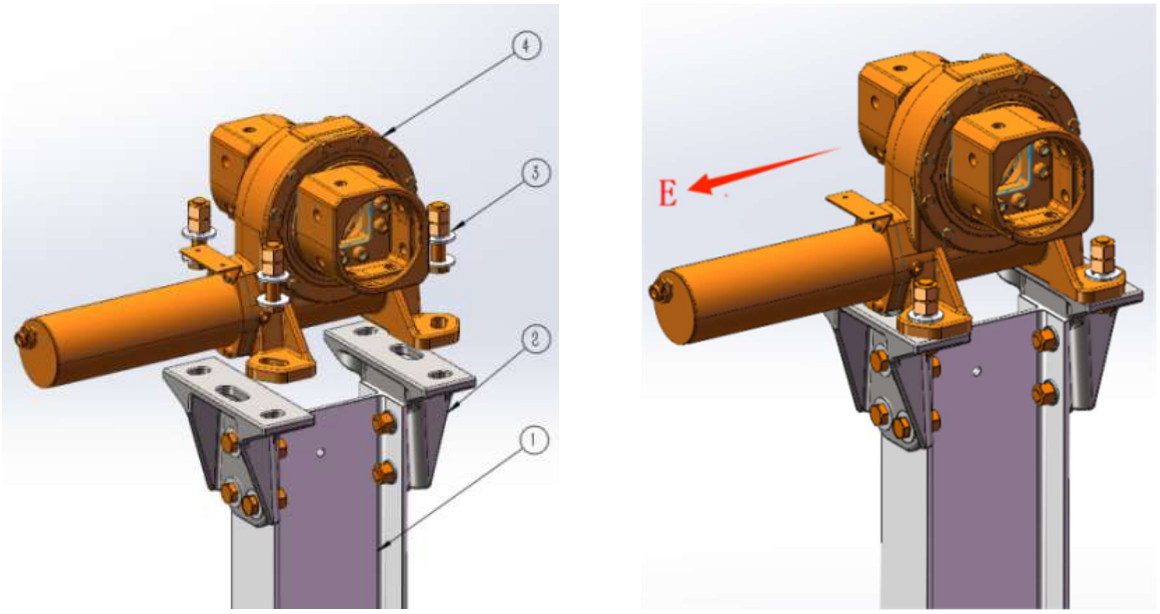


Figure 5.4-2: Explosion diagram of reducer installation

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5.5 Installation of bearing housing (Step 5)

1. Prepare bearing housing parts and check the dimensions and quantity according to the product BOM and the engineering drawings;
2. Prepare installation tools: 16mm /24mm open end wrenches, and related supporting equipment.
3. Install bearing housing with bearing base and open the upper cover first;
4. After the torque tube is installed, slide the bearing onto it, push it to the position of the bearing housing until it seats into the housing, and finally secure the upper cover of the bearing housing.
5. ☆ The installation direction of the bearing housing refers to Figure 5.5-4.
6. ☆Two types of bearing housings of different sizes are compatible with different models of posts respectively.，refers to Figure 5.5-2.

Table 5.5-1: Materials required for bearing housing installation

NO.	Description	Specifications	Remarks
1	Bearing housing	BP094-200	
2	Bearing housing cover	BP090	
3	Hex bolt with pin	M10 * 90 * 26 (1 nut, 1 cotter pin 2.5 * 18)	
4	Hex bolt	M16 * 45 * 30 (1 big flat washer 1 double self-lock gasket 1 nut)	Torque 145~193 N.m
5	Bearing base	BP036H	
6	Polymer bearing	BP092	



Figure 5.5-2 bearing housing

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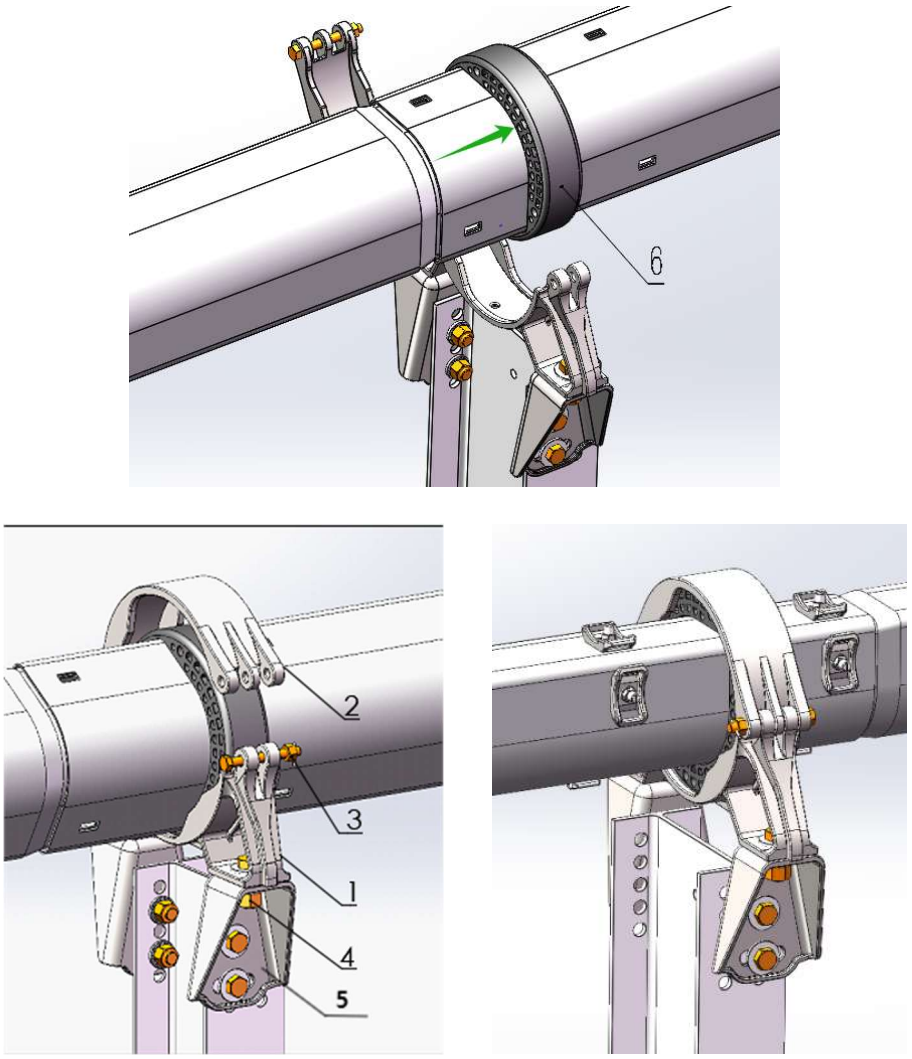


Figure 5.5-3 Installation explosion of the bearing housing

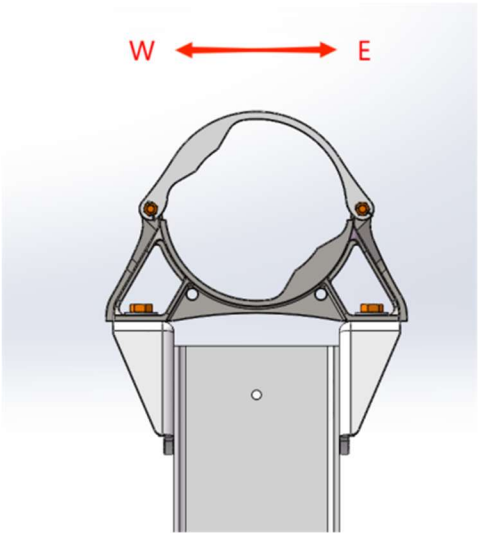


Figure 5.5-4 Mounting direction of the bearing housing

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5.6 Installation of the torque tubes (Step 6)

5.6.1 Installation of the torque tubes on both sides of the reducer

1. Prepare the torque tube installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 24mm open end wrench, leveling instrument, rubber hammer, and related supporting equipment;
3. Insert the torque tube connecting the reducer from one side, install it to the reducer and tighten all bolts to the corresponding torque.

Remarks:

① When connecting the torque tube to the reducer, pay attention to the installation direction, the end with 8 φ 18 round holes will be connected to the reducer;

② The installation sequence of M16 bolts is shown in Figure 5.6.1-3;

③ ☆ The torque tube weld is installed on the lower side, a slot on the reducer shaft is left for the installation of the torque tube weld, as shown in Figure 5.6.1-4;

④ ☆ The torque tubes on both sides of reducer must be in same plane. Check the parallelism of the torque tubes on both sides with an angle instrument. After installation, the parallelism of the torque tubes on both sides of the reducer should not be greater than 0.5°, as shown in Figure 5.6.1-5;

⑤ Refer to the **Appendix 2.2 Torque tube installation process** for details on the installation locations of other torque tubes with different part numbers.;

⑥ After the installation of the solar tracker and the normal operation for three months, all the bolts connecting the torque tube should be inspected, and the bolts loosed must be tightened again.

⑦ Take the following model as an example.

Table 5.6.1-1: Materials required for torque tube installation

NO.	Description	Specifications	Remarks
1	Slewing reducer	Refer to **Appendix 5: BOM List**	
2	Torque tube reinforcing pads	BP099	
3	Torque tube	Refer to **Appendix 5: BOM List**	
4	Hex bolt	M16 * 30 * 30 (1 double stack self-lock gasket)	Torque 145~193 N.m

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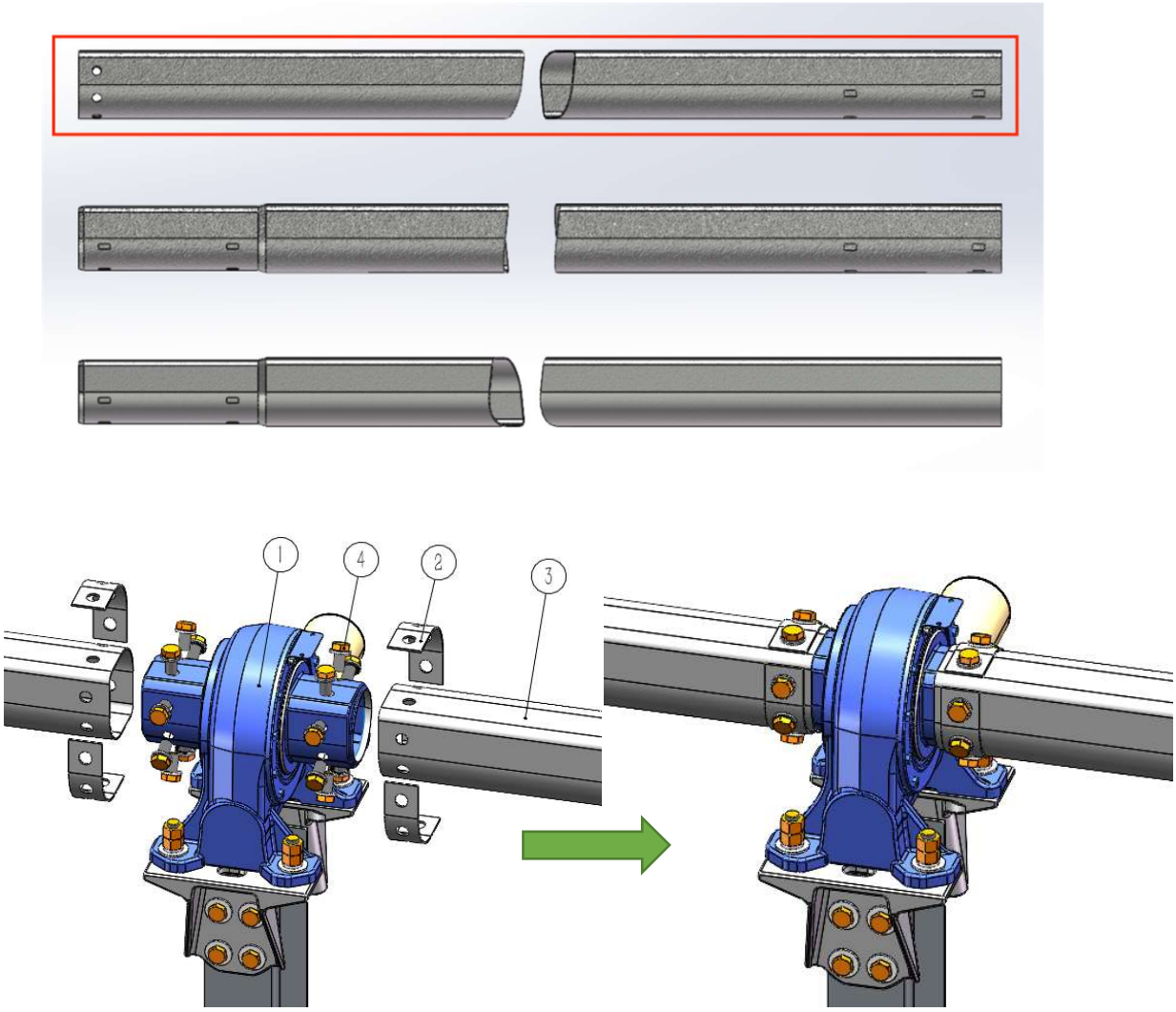


Figure 5.6.1-2: Explosion diagram and schematic diagram of the torque tube install to the reducer

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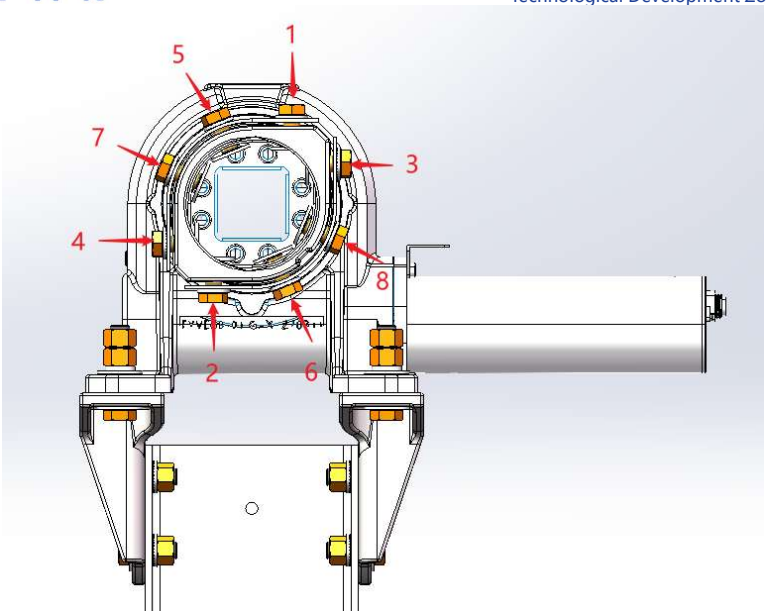


Figure 5.6.1-3: Schematic diagram of the installation sequence of the reducer bolts

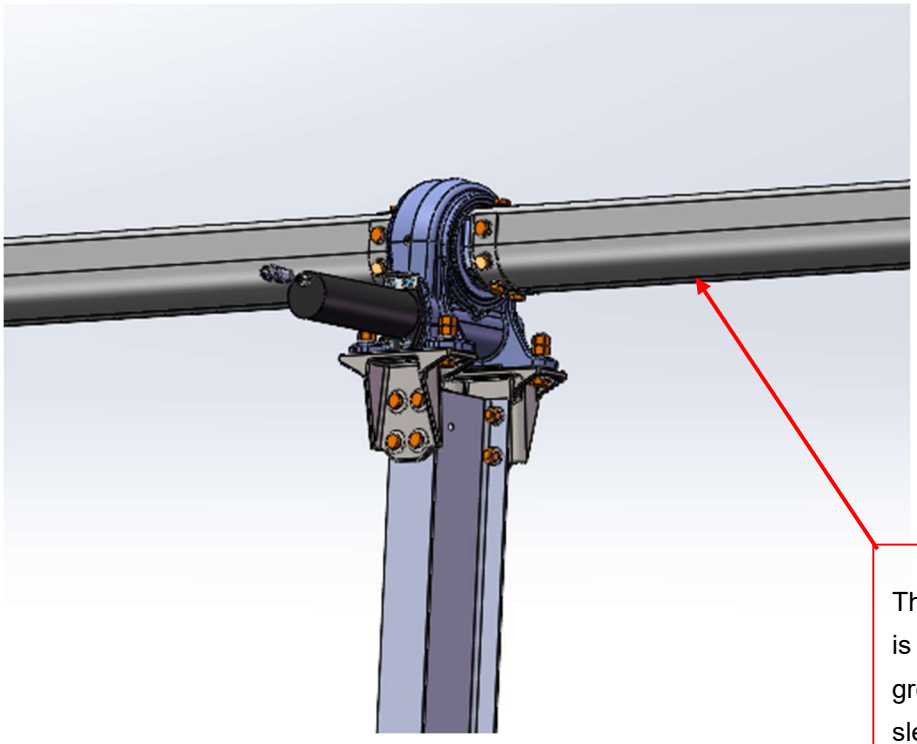


Figure 5.6.1-4: Installation diagram of torque tube weld

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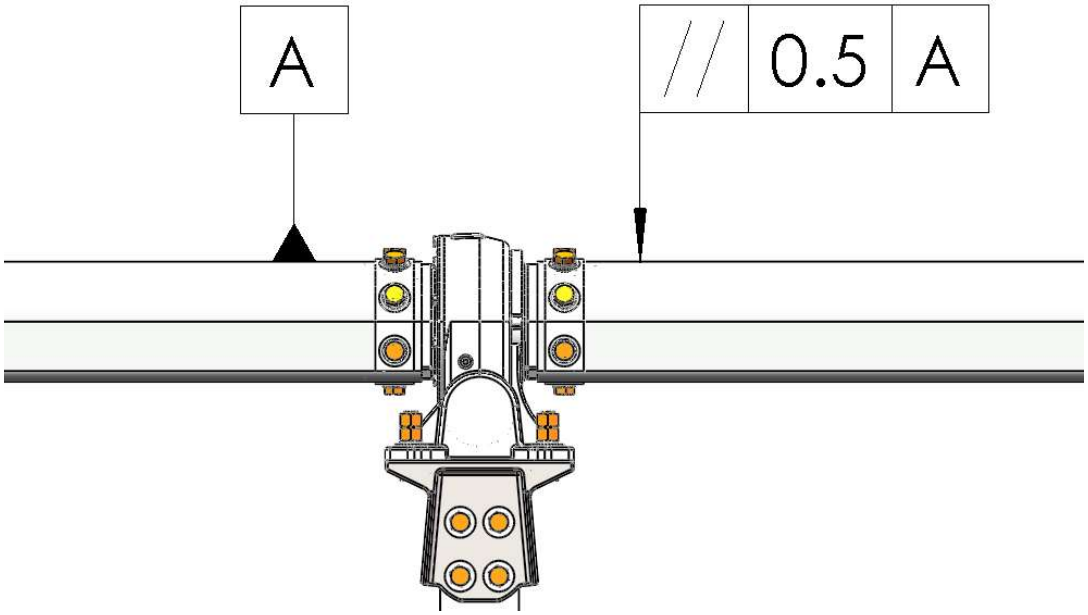


Figure 5.6.1-5: Schematic diagram of the parallelism of the torque tubes

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5.6.2 Installation of diameter reduced torque tubes

1. Prepare the installation parts such as the torque tubes and carriage bolt group, and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: 16mm socket wrench, leveling instrument, rubber hammer, tape, and related equipment
3. The two torque tubes are connected by neck reduced method. Firstly, the carriage bolts are pre-installed on the shear guide gasket, then the carriage bolt group is mounted into the square holes formed by the two torque tubes, and finally tighten the carriage bolts.

Remarks:

① The installation sequence and corresponding part numbers of the torque tube refer to **Appendix 2.2:Torque tube Installation Process**.

② When installing the M10 carriage bolt, the double stacked self-locking washer and nut should be pre-installed on the bolt, then the carriage bolt group is pre-installed on the shear guide gasket, then put the T-type head bolt into the rectangular hole of the torque tube connection, tighten it clockwise, so that the T-type head bolt is stuck in the square hole of the shear guide gasket, and finally fastened with electric wrench, as shown in Figure 5.6.2-4.

③ Before installation, the '—' slot of the bolt is parallel to the front and rear block of the gasket, and be perpendicular to the front and rear block after fastening, as shown in Figure 5.6.2-5.

④ When installing the carriage bolts on the torque tube connection, tighten it twice. The installation sequence is shown in Figure 5.6.2-6.

⑤ ☆ All bolts must be fastened to the corresponding torque value.

⑥ ☆ The torque tubes on both sides of reduced torque tube must be in the same plane. Check the parallelism & torsion angle of the torque tubes on both sides with an angle instrument. After installation,the parallelism between adjacent torque tubes shall not exceed 0.5°. The cumulative torsion angle error of the entire row shall be within ±2°,as shown in Figure 5.6.2-7.

⑦ ☆ Due to the cumulative tolerance of post spacing, when interference occurs between the Bearing Housing and Carriage Bolt, the positions of the mounting holes of the Bearing Base shall be adjusted, as shown in Figure 5.6.2-8.

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Table 5.6.2-1: Materials required for torque tube connection

NO.	Description	Specifications	Remarks
1	Torque tube	Refer to **Appendix 2.2:Torque tube Installation Process**	
2	Carriage bolt	MC10 * 31 * 29 (1 double-stack self-lock gasket 1 nut)	Torque 33~45 N.m

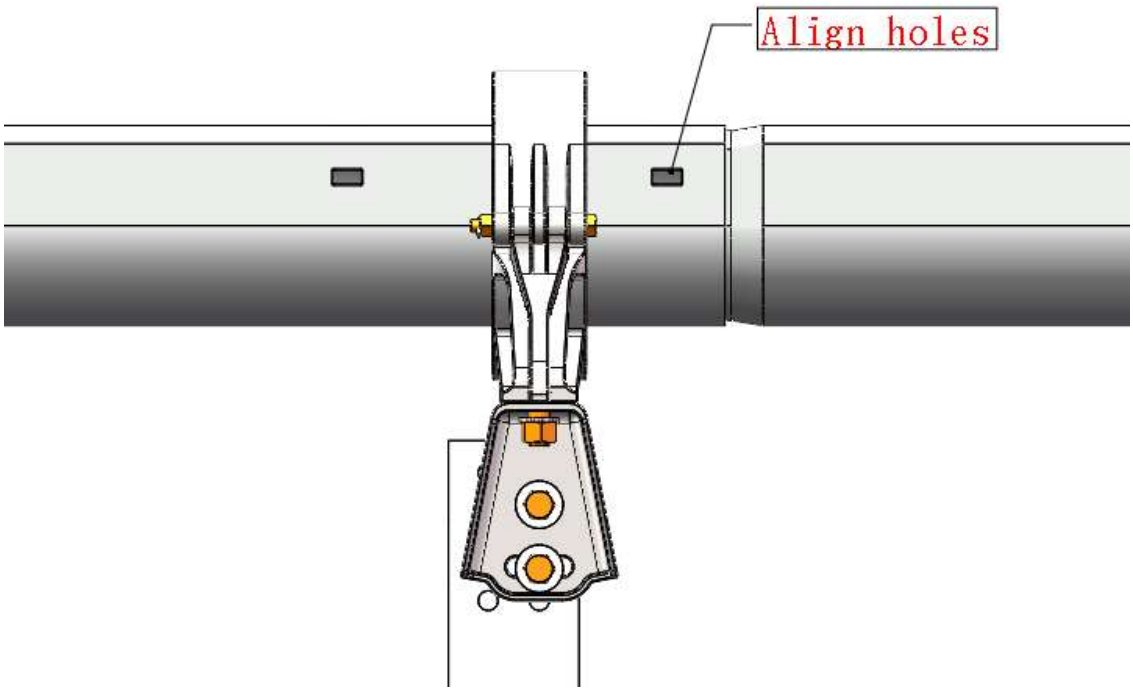


Figure 5.6.2-2: Holes position diagram of diameter reduced torque tubes

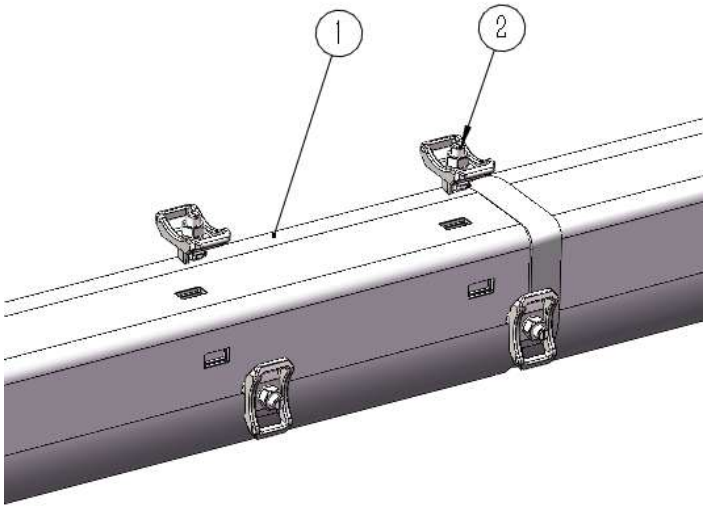


Figure 5.6.2-3: Explosion diagram of diameter reduced torque tubes installation

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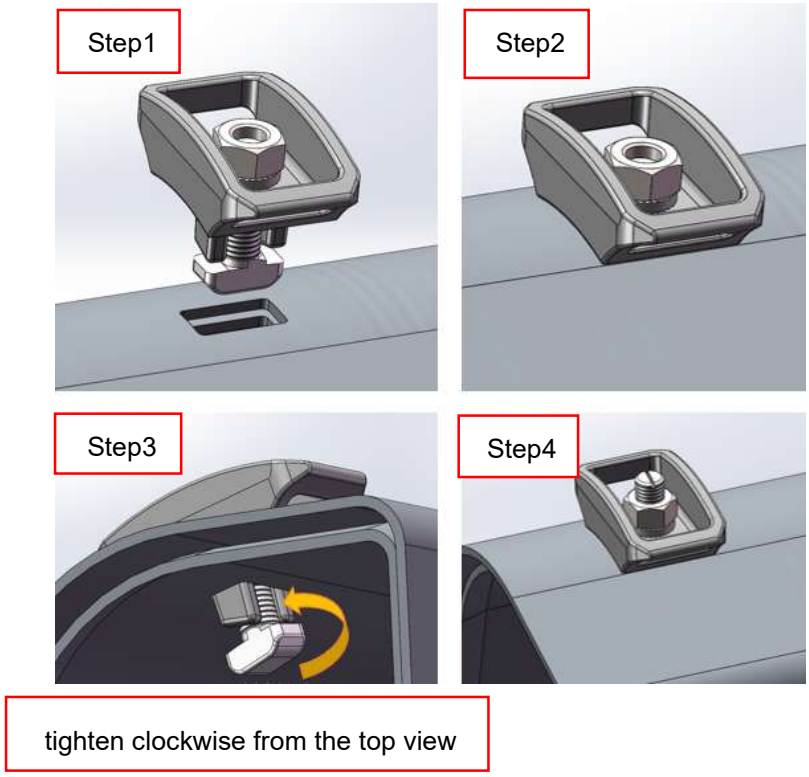


Figure 5.6.2-4: Installation diagram of carriage bolts

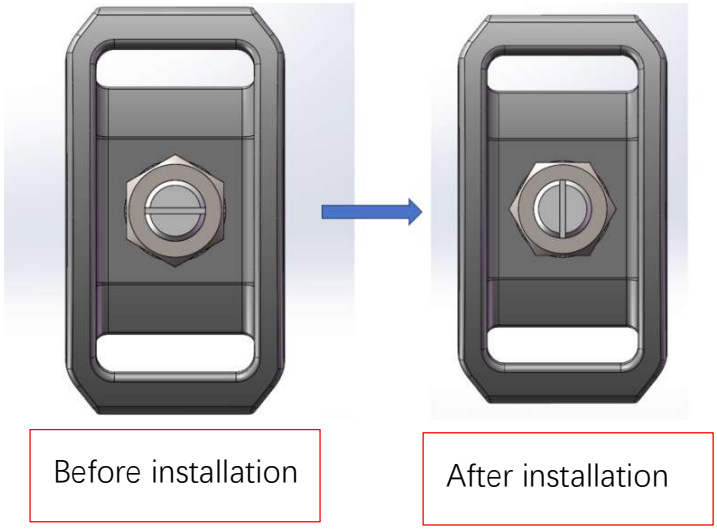


Figure 5.6.2-5: Installation diagram of carriage bolts

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Table 5.6.2-6: Materials required for torque tube connection for 1Px70

NO.	Description	Specifications	Remarks
1	Torque tube	S156	
2	Torque tube hoop	-4.5xS156	
3	Hoop spacers	-3x37	
4	Hex bolt	M12 * 50 * 40 (1 flat washer 1 double self-lock gasket 1 nut)	
5	Carriage bolt	MC12 * 35 * 30 (1 big flat washer 1 spring washer 1 nut) Big flat washer: -3x ϕ 30x ϕ 13.5	

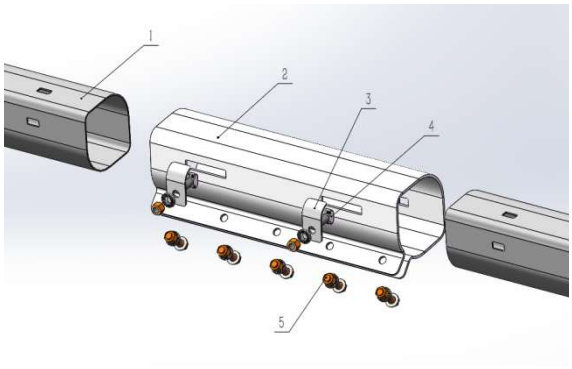


Figure 5.6.2-7.: Explosion diagram of torque tubes installation with hoop

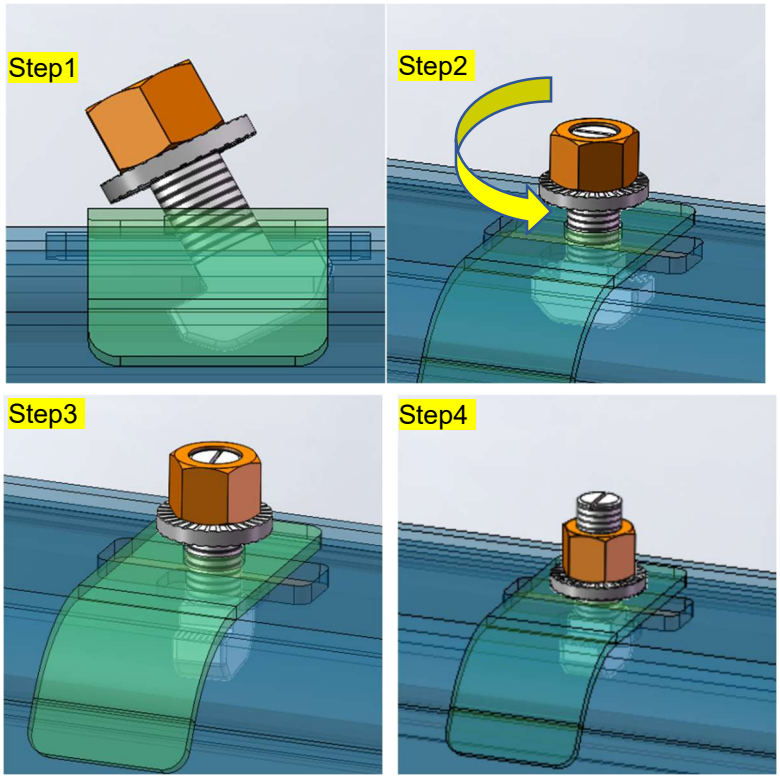


Figure 5.6.2-8.: Installation diagram of carriage bolts

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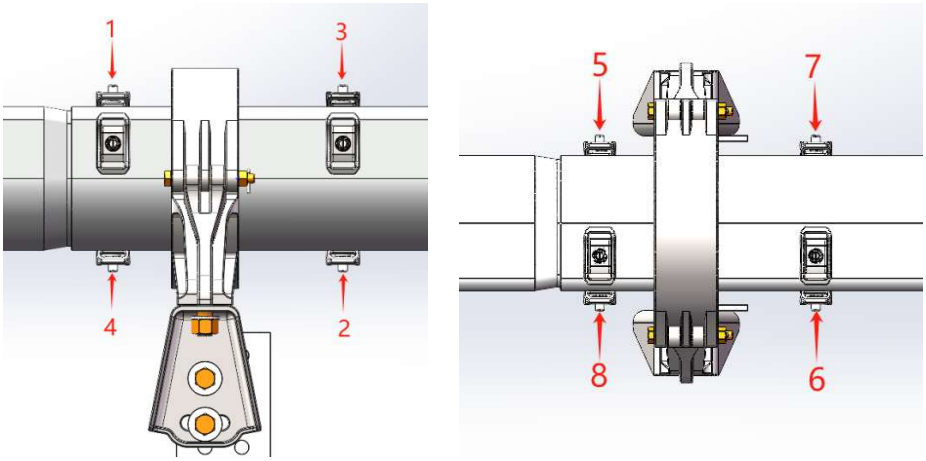


Figure 5.6.2-9: Schematic diagram of the installation sequence of the carriage bolts

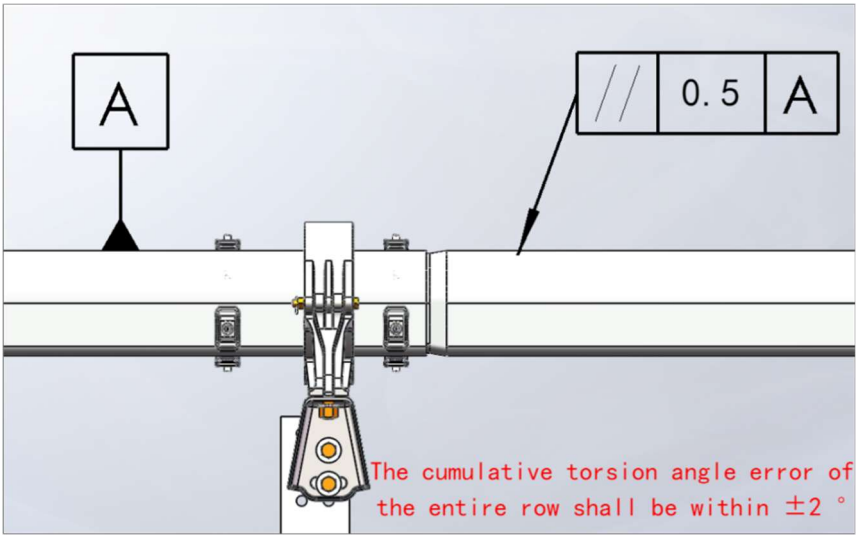


Figure 5.6.2-10: Schematic diagram of the parallelism of the torque tubes

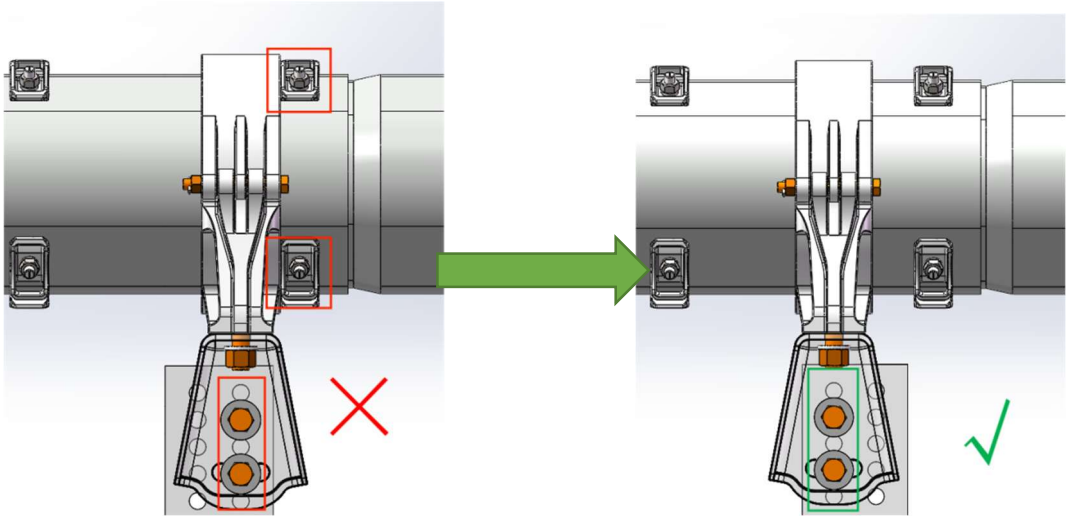


Figure 5.6.2-11: Adjust the positions of the mounting holes when the Bearing Housing and Carriage Bolt interfere.

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5.7 Installation of purlins (Step 7)

1. Prepare the installation parts, and check the size and quantity according to the product BOM and engineering drawings in the attached documents;
2. Prepare installation tools: 16mm open wrench*2, leveling instrument, rubber hammer, tape and related equipment;
3. Purlins shall be installed in sequence from the middle of the bracket row to both sides. During installation, their angle shall be adjusted and secured.

Remarks:

① Purlin assemblies are installed onto the torque tube and secured, for the subsequent installation of PV modules,Mounting dimensions refer to **Appendix 2.5: Installation of purlins and PV modules**.

② Before and after tightening, it is necessary to ensure that the length direction of the purlin is perpendicular to the axial direction of the tube.

③ When tightening the U bolt, the bolts on both sides shall be tightened alternately and gradually. The bolts on one side shall not be fully tightened directly to the specified torque value, so as to prevent the U bolt from shifting to one side.

④ The outermost purlins at the north and south ends shall extend beyond the PV modules, and the modules at these positions shall be installed in the bolt holes on the inner side of the respective purlins.Refer to Figure 5.7-14.

Table 5.7-9: List of materials for tracker purlin installation

No.	Description	Specifications	Remark
1	Purlin	Refer to **Appendix 2.5: Installation of purlins and PV modules**	
2	U bolt	UM10*166*188*30	
3	Purlin pad block	BP095	
4	Purlin reinforcement gasket	- 3*22*35	
5	Flange nut	M10	Torque 25~32 N.m

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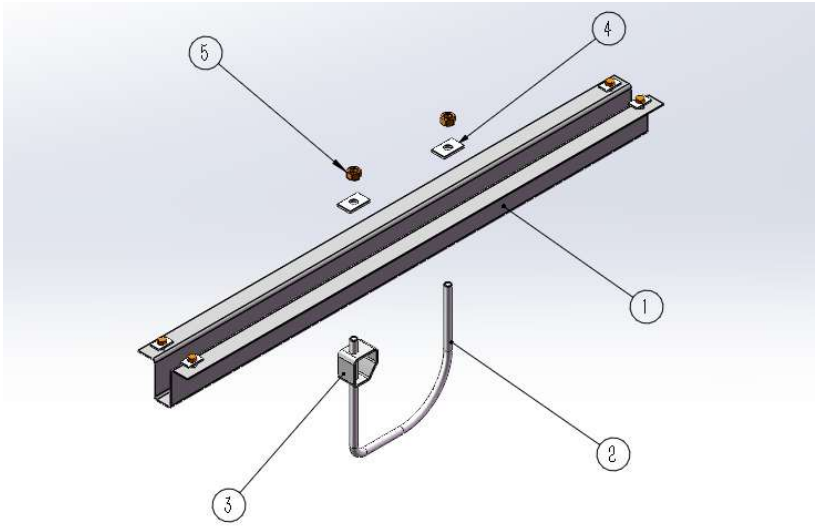
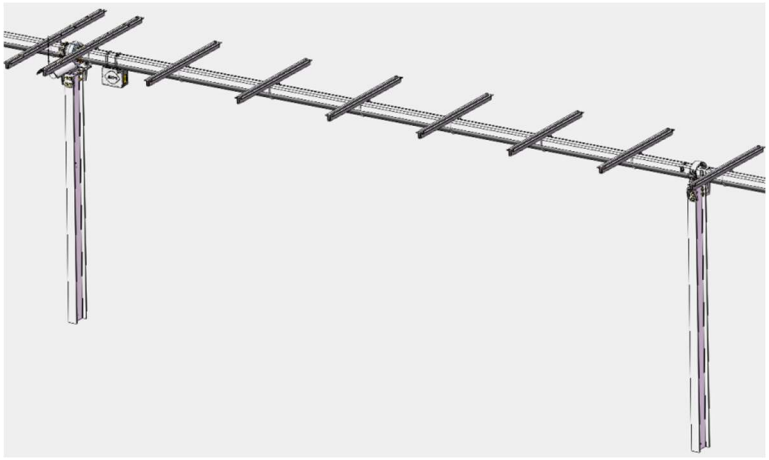


Figure 5.7-10: Installation and explosion diagram of purlin system



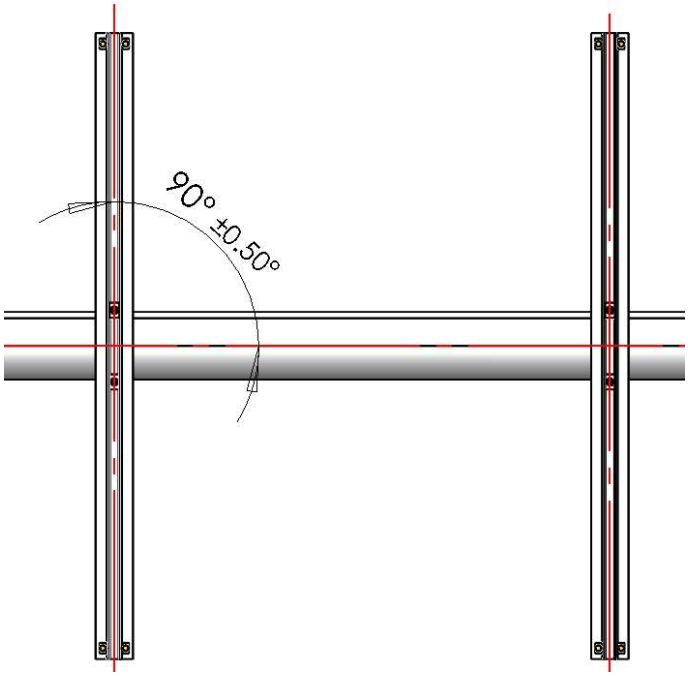
Figure 5.7-11: Pre-installation drawing of purlin system



1. The picture does not mean the real quantity of purlins.
2. Purlins shall be installed in sequence from the middle of the bracket row to both sides. During installation, their angle shall be adjusted and secured.

Figure 5.7-12: Installation drawing of purlin assemblies

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1. Before and after tightening,it is necessary to ensure that the length direction the purlin is perpendicular to the axial direction of the tube, the permissible tolerance range is as shown in Figure 5.7-13.

Figure 5.7-13: Purlin installation diagram

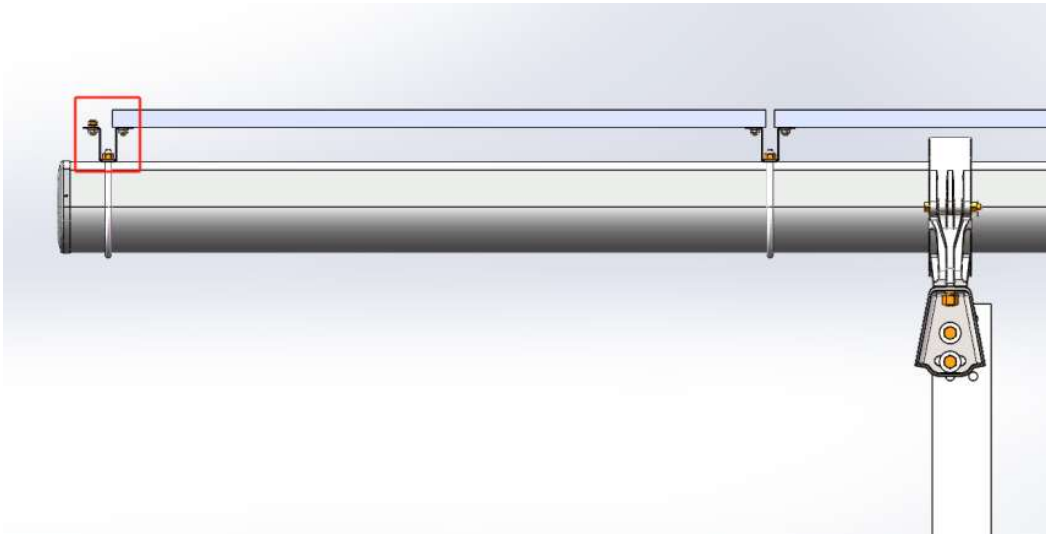
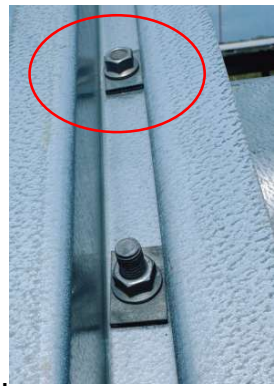


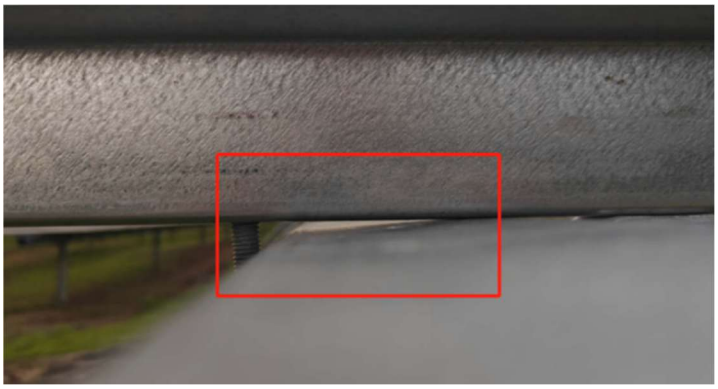
Figure 5.7-14: End purlin installation diagram

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Troubleshooting:

These bolt fastening methods shown in the images below are not permitted, as they reduce the contact area between the purlin and the torque tube (as illustrated in the image below), which in turn weakens the overall structural strength of the tracker.





Adjustment Methods:

For trackers without PV modules installed:

Use an electric wrench to adjust the nuts on the U-bolts. After adjustment, the exposed thread depth on both nuts should be greater than 2 turns.

The image below shows a correct installation example:



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5.8 Installation of the dampers (Step 8)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: 18mm open wrench * 2, 21mm open wrench * 2, leveling instrument, needle-nose pliers, and related supporting equipment
3. Install the damper lower fixture on the damper post; install the damper hoop on the torque tube, and tighten the nut by hand first; install the damper between the damper hoop and lower fixture, then fix the bolt with a pin (see Figure 5.8-6、5.8-7)

Remarks:☆ When installing dampers: the large diameter end is on the upper side.

① ☆ The dampers are symmetrically installed.

② ☆ Note that the fixing positions of the damper lower fixture differ between the damper and damper-S. Relevant requirements shall refer to Figure 5.8-3.

③ ☆ When installing dampers at positions other than the connection nodes, prioritize tightening the bolts at the damper lower fixture first. Then, confirm the installation position of the damper hoop based on the length of the damper upon delivery, and complete the installation after confirming the position (refer to Figure 5.8-4).

④ At the torque tube connection nodes, the installation position of the damper refers to Figure 5.8-5.

⑤ ☆ Insert the cotter pin into the hole of the bolt, split the pin and the angle of the pin is bigger than 180°, as shown in Figure 5.8-6.

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Table 5.8-1: Materials required for damper installation

NO.	Description	Specifications	Remarks
1	Damper post	Refer to **Appendix 5: BOM List**	
2	Damper fixture	BP168	
3	Round tube of damper	Φ20*2.0	
4	Damper	/	
5	U Bolt	UM12*168*195*35	Torque 45~55 N.m
6	Long Purlin pad block	BP095L	
7	Damper lower fixture	BP093	
8	Hex bolt	M12 * 40 * 25 (1 double stack self-lock gasket 1 nut)	Torque 58~78 N.m
9	Hex bolt	M14 * 80 * 25 (1 slot nut 1 cotter pin 4 * 32)	

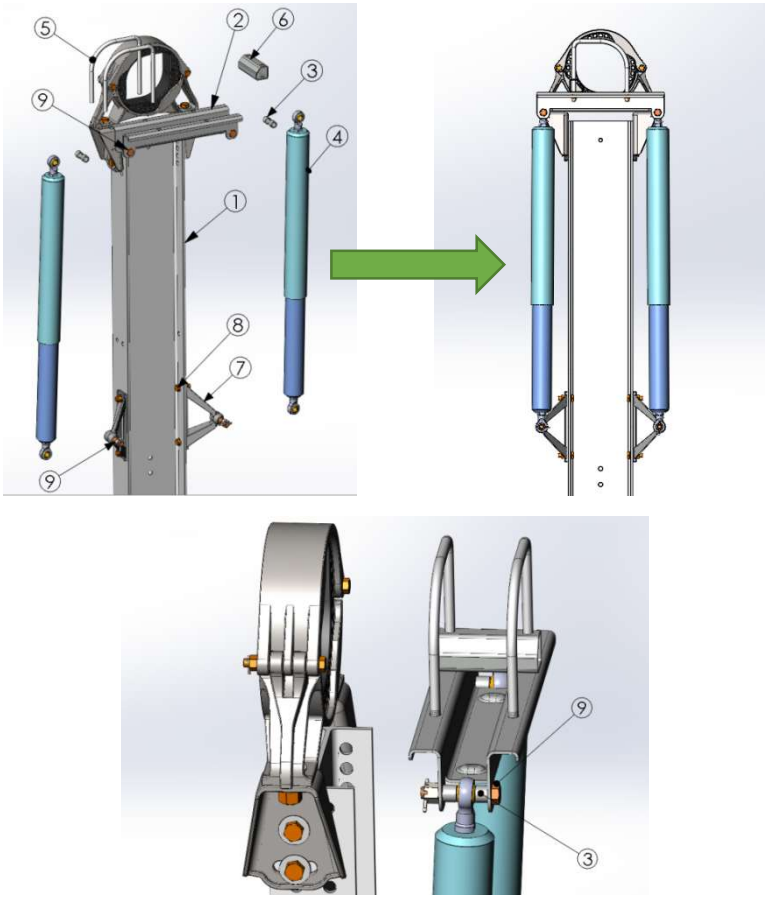


Figure 5.8-2: Schematic diagram of the damper installation

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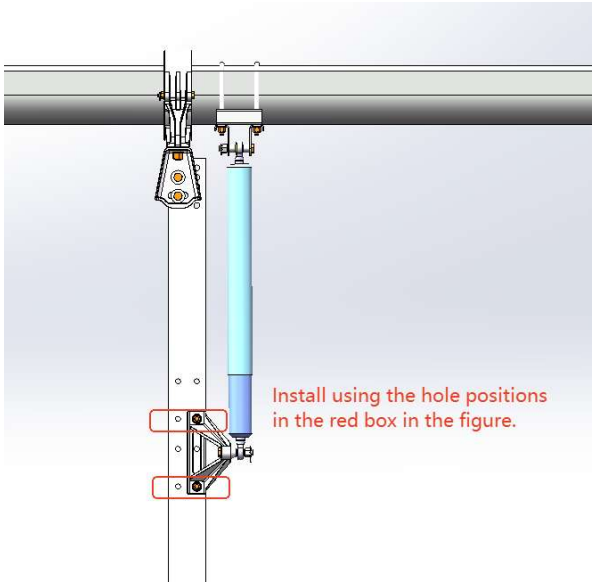


Figure 5.8-3: Schematic Showing the Differences in Installation Positions of the Damper Lower Fixture

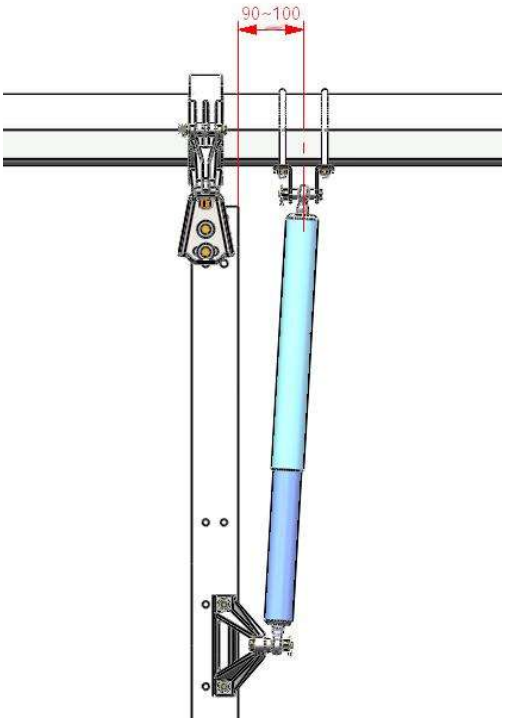
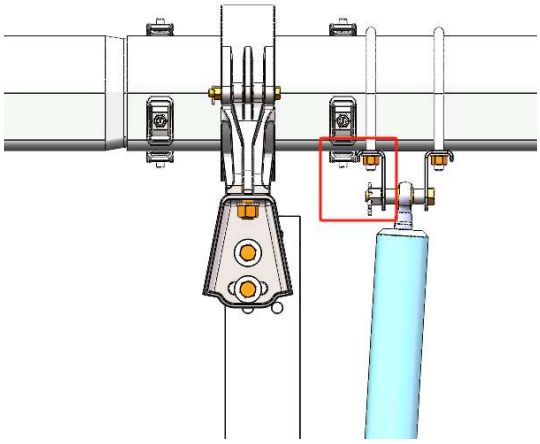


Figure 5.8-4: Schematic Diagram of Damper Installation at Positions Other Than the Connection Node of the Torque Tube

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1. The edge of the damper fixture shall fit closely against the edge of the carriage bolt.

Figure 5.8-5: Schematic Diagram of Damper Installation at the Connection Node of the Torque tube

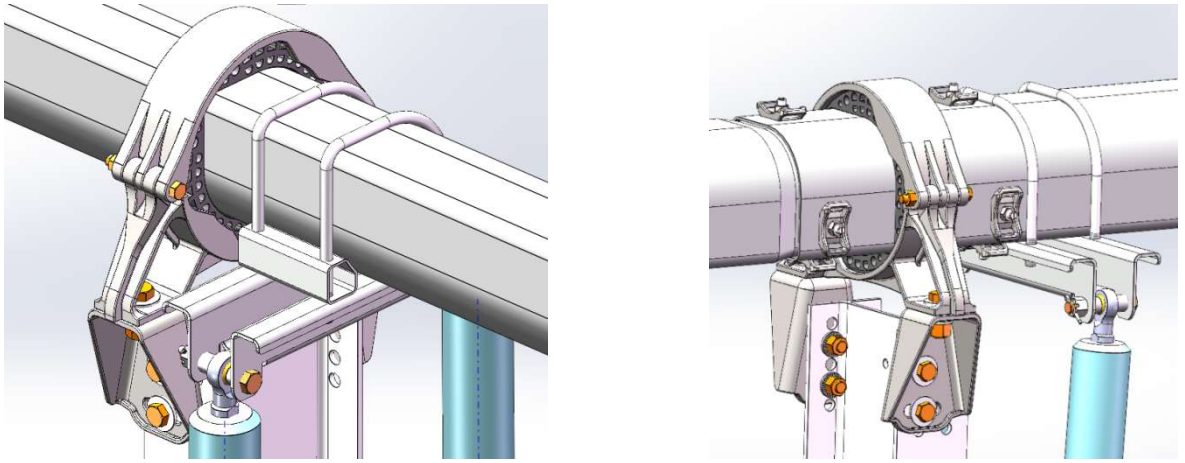


Figure 5.8-6: Diagram of damper hoop installation

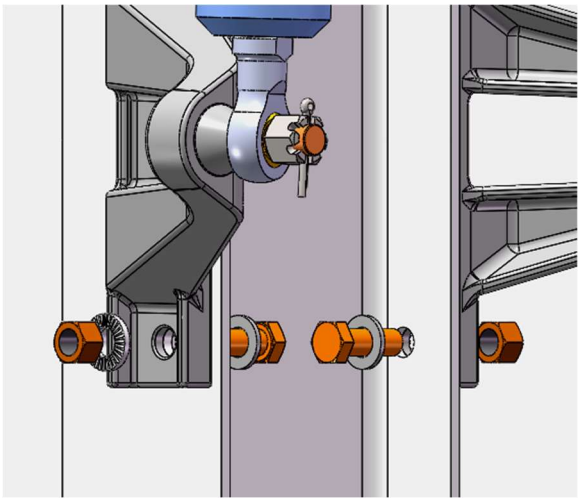


Figure 5.8-7: The connection diagram of the damper lower fixture

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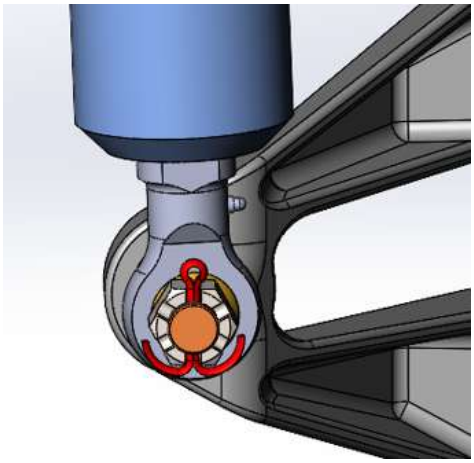


Figure 5.8-8: Drawing of cotter pin installation

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5.9 Installation of TCU and wireless receiver (Step 9)

5.9.1 Installation of TCU

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents.
2. Prepare the installation tools: 6mm Allen wrench, and related supporting equipment
3. First, install the TCU hoop onto the torque tube, and then fasten the M8 bolts on the hoop to fix the TCU onto the torque tube.
4. The distance between the center of the TCU and the center of reducer should be controlled at 600 mm.

Remarks:

① Take waterproof, moisture-proof and other protection measures before the installation of the TCU;

② When installing the TCU, please pay attention to the label on the back of the TCU. The channel address has been pre-configured before the factory. During installation, ensure that the code on the TCU corresponds with the code marked in the Layout drawing；

③ The control box（TCU）is usually installed on the north side of the reducer, the 'N' symbol on the shell points to the north;

④ After the installation, fix the motor cable on the motor; the length of the cable should be slightly longer considering the rotation of the tracker.

Table 5.9-1: Materials required for installation of TCU

NO.	Description	Specifications	Remarks
1	Controller box（TCU）	Refer to **Appendix 5: BOM List**	
2	Hoop		

5.9.2 Installation of the wireless receiver

1. For the installation method of the wireless receiver, refer to the **Appendix 2.4 Installation of the wireless receiver.**

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5.10 Installation of PV modules (Step 10)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: 10mm socket wrench, and related supporting equipment
3. Locate and tighten purlins. The interval from the center of the reducer to the purlin is L1, the interval from the purlin to the end of the torque tube is L2, and the interval between adjacent purlins is L3 and L4, refer to **Appendix 2.5.2: Installation of PV modules**. Screw the bolts on by hand first, and then tighten them with a power tool to the specified torque value when installing PV modules later (refer to Appendix 1)
4. Install the power supply module directly above the reducer, with their fixing purlins located on the north and south sides of the reducer respectively
5. Taking the reducer as the starting point, install the PV modules one by one from the center to both sides, with the spacing between the modules on both sides of the reducer as L4, the spacing between the end of the torque tube and the unilateral module as L5, and the spacing between the two PV modules as L6, refer to **Appendix 2.5.2: Installation of PV modules**.

Remarks:

① ☆ Installation method ： 2 people lift and 1 people install, bolted with hand first and then tighten them with electric wrench to the corresponding torque later.

② ☆ When fix the PV modules, ensure that the square gasket is installed on the PV module side . (see Figure 5.10-4)

③ ☆ The installation direction of the positive and negative poles of the PV module is "the positive pole is in the south".

④ ☆Refer to the Project Corresponding Installation Documents for the specific installation dimensions of L1~L7,refer to **Appendix 2.5.2: Installation of PV modules.**

⑤ After installation of PV modules, the tracker should be rotated to the 60° (wind protection angle), as shown in Figure 5.10-5.

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Table 5.10-1: List of materials required for the installation of PV modules

NO.	Description	Specifications	Remarks
1	PV modules	/	
2	Stainless Steel Hex Bolt	M6 * 14 * 14 (1 piercing washer 1 big flange nut)	Torque 8 N.m
3	Stainless Steel Hex Bolt	M8 * 16 *16 (1 piercing washer 1 big flange nut)	Torque 16 N.m

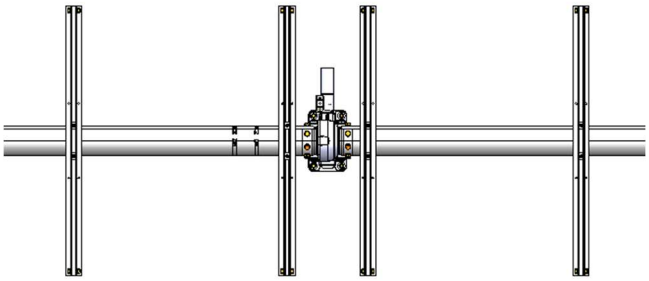


Figure 5.10-2: Schematic diagram of power supply module

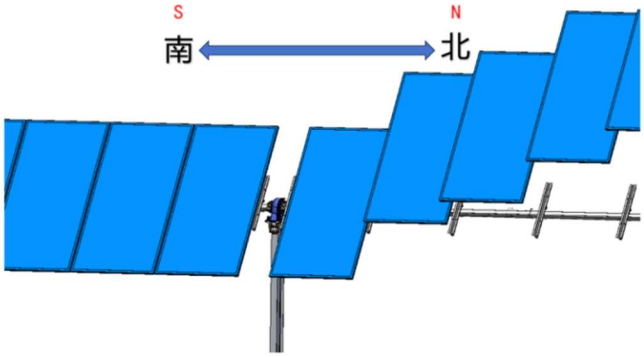


Figure 5.10-3: Schematic diagram of PV modules

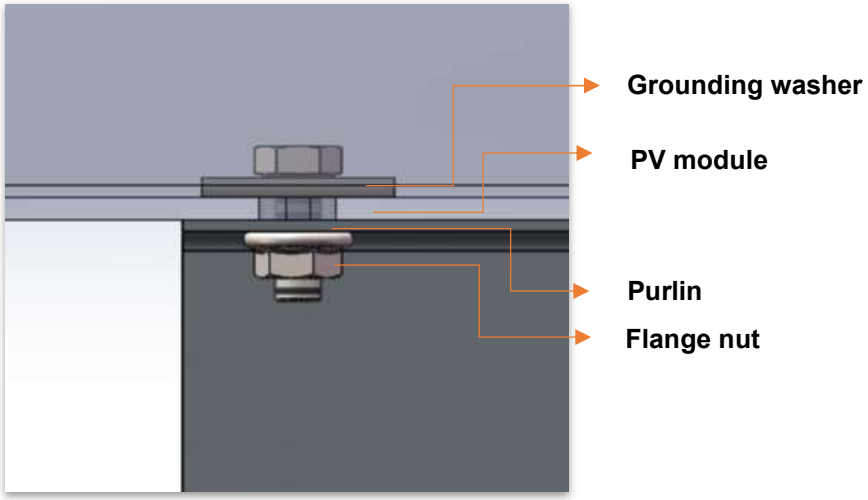


Figure 5.10-4: Installation diagram of PV module installation bolts

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5.11 Wiring of PV system (Step 11)

General rules for Wiring routing

1. The cable wiring should be beautiful and neat, the fixing should be firm, the reserved length should be moderate, and the cable should not be stressed when the bracket rotates.
2. Cables exposed to sunlight require a bellows.
3. When burying wiring, the buried depth is required to be not less than 0.8m.
4. Cables must be protected when entering and exiting the cable tray, conduit, and holes through metal partitions.
5. Workers must carefully and strictly adhere to the module manufacturer safety information for the proper installation, maintenance, and handling of the modules. All module manufacturer instructions and requirements must be read and understood before attempting to install, wire, and operate and/or service the module and accompanying electrical equipment.
6. This procedure provides simplified, schematic examples of recommended wire management configurations to avoid contact with tracker mounting components. All wire management must comply with IEC standards and local policy requirements for maintaining wires (service loops, cable trays, raceways, and so on).
7. Do not attach cable trays or shields to the underside of the panels. The maximum weight capacity attached to the back of modules is not to exceed 0.8 kg per meter.
8. ☆ For the arrangement and details of solar cables in the tracker, refer to the document ***SAU1004-G000-MQA-&ETL-00002 SOLAR CABLE ARRANGEMENT AND DETAILS IN TRACKER***

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5.12 Installation of grounding wire (Step 12)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Prepare installation tools: socket wrench, cross screwdriver, and related supporting equipment
3. Attach one end of the yellow-green ground wire to the web of the column and the other end to the Grounding Clamp, then tighten the nuts at both connections,refer to Figure 5.12-4.
4. All Grounding Clamps are installed next to the first column at both the north and south ends, regardless of whether there is a damper at this location, refer to Figure 5.12-1.
5. The ground wire of TCU is integrated in motor wire, when motor wire is connected with motor, grounding function has equipped directly.

Table 5.12-1 :List of Materials Required for the Installation of Yellow-Green Ground Wires

NO.	Description	Specifications	Remarks
1	Grounding Cable	S156	
2	thin nut	M12	
3	Hex bolt	M10 * 25 (1 flat washer, 1 nut)	Secure the Grounding Cable

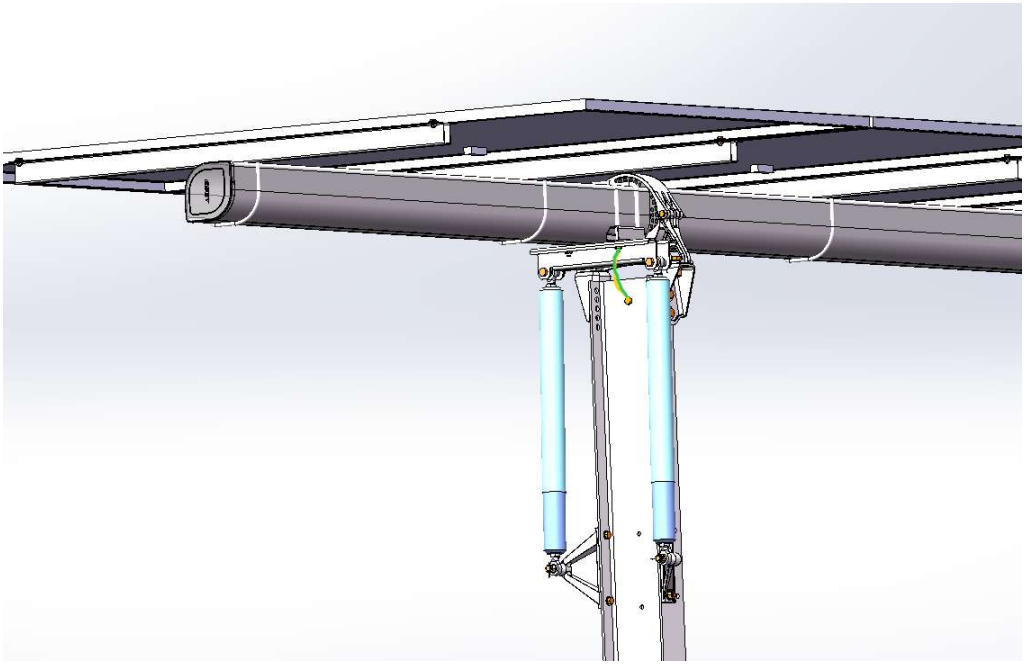


Figure 5.12-2:Installation schematic for tracker grounding clamp

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5.13 Installation of torque tube cover (Step 13)

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents；
2. After the torque tube is installed, finally install the cover of the torque tube at both ends, as shown in Figure 5.13-2.

Table 5.13-1: Materials required for torque tube cover installation

NO.	Description	Specifications for S156	Remarks
1	Torque tube cover	BP098	

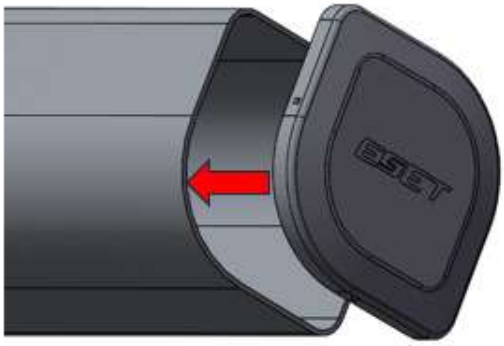


Figure 5.13-2: Installation diagram of torque tube cover

Appendix 1: Description of the tightening torque of the bolts

Bolt strength grade	Yield strength	Bolt nominal diameter / mm			
		6	8	10	12
	N/mm²	Tightening torque of M / N.m			
4.6	240	4 ~ 5	10 ~ 12	20 ~ 25	36 ~ 45
6.8	300	5 ~ 7	12 ~ 15	25 ~ 32	45 ~ 55
8.8	480	7 ~ 9	17 ~ 23	33 ~ 45	58 ~ 78
Bolt strength grade	Yield strength	Bolt nominal diameter / mm			
		14	16	18	20
	N/mm²	Tightening torque of M / N.m			
4.6	240	55 ~ 70	90 ~ 110	120 ~ 150	170 ~ 210
6.8	300	70 ~ 90	110 ~ 140	150 ~ 190	210 ~ 270
8.8	480	93 ~ 124	145 ~ 193	199 ~ 264	282 ~ 376
Bolt strength grade	Yield strength	Bolt nominal diameter / mm			
		22	24	27	30
	N/mm²	Tightening torque of M / N.m			
4.6	240	230 ~ 290	300 ~ 377	450 ~ 530	540 ~ 680
6.8	300	290 ~ 350	370 ~ 450	550 ~ 700	680 ~ 850
8.8	480	384 ~ 512	488 ~ 6 50	714 ~ 952	969 ~ 1293

Notes：

1. **After the tracker is installed, the bolts must be tested one by one with a torque wrench to see if the tightening torque value is reached and marked with a line.**
2. **If there is damage to the zinc layer on the surface after the bolt is tightened, surface rust prevention is required.**
3. **After three months of normal operation of the tracker, all the bolts connecting the torque tube need to be inspected and the relevant bolts need to be tightened for a second time.**

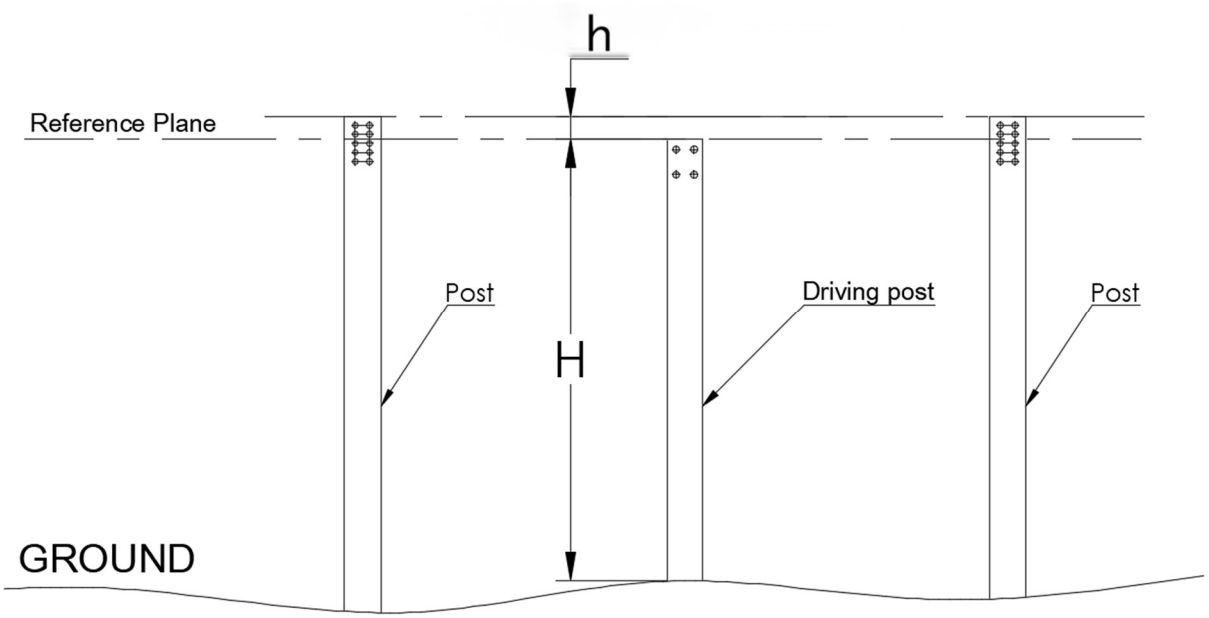
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Appendix 2: Project-specific installation dimensions

Appendix 2.1: Installation of posts

① The installation direction of posts is the north and south;
② Refer to Table a2.1-1 to Figure a2.10 for foundation height;
③ Refer to Table a2.1-2 to Figure a2.1-10 for post installation location and spacing.
④ *Take the following model as an example for reference.

Table a2.1-1: Schematic chart of foundation depth and height from the ground



No.	1x28-Exterior	1x42-Exterior	1x56-Exterior	1x70-Exterior
	5 Inch	6 Inch	7 Inch	5 Inch
H	1150±100 (mm)	1134±100 (mm)	1124±100 (mm)	1150±100 (mm)
h	46 (mm)	62 (mm)	72 (mm)	46 (mm)

Remark:

* With **H** as the reference, the remaining dimensions shall move accordingly.

* Taking **h** as the reference dimension for the ground clearance of the top surface of the Driving Post, this height dimension will vary within ±100mm with slope variations, and the remaining

dimensions shall be adjusted accordingly.

Table a2.1-3: Materials required for the installation of 1x28 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 1	W8*10	QZ1-3050	1	
2	Non-driving post 1	W8*10	FZ1-3096	3	
3	Damper post 1	W8*10	NZ1-3096	2	

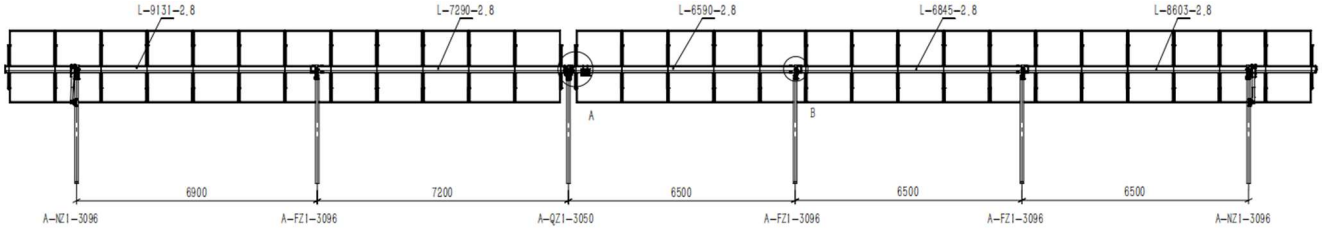


Figure a2.1-4: Schematic diagram of the installation position of 1P*28 **Exterior Tracker** posts with different part number.

Table a2.1-5: Materials required for the installation of 1x42 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 2	W8*13	QZ2-3034	1	
2	Non-driving post 1	W8*10	FZ1-3096	6	
3	Damper post 1	W8*10	NZ1-3096	2	

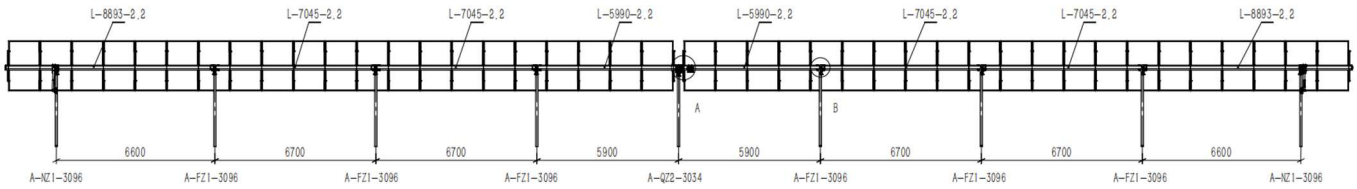


Figure a2.1-6: Schematic diagram of the installation position of 1P*42 **Exterior Tracker** posts with different part number.

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Table a2.1-7: Materials required for the installation of 1x56 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 3	W8*15	QZ3-3024	1	
2	Non-driving post 1	W8*10	FZ1-3096	6	
3	Non-driving post 2	W8*13	FZ2-3096	2	
4	Damper post 1	W8*10	NZ1-3096	2	

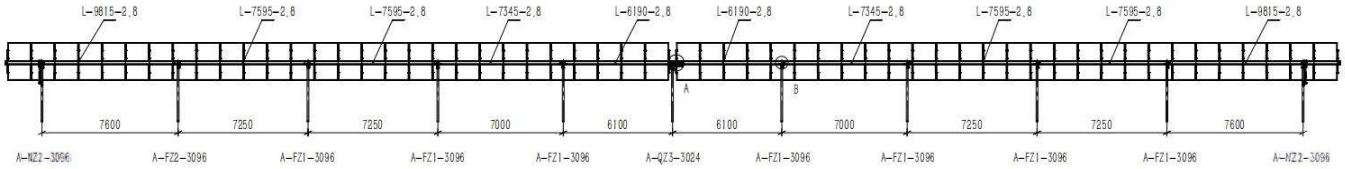


Figure a2.1-8: Schematic diagram of the installation position of 1P*56 **Exterior Tracker** posts with different part number.

Table a2.1-9 Materials required for the installation of 1x70 **Exterior Tracker**

NO.	Description	Specifications and models			Remarks
		Specification	Part number	Quantities	
1	Driving post 4	W8*13	QZ4-3050	2	
2	Non-driving post 1	W8*10	FZ1-3096	9	
3	Damper post 1	W8*10	NZ1-3096	3	

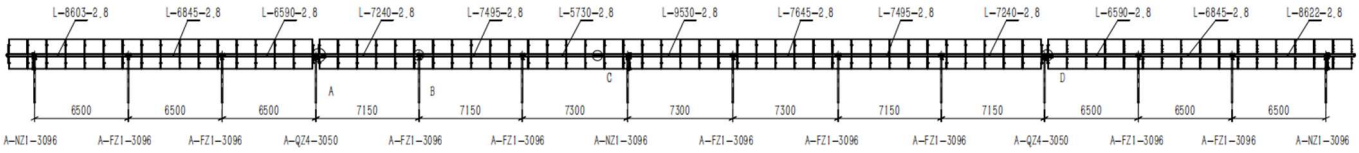


Figure a2.1-10: Schematic diagram of the installation position of 1P*70 **Exterior Tracker** posts with different part number.

For detail information, please check Pile foundation construction guidance.

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Appendix 2.2: Torque tube installation process

① See Figure a2.2-1 to a2.2-22 for details of torque tube installation locations with different part number;

Table a2.2-1: Materials required for the **Torque Tube** installation of 1P*28 **Exterior Tracker**

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A1	S156x2.8	L-7290-2.8	1	
2	Torque tube A2	S156x2.8	L-6590-2.8	1	
3	Torque tubeB1	S156x2.8	L-6845-2.8	1	
4	Torque tube C1	S156x2.8	L-8603-2.8	1	
5	Torque tube C2	S156x2.8	L-9131-2.8	1	

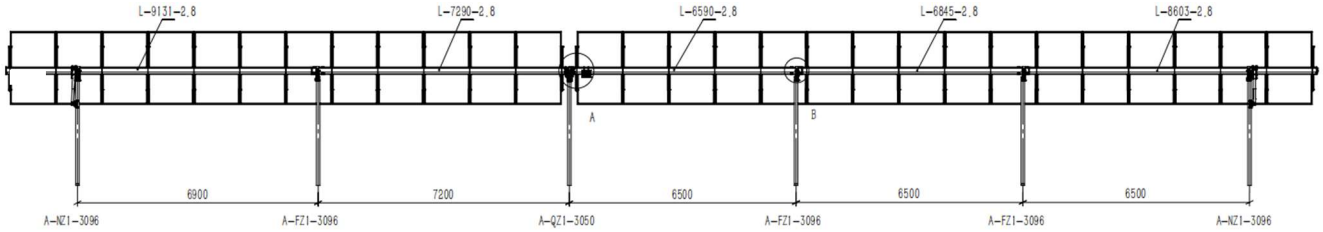


Figure a2.2-2: Schematic diagram of 1P*28 **Exterior Tracker** torque tube mounting position with different part number.

Table a2.2-3: Materials required for the **Torque Tube** installation of 1P*42 **Exterior Tracker**.

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A5	S156x2.2	L-5990-2.2	2	
2	Torque tube B6	S156x2.2	L-7045-2.2	4	
3	Torque tube C5	S156x2.2	L-8893-2.2	2	

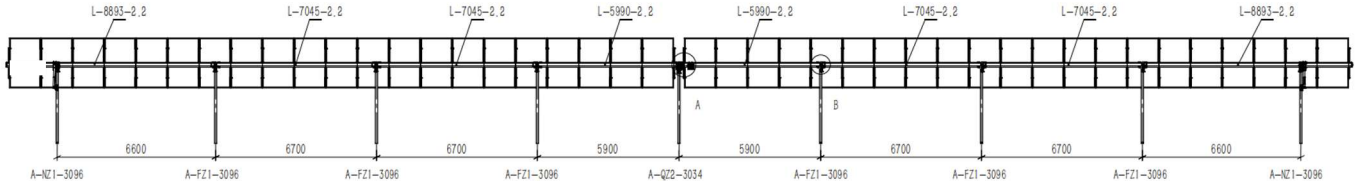


Figure a2.2-4: Schematic diagram of 1P*42 **Exterior Tracker** torque tube mounting position with different part number.

Table a2.2-5: Materials required for the **Torque Tube** installation of 1P*56 **Exterior Tracker**.

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A3	S156x2.8	L-6190-2.8	2	
2	Torque tube B2	S156x2.8	L-7345-2.8	2	
3	Torque tube B3	S156x2.8	L-7595-2.8	4	
4	Torque tube C3	S156x2.8	L-9815-2.8	2	

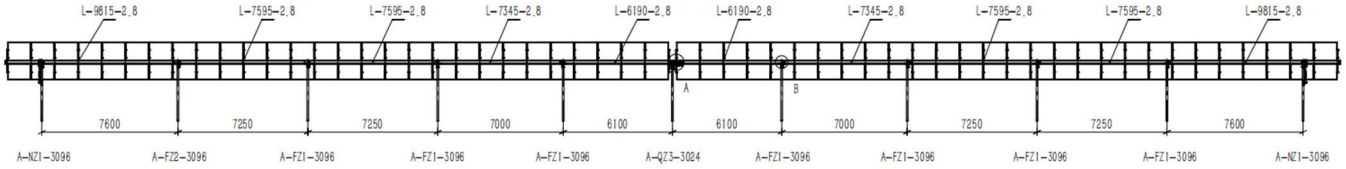


Figure a2.2-6: Schematic diagram of 1P*56 **Exterior Tracker** torque tube mounting position with different part number.

Table a2.2-7: Materials required for the **Torque Tube** installation of 1P*70 **Exterior Tracker**.

NO.	Description	Spec.	Item No.	Quantities	Remark
1	Torque tube A2	S156x2.8	L-6590-2.8	2	
2	Torque tube A4	S156x2.8	L-7240-2.8	2	
3	Torque tube B1	S156x2.8	L-6845-2.8	2	
4	Torque tube B4	S156x2.8	L-7495-2.8	2	
5	Torque tube B5	S156x2.8	L-7645-2.8	1	
7	Torque tube C4	S156x2.8	L-8613-2.8	2	
8	Torque tube D1	S156x2.8	L-5730-2.8	1	
9	Torque tube D2	S156x2.8	L-9530-2.8	1	

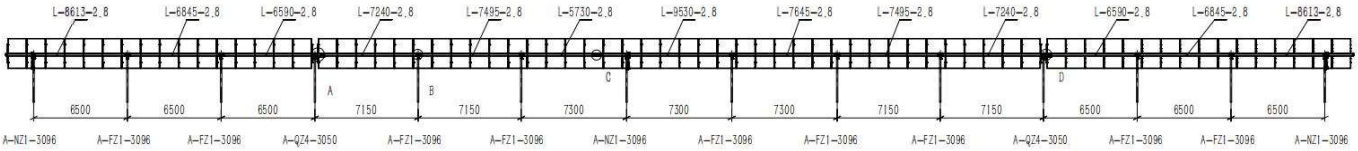


Figure a2.2-8: Schematic diagram of 1P*70 **Exterior Tracker** torque tube mounting position with different part number.

Appendix 2.3: Installation of TCU and wireless receiver

Appendix 2.3.1 Installation of TCU

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents.
2. Prepare the installation tools: 6mm Allen wrench, and related supporting equipment
3. First, install the TCU hoop onto the torque tube, and then fasten the M8 bolts on the hoop to fix the TCU onto the torque tube.
4. The distance between the center of the TCU and the center of reducer should be controlled at 600 mm.

Remarks:

① Take waterproof, moisture-proof and other protection measures before the installation of the TCU;

② The control box（TCU）is usually installed on the north side of the reducer, the 'N' symbol on the shell points to the north;

③ After the installation, fix the cable of the motor on the post; the length of the cable should be slightly longer considering the rotation of the tracker.

Table a2.3.1-1: Materials required for installation of TCU

NO.	Description	Specifications	Remarks
1	Controller box（TCU）	/	
2	Hoop	L=530	

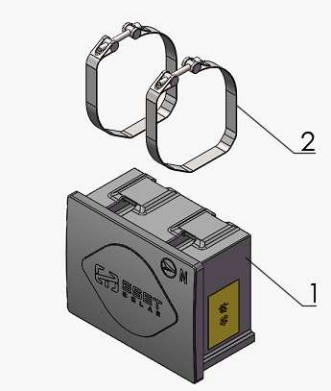


Figure a2.3.1-2: Installation and explosion diagram of the TCU-M

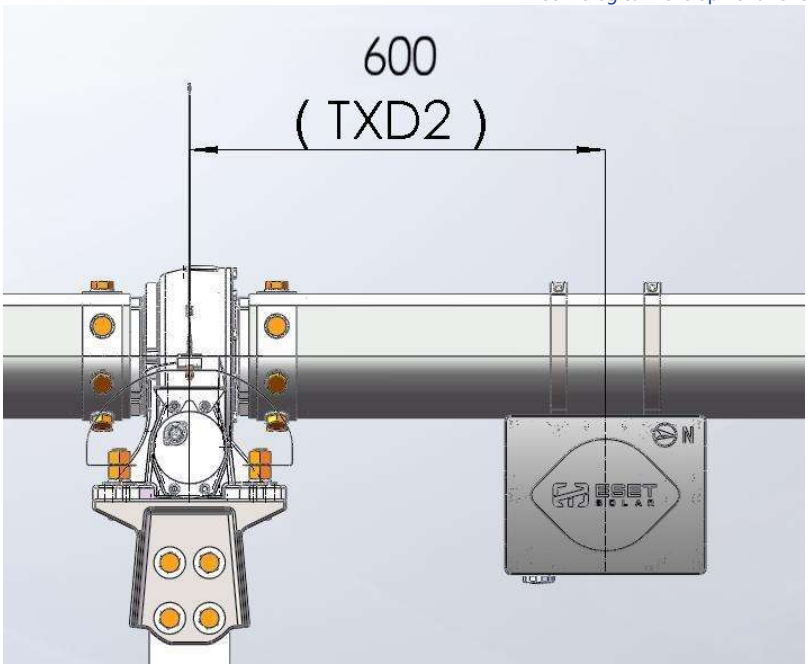


Figure a2.3.1-3: Diagram of installation position of TCU-M

Table a2.3.1-4: Materials required for installation of TCU-S

NO.	Description	Specifications	Remarks
1	Controller box（TCU）	/	
2	Hoop	L=530	

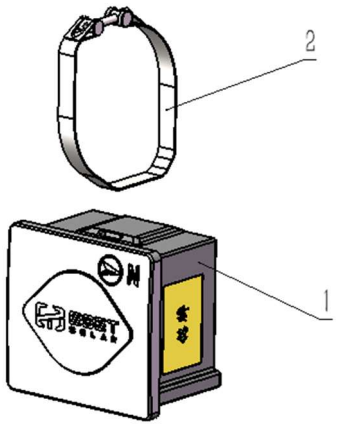


Figure a2.5.1-5: Installation and explosion diagram of the TCU-S

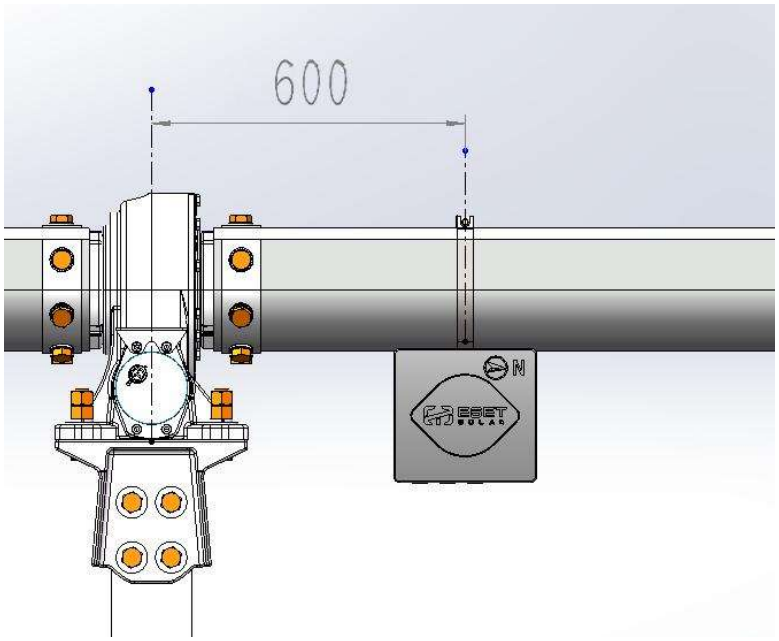


Figure a2.5.1-6: Diagram of installation position of TCU-S

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Appendix 2.3.2 Battery Power Description of TCU

1 Daytime

Solar panel modules power the controller, with the battery on standby. Battery power will only be used in high wind mode or during continuous rainy weather. The battery capacity is 5Ah, and it is charged to 90% during the day. Considering a certain amount of loss, the usable capacity is calculated as 4Ah.

In the scenario where the modules generate no electricity and only the battery supplies power:

- The controller's operating current is 1A,the controller operates for approximately 0.5h per day under normal tracking state.
- Battery's available operating duration for the controller = $\frac{4Ah}{1A}$ =4h
- In high wind protection mode:Solar panel modules provide 85% of the power, and the battery provides 15%.

2 Nighttime

Solar panel modules are not in operation, and only the battery supplies power. The battery capacity is 5Ah, and it is charged to 90% during the day. Considering a certain amount of loss, the usable capacity is calculated as 4Ah.

① Without high wind:

- The controller is parked at a fixed position.
- The battery only supplies power for the controller's communication, with a communication power consumption current of 0.024A.
- Battery's available communication duration for the controller = $\frac{4Ah}{0.024A}$ =166.7h

② In high wind protection mode:

- Considering the controller's maximum operating angle of 120°, the controller's operating current is 8A.
- Battery's available operating duration for the controller in high wind mode = $\frac{4Ah}{8A}$ =0.5h
- Required duration for the bracket to operate 120° in high wind mode =$\frac{120}{360*0.06}$ =5.6min=0.09h

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Period		Scenario	Available Battery Capacity (Ah)	Operating Current (A)	Theoretical Battery Power Supply Duration (h)
Daytime	Automatic tracking (battery-only power)	4	1	4	0.5
Daytime	High wind protection mode	4	8 (solar panel: 6.8; battery: 1.2)	/	/
Nighttime	Automatic tracking (communication-only)	4	0.024	166.7	12
Nighttime	High wind protection	4	8	0.5	0.09

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Appendix 2.3.3 Installation of the wireless receiver

1. Prepare the installation parts and check the size and quantity according to the product BOM and engineering drawings in the attached documents
2. Place the wireless receiver on the wireless receiver bracket (on the slew drive cover) and fix it with M5 * 15 bolts

Table a2.3.3-1: List of materials required for the installation of the wireless receiver

NO.	Description	Specifications	Remarks
1	Wireless receiver bracket2	-2x55x90	
2	Wireless receiver	/	
3	Slew drive cover	-1x120x260	

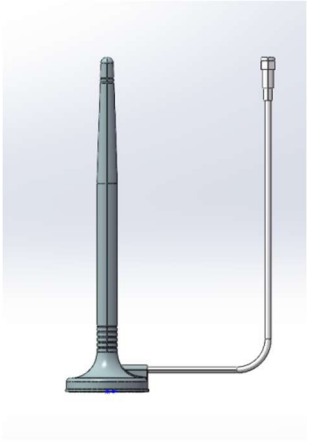


Figure a2.3.3-2: Schematic diagram of the wireless receiver

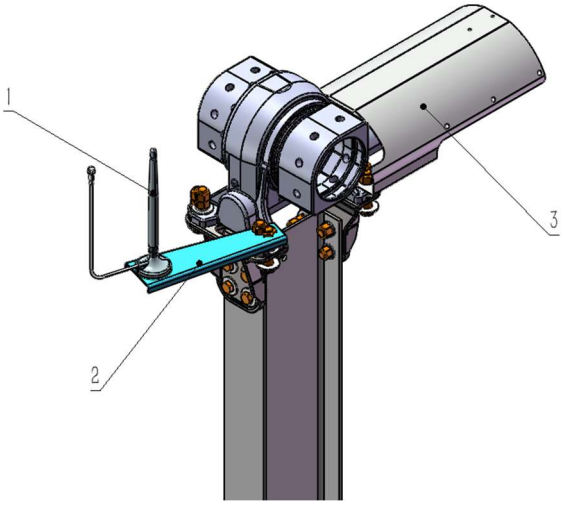
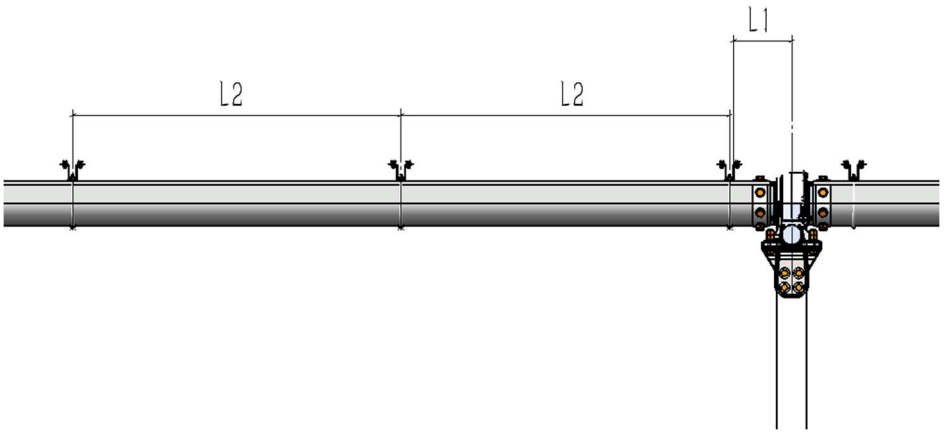


Figure a2.3.3-3: Installation diagram of the wireless receiver

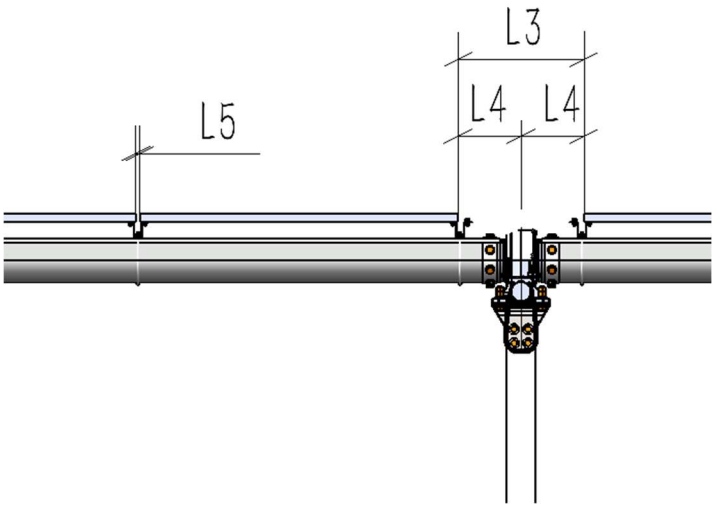
Appendix 2.4: Installation of purlins and PV modules

Appendix 2.4.1 Installation of purlins



Pos.	Dimension(mm)
L1	204
L2	1318

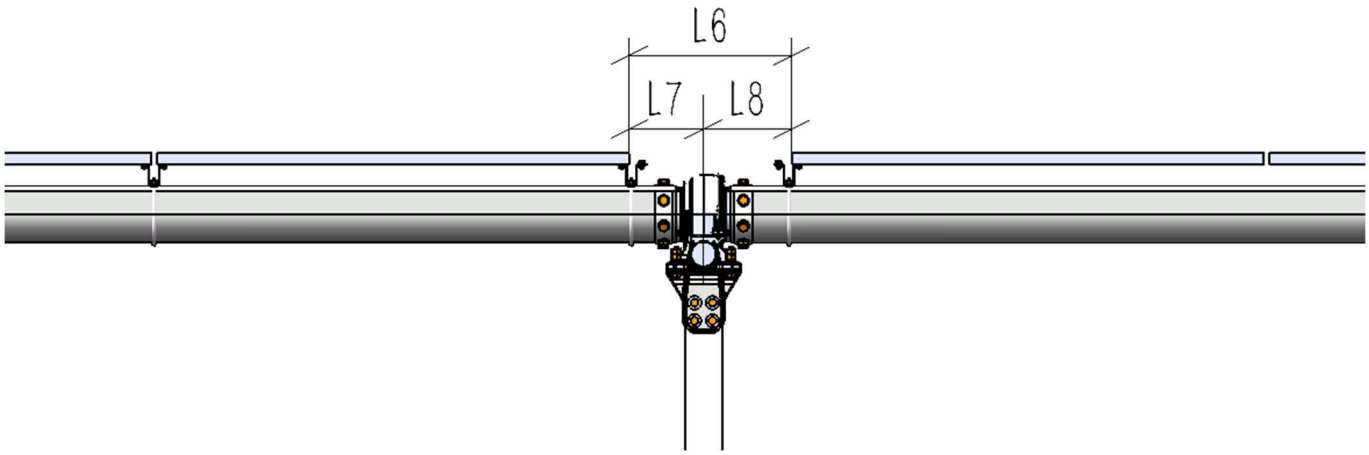
Figure a2.4.1-1: Schematic diagram of purlin installation distance(1x28，1x42，1x56，1x70)



Pos.	Dimension(mm)
L3	450
L4	225
L5	15

Figure a2.4.1-2: Schematic diagram of the installation distance of PV modules (1x28，1x42，1x56，1x70-TCU-M)

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Pos.	Dimension(mm)
L6	450
L7	216
L8	234

Figure a2.4.1-2: Schematic diagram of the installation distance of PV modules (1x70-TCU-S)

Table a2.4.1-3: List of materials for 1x28 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.8	30	LT1-1450	
2	U bolt	UM10*166*188*30	30	/	
3	Purlin pad block	28*43.5*42	30	/	
4	Purlin reinforcement gasket	- 3*22*35	60	/	
5	Flange nut	M10	60	/	Torque 25-32 N.m

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Table a2.4.1-4: List of materials for 1x42 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.8	44	LT1-1450	
2	U bolt	UM10*166*188*30	44	/	
3	Purlin pad block	28*43.5*42	44	/	
4	Purlin reinforcement gasket	- 3*22*35	88	/	
5	Flange nut	M10	88	/	Torque 25-32 N.m

Table a2.4.1-5: List of materials for 1x56 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.8	58	LT1-1450	
2	U bolt	UM10*166*188*30	58	/	
3	Purlin pad block	28*43.5*42	58	/	
4	Purlin reinforcement gasket	- 3*22*35	116	/	
5	Flange nut	M10	116	/	Torque 25-32 N.m

Table a2.4.1-6: List of materials for 1x70 **Exterior Tracker** purlin installation

No.	Description	Spec.	Quantities	Item No.	Remark
1	Purlin 1	J60x30x1.6	73	LT1-1450	
2	U bolt	UM10*166*188*30	73	/	
3	Purlin pad block	28*43.5*42	73	/	
4	Purlin reinforcement gasket	- 3*22*35	146	/	
5	Flange nut	M10	146	/	Torque 25-32 N.m

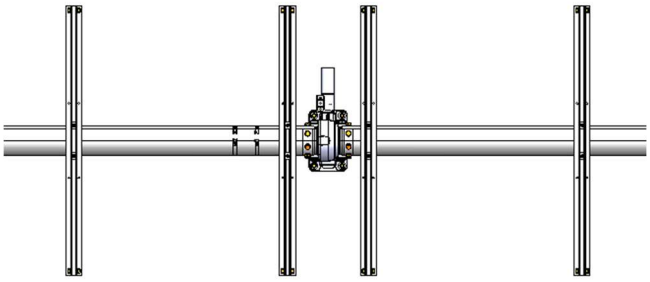
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Appendix 2.4.2 Installation of PV modules

1. All PV modules shall be fastened with four bolts during installation.

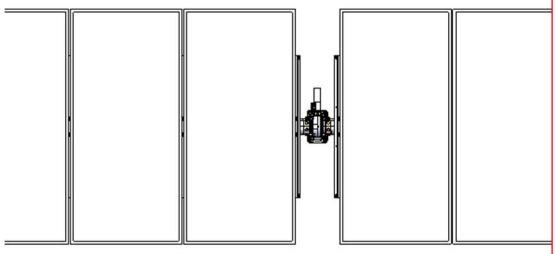
Table a2.4.2-1: List of materials required for the installation of PV modules

NO.	Description	Specifications	Remarks	Remarks 2
1	PV modules	/		Refer to **Appendix 5: BOM List**
2	Hex bolt	M6 * 14 * 14 (1 Grounding washer 1 big flange nut)	Torque 8 N.m	
3	Hex bolt	M8 * 16 *16 (1 Grounding washer 1 big flange nut)	Torque 16 N.m	



1. The installation position of the power supply module is above the reducer and on the opposite side of the drive motor.

Figure a2.4-16: Schematic diagram of power supply module

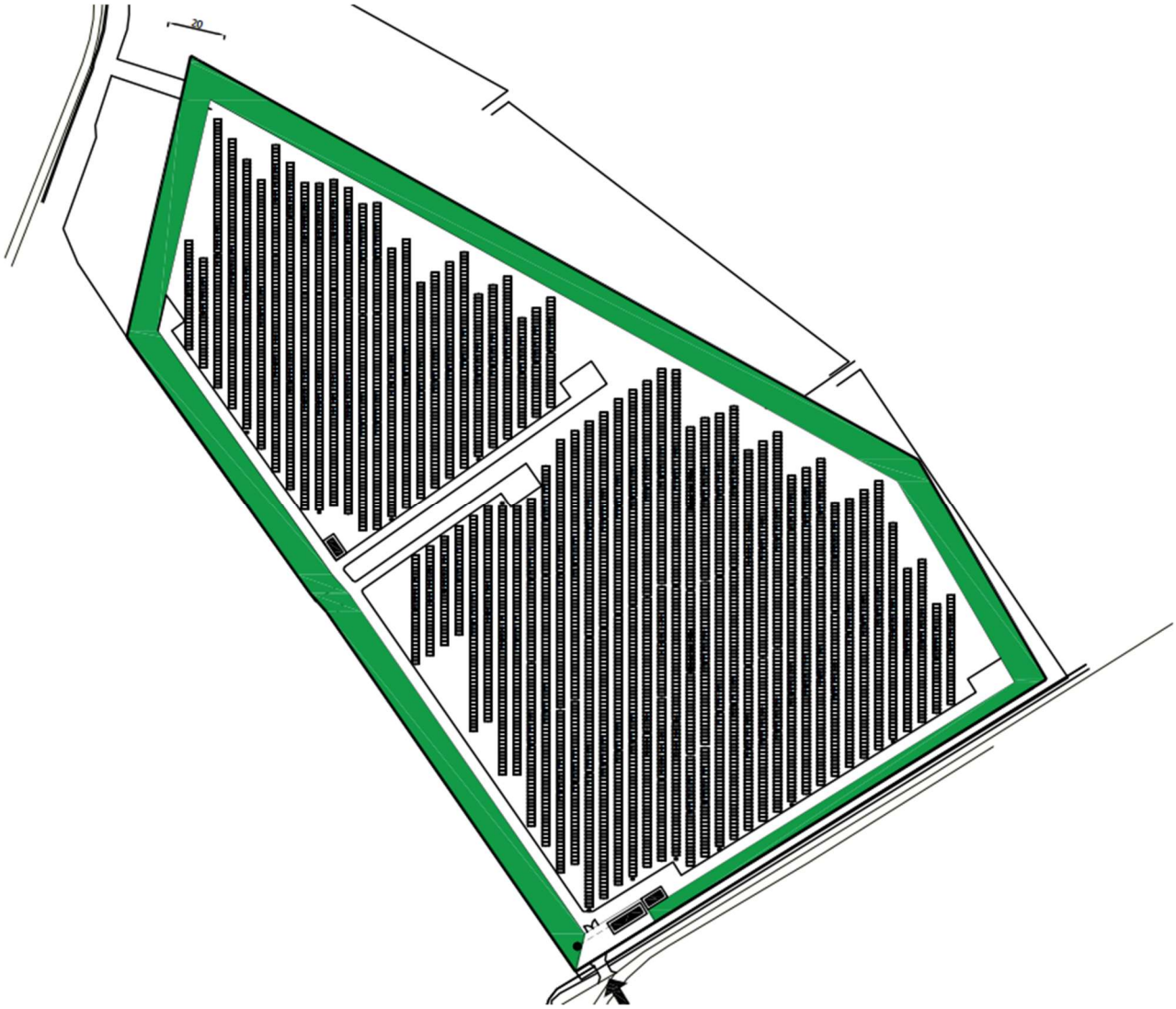


1. The end faces of the power supply module shall be flush with those of the PV modules.

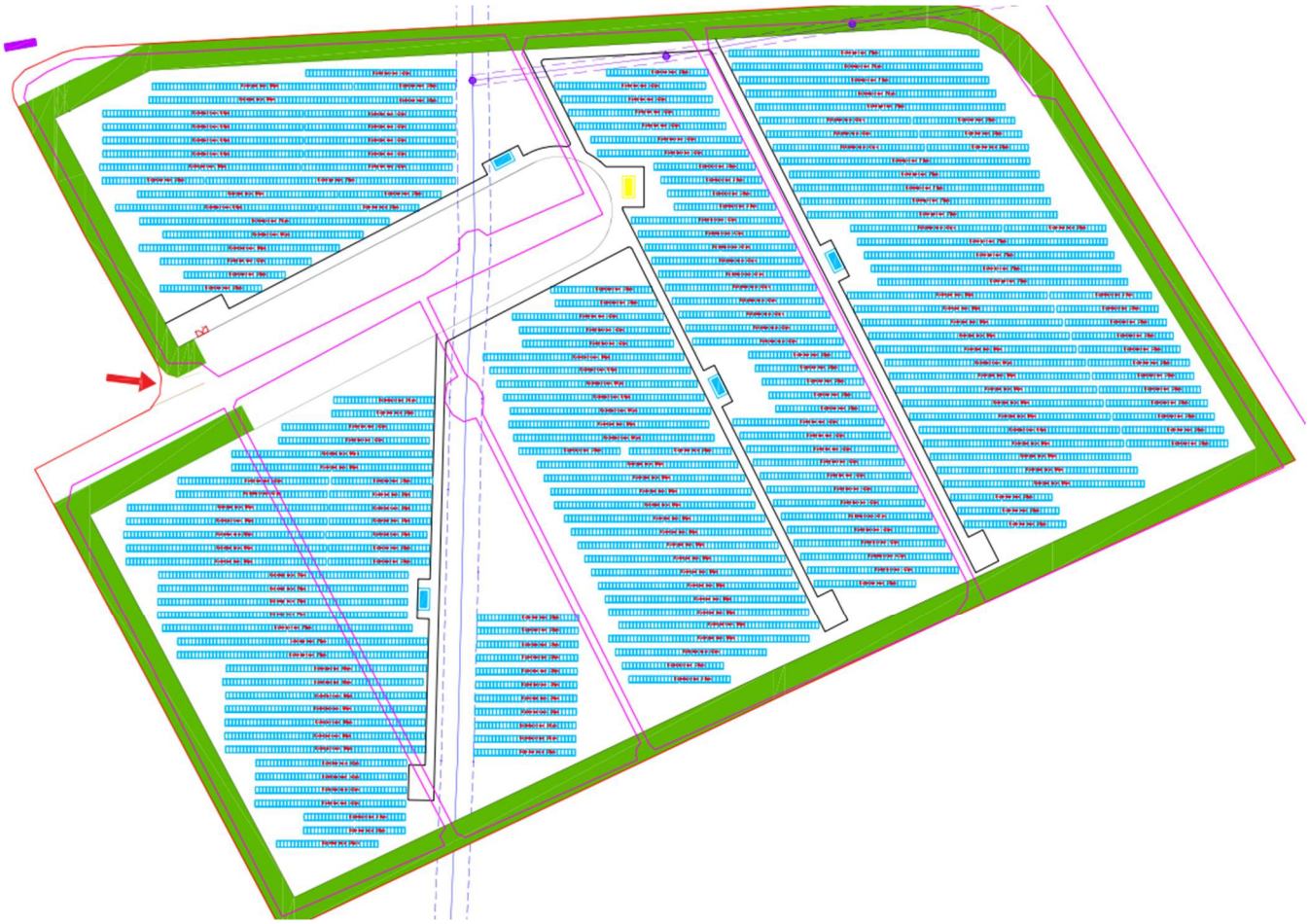
Figure a2.6-17: Schematic diagram of power supply module

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Appendix 3: Layout drawing



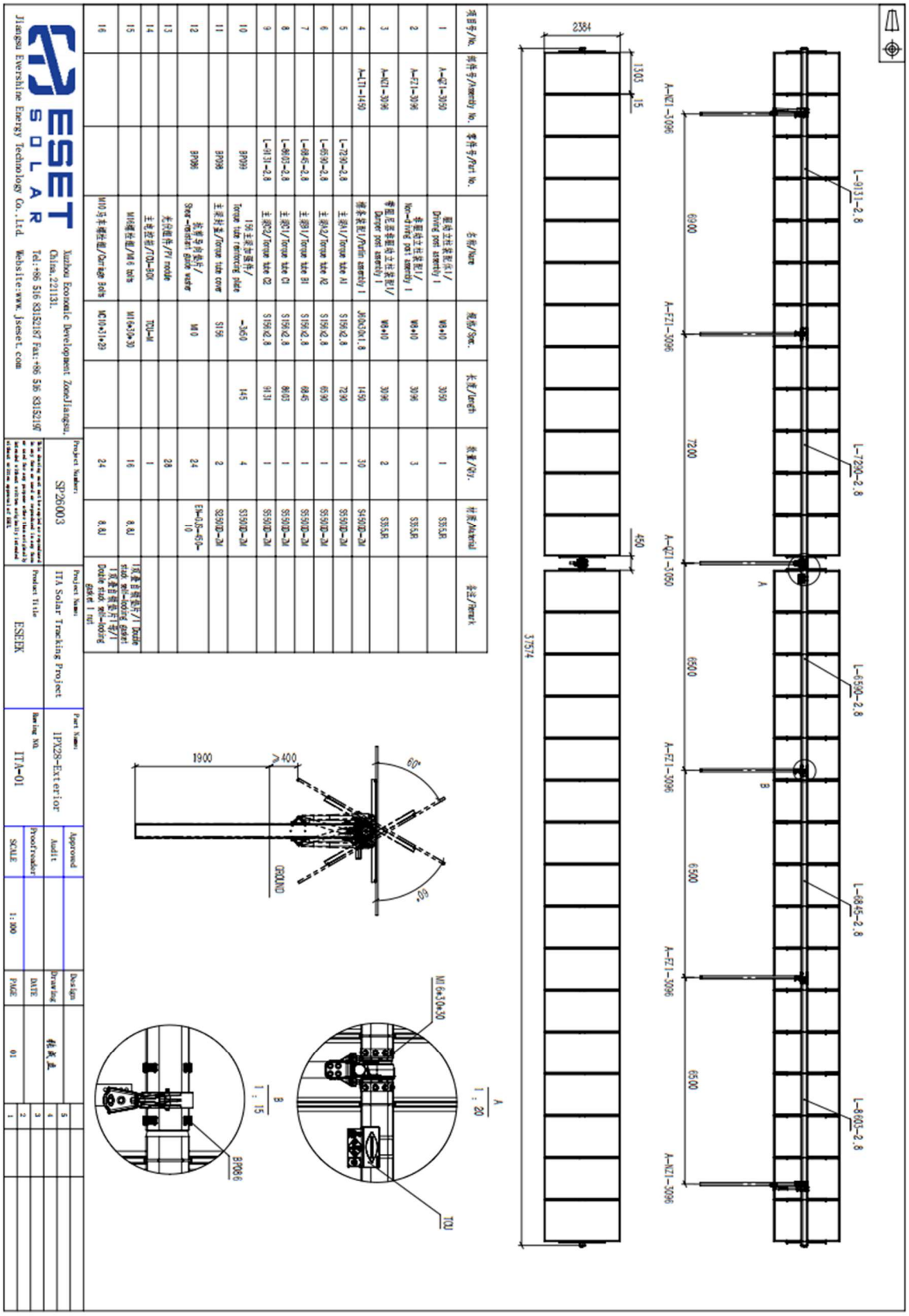
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Appendix 4: General design drawing

1PX28-Exterior

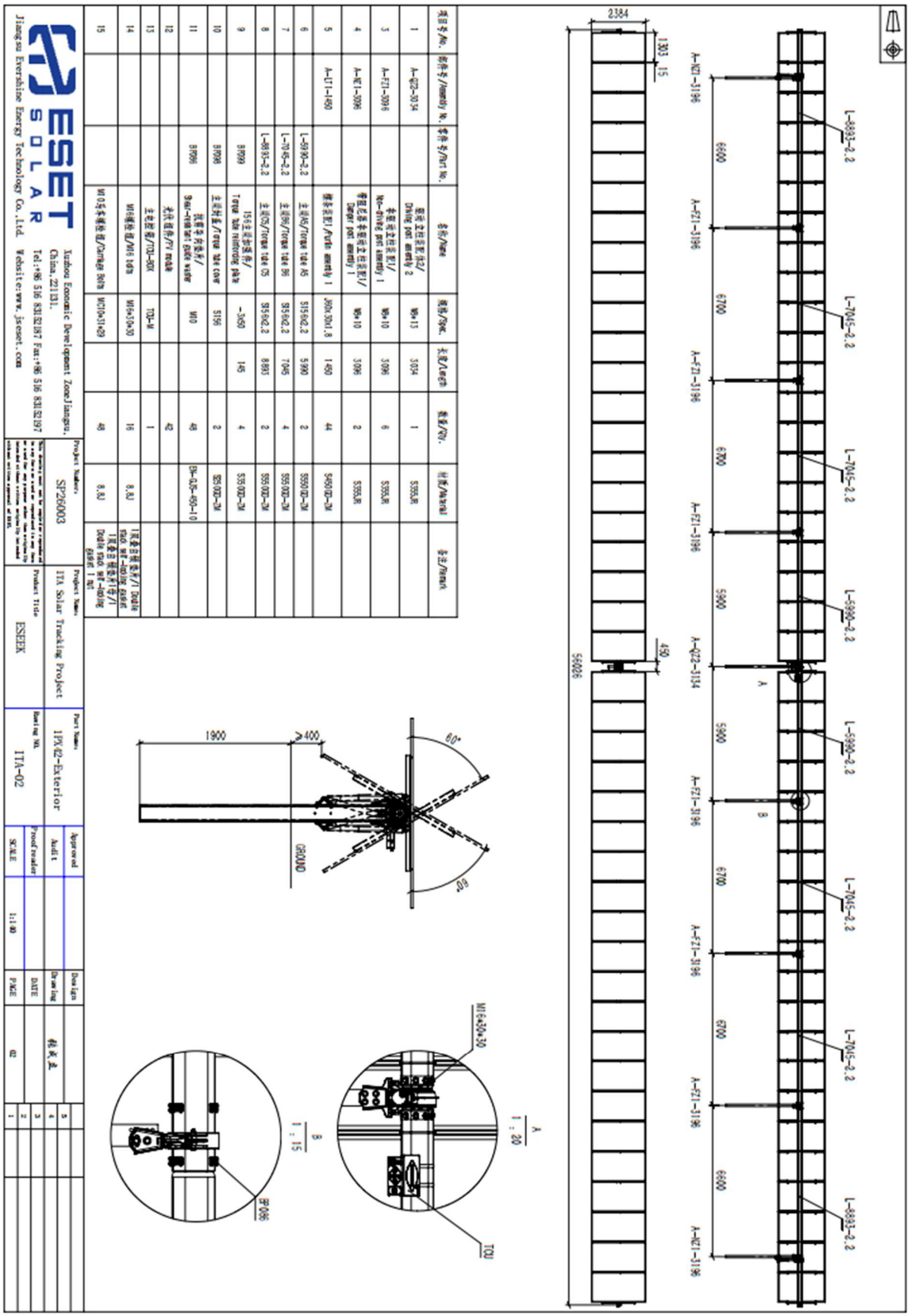


项目号/No.	组件号/Assembly No.	零件号/Part No.	名称/Name	规格/Spec.	长度/Length	数量/Qty.	材质/Material	备注/Remark
1	A-QZ1-3060		驱动立柱装配体/ Driving post assembly 1	M8×40	3060	1	S355JR	
2	A-FZ1-3096		非驱动立柱装配体/ Non-driving post assembly 1	M8×40	3096	3	S355JR	
3	A-NZ1-3096		带扭簧非驱动立柱装配体/ Damper post assembly 1	M8×40	3096	2	S355JR	
4	A-LT1-1450		横条装配体/Purlin assembly 1	JR80×61.8	1450	30	S450QD+ZM	
5		L-7230-2.8	主梁1/Torque tube N1	S150×2.8	7230	1	S550QD+ZM	
6		L-6590-2.8	主梁2/Torque tube N2	S150×2.8	6590	1	S550QD+ZM	
7		L-6845-2.8	主梁3/Torque tube N1	S150×2.8	6845	1	S550QD+ZM	
8		L-8603-2.8	主梁1/Torque tube C1	S150×2.8	8603	1	S550QD+ZM	
9		L-9131-2.8	主梁2/Torque tube C2	S150×2.8	9131	1	S550QD+ZM	
10	BP099		150主梁加强板/ Torque tube reinforcing plate	-360	145	4	S350QD+ZM	
11	BP098		主梁封盖/Torque tube cover	S150		2	S350QD+ZM	
12	BP086		热镀锌外露扁形/ Shear-resistant plank washer	M10		24	EN-GJS-400-18-LT	
13			尤尼斯特/T1 nozzle	TOU-M		28		
14			主梁螺栓/TOU-BOLT			1		
15			M16螺栓组/M16 bolts	M16×30×30		16	8.8U	150主梁加强板/ Double stub mid-hooting gasket
16			M10马车螺栓组/Carriage Bolts	M10×31×20		24	8.8U	150主梁加强板/ Double stub mid-hooting gasket

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Project Number: SP26003
Product Title: ESEEK
Project Name: ITA Solar Tracking Project
Drawing No.: ITA-01
Part Name: 1PX28-Exterior
SCALE: 1:300
Proofreader
Approved
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Design
Drawing
DATE
PAGE: 01

1PX42-Exterior

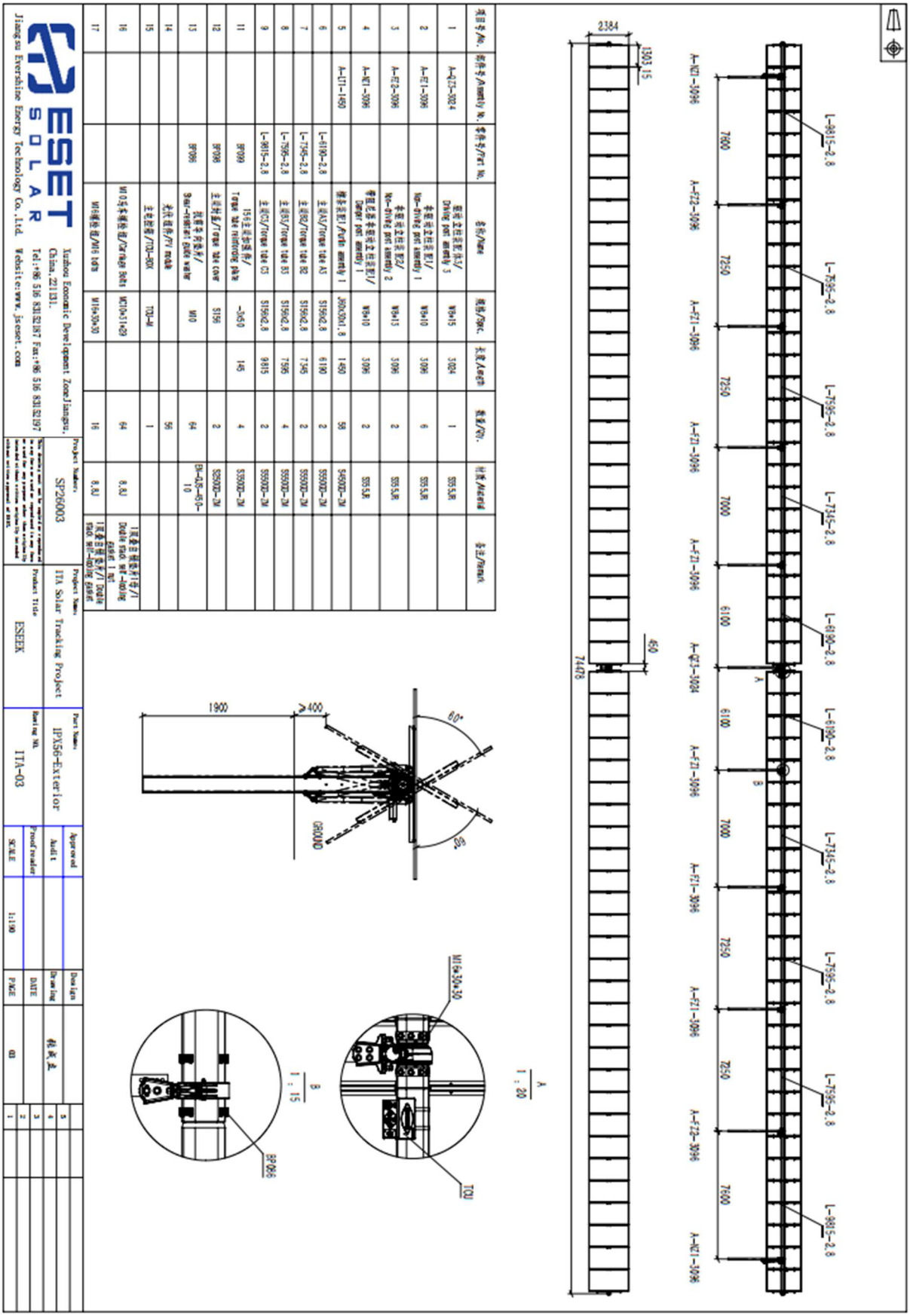


项目号/No.	零件号/Assembly No.	零件号/Part No.	名称/Name	规格/Spec.	长度/Length	数量/Qty.	材质/Material	备注/Remark
1	A-U2I-3154		驱动立柱总成2/Driving post assembly 2	M24-13	3054	1	S355JR	
3	A-F2I-3196		非驱动立柱总成1/Non-driving post assembly 1	M24-10	3196	6	S355JR	
4	A-WI-3196		带组串安装板立柱总成1/Tracker post assembly 1	M24-10	3196	2	S355JR	
5	A-LI-1450		横条总成1/Purlin assembly 1	J80×30×1.8	1450	44	S450QD+ZM	
6		L-8980-2.2	主梁B/Primary tube B	S355GD.2	5980	2	S550QD+ZM	
7		L-7045-2.2	主梁B/Primary tube B	S355GD.2	7045	4	S550QD+ZM	
8		L-8883-2.2	主梁C/Primary tube C	S355GD.2	8883	2	S550QD+ZM	
9		81009	150主梁加强件/Torque tube reinforcing parts	-3650	145	4	S350GD+ZM	
10		81008	主梁封盖/Primary tube cover	S156		2	S250GD+ZM	
11		81006	组串安装板/Quick-release panel clamp washer	M10		48	EN-GJS-450-10	
12			无线模块/TTI module			42		
13			主电控箱/TQU-80K	TQU-M		1		
14			M16螺栓组/M16 bolts	M16×30×30		16	8.8U	
15		M10头部螺栓/Carriage Bolts	MC10×30×20			48	8.8U	

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Tuzhou Economic Development Zone,Jiangsu, China, 221131.

Project Number:	Project Name:	Part Name:	Approved	Design
SP260003	ITA Solar Tracking Project	1PX42-Exterior	Audit	
Product Title:	Bowing NO.	Proofreader	Drawing	
ESEEK	ITA-02	SCALE 1:100	DATE	PAGE 02

1PX56-Exterior



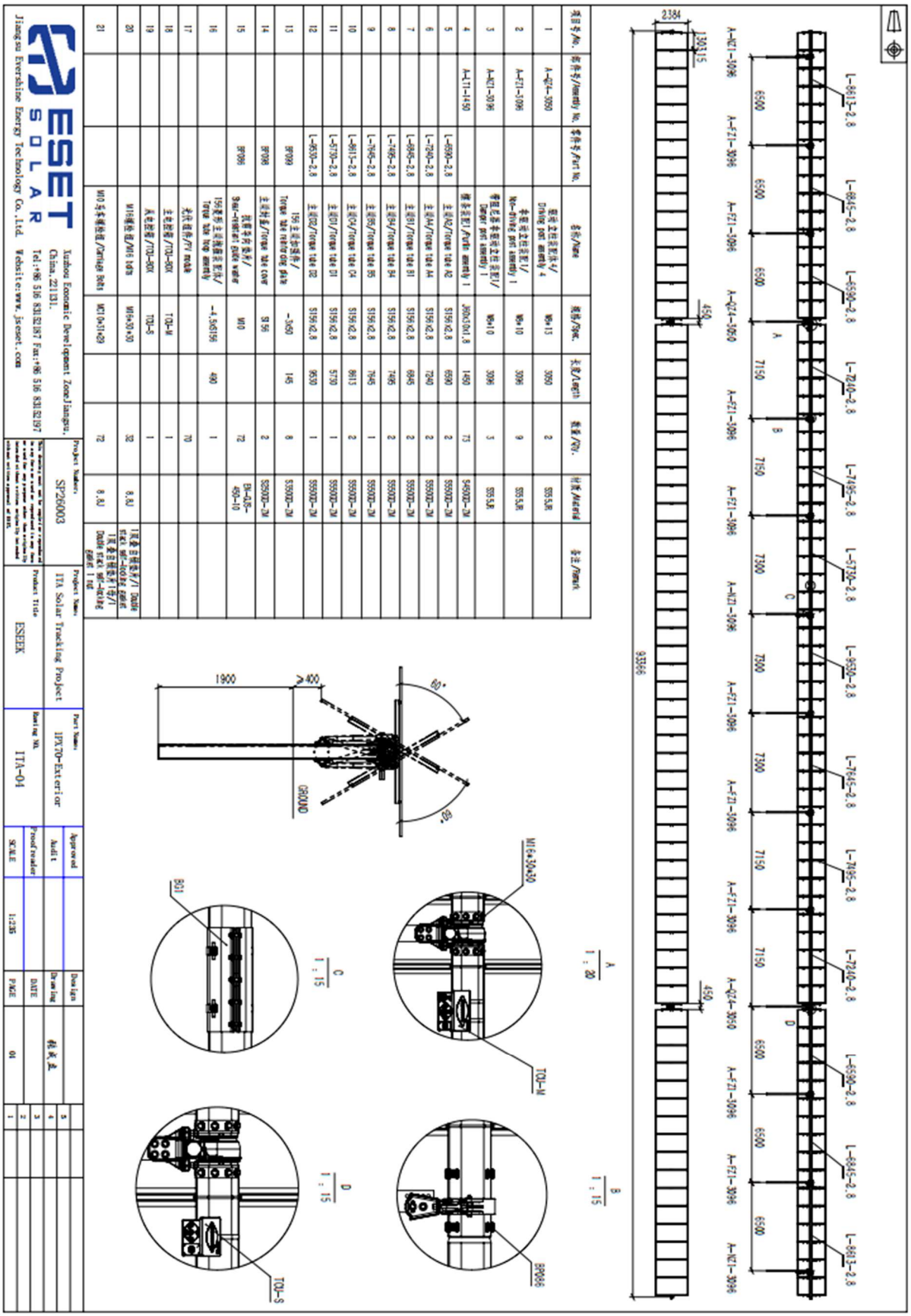
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1	A-A2S-3024		驱动立柱总成/Driving post assembly 3	W8415	3024	1	SS5.5R	
2	A-A21-3096		非驱动立柱总成/Non-Driving post assembly 1	W8410	3096	6	SS5.5R	
3	A-A22-3096		非驱动立柱总成/Non-Driving post assembly 2	W8413	3096	2	SS5.5R	
4	A-W21-3096		带组合导轨的立柱总成/Upright post assembly 1	W8410	3096	2	SS5.5R	
5	A-LT1-1450		横梁总成/Purlin assembly 1	J60x30x1.8	1450	58	S450D-ZM	
6		L-6190-2.8	主横梁/Torque tube A3	S150G2.8	6190	2	S550D-ZM	
7		L-7345-2.8	主横梁/Torque tube B2	S150G2.8	7345	2	S550D-ZM	
8		L-7595-2.8	主横梁/Torque tube B3	S150G2.8	7595	4	S550D-ZM	
9		L-9815-2.8	主横梁/Torque tube C3	S150G2.8	9815	2	S550D-ZM	
11		BP099	150主动加强件/Torque tube reinforcing plate	-3650	145	4	S350D-ZM	
12		BP098	主横梁盖/Torque tube cover	S156		2	S350D-ZM	
13		BP096	双面卡扣弹簧/Dual-reinforced guide washer	M10		64	65Mn-BH-A2S-65-10	
14			光伏组件/PV module			56		
15			主拉杆管/TQU-80X	TQU-W		1		
16		M10头部螺栓组/Carriage Bolt	MC10x31x20			64	8.8U	1.双面自锁螺母带平垫/1 Double nuts self-locking gasket 1 set
17		M16螺栓组/M16 bolt	M16x30x30			16	8.8U	1.双面自锁螺母带平垫/1 Double nuts self-locking gasket

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Jiangsu Evershine Energy Technology Co.,Ltd.

Project Number	SF260003
Project Name	ITA Solar Tracking Project
Product Title	ESEEK
Part Name	1PX56-Exterior
Drawing NO.	ITA-03
Approved	
Audit	
Proofreader	
SCALE	1:100
PAGE	A3
Drawing	机械类
DATE	
Design	

1PX70-Exterior



项目号/No.	部件号/Assembly No.	零件号/Part No.	名称/Name	规格/Spec.	长度/Length	数量/Qty.	材质/Material	备注/Remark
1	A-A2T-3350		驱动立柱组装件4/ Driving post assembly 4	M24-13	3350	2	S355JR	
2	A-A2T1-3096		非驱动立柱组装1/ Non-driving post assembly 1	M24-10	3096	9	S355JR	
3	A-A2T1-30.96		减震立柱组装件1/ Damper post assembly 1	M24-10	3096	5	S355JR	
4	A-A1T1-1450		横条支架1/Purlin assembly 1	300x100x1.8	1450	73	S450QD-ZM	
5		L-8590-2.8	主梁A2/Torque tube A2	S150x2.8	6590	2	S550QD-ZM	
6		L-7240-2.8	主梁A1/Torque tube A1	S150x2.8	7240	2	S550QD-ZM	
7		L-6845-2.8	主梁B1/Torque tube B1	S150x2.8	6845	2	S550QD-ZM	
8		L-7495-2.8	主梁B4/Torque tube B4	S150x2.8	7495	2	S550QD-ZM	
9		L-7545-2.8	主梁B5/Torque tube B5	S150x2.8	7545	1	S550QD-ZM	
10		L-8611-2.8	主梁C4/Torque tube C4	S150x2.8	8611	2	S550QD-ZM	
11		L-5730-2.8	主梁D1/Torque tube D1	S150x2.8	5730	1	S550QD-ZM	
12		L-8630-2.8	主梁D2/Torque tube D2	S150x2.8	8630	1	S550QD-ZM	
13		BF099	150主梁连接套件/ Torque tube reinforcing plate	-1560	145	8	S550QD-ZM	
14		BF098	主梁封盖/Torque tube cover	S136		2	S250GD-ZM	
15		BF096	抗弯套件组件/ Shear-resistant plate washer	M10		72	EN-GJS-450-10	
16			150主梁连接套筒组/ Torque tube loop assembly	-4.5x5150	400	1		
17			光伏组件/PV module			70		
18			主电缆卡/T20-BOX	T20-M		1		
19			末电缆卡/T20-BOX	T20-S		1		
20			M16螺栓组/M16 bolts	M16x30x30		32	8.8J	
21		M10.9高强螺栓/Carriage Bolts	M10x31x29			72	8.8J	

1								
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Project Number: SP260003 | Project Name: ITA Solar Tracking Project | Part Name: 1PX70-Exterior

Product Title: ESEEK | Bering No.: ITA-04 | SCALE: 1:235 | Proofreader | Audit | Approved

PAGE: 04 | DATE | Drawing | Design

Appendix 5: BOM List

江苏新恒源能源技术有限公司

产品BOM清单

编码：XHY-FM-RD-29　　0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（DESMO）	变更内容：	1版：
型 号：	1Px28-外围	订单套数：	27		2版：
组件功率W：	730	行数：	1	单套功率KW：20.44	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	28	总容量MW：0.55188	基础类型：	桩柱一体

序号	内容				数量		材质	防腐要求	零件号	备注
	名称	规格	长度(mm)	单位	单数	总数				
1	驱动立柱1 Driving post 1	W8*10	3050	件	1	27	S355JR	热镀锌 平均锌层≥85μm	QZ1-3050	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	3	81	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	2	54	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
4	150底座 Slew drive base-150	77x206	177	件	2	54	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP088	铸造件
5	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	5	135	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
6	156轴承座上 Bearing housing cover	R116x73	288	件	5	135	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
7	重型牛腿 Bearing base	75x124	163	件	10	270	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
8	156阻尼支臂 156 Damper fixture	81.5x131	400	件	2	54	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
9	阻尼器套管 Round tube of damper	ϕ20x2.0	17	件	8	216	S250GD-ZM	锌铝镁 双面310g/m²	TG	
10	阻尼器下固定件-轮型 Damper lower fixture	102x130	220	件	4	108	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
11	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	24	648	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
12	主梁A1 Torque tube A1	S156x2.8	7290	件	1	27	S550GD-ZM	锌铝镁 双面310g/m²	L-7290-2.8	
13	主梁A2 Torque tube A2	S156x2.8	6590	件	1	27	S550GD-ZM	锌铝镁 双面310g/m²	L-6590-2.8	
14	主梁B1 Torque tube B1	S156x2.8	6845	件	1	27	S550GD-ZM	锌铝镁 双面310g/m²	L-6845-2.8	
15	主梁C1 Torque tube C1	S156x2.8	8603	件	1	27	S550GD-ZM	锌铝镁 双面310g/m²	L-8603-2.8	
16	主梁C2 Torque tube C2	S156x2.8	9131	件	1	27	S550GD-ZM	锌铝镁 双面310g/m²	L-9131-2.8	
17	檩条1 Purlin 1	J60x30x1.8	1450	件	30	810	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
18	156檩条垫块 156 Purlin pad block	42x43.5	28	件	30	810	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095	
19	156长垫块 156 Long purlin pad block	42x43.5	121	件	2	54	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095L	
20	加强垫板 Purlin reinforcement gasket	-3x22	35	件	60	1620	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
21	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	4	108	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
22	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	27	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
23	156主梁封盖 Torque tube cover	-1x25	158	件	2	54	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
24	减速机罩 Slew drive cover	-1x120x260	460	件	1	27	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
25	喉箍 Hoop	20	530	件	2	54	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
螺栓配件										
25	外六角螺栓 Outer Hex Bolt	M16*75*50	8.8J	套	4	108	2大平1弹2母 大平垫尺寸：-3xϕ40xϕ17.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
26	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	16	432	1双叠自锁垫片 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
27	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	8	216	1大平1双叠自锁垫片1母 大平垫尺寸：-3xφ37.6φ15.5 双叠垫片尺寸：-3.4xφ30.7xφ15.2	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
28	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	10	270	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
29	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	10	270	1大平1双叠自锁垫片1母 大平垫尺寸：-3xΦ40xΦ17.5 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
30	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	20	540	1大平1双叠自锁垫片1母 大平垫尺寸：-3xφ37.6φ15.5 双叠垫片尺寸：-3.4xφ30.7xφ15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
31	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	8	216	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
32	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	8	216	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
33	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	24	648	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xϕ16.6xϕ10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
34	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	4	108	2双叠自锁垫片2母 双叠垫片尺寸：-3.4xφ25.4xφ13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
35	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	30	810	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
36	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	112	3024	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径φ9	不锈钢		组件固定螺栓
37	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	81	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
38	薄螺母 thin nut	M12	C级	件	2	54	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
39	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	81	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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ESET SOLAR 江苏新恒源能源技术有限公司

编码：XHY-FM-RD-29

产品BOM清单

0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（DESMO）	变更内容：	1版：
型 号：	1Px42-外围	订单套数：	20		2版：
组件功率W：	730	行数：	1	单套功率KW：30.66	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	42	总容量MW：0.6132	基础类型：	桩柱一体

序号	内容				数量		材质	防腐要求	零件号	备注
	名称	规格	长度(mm)	单位	单数	总数				
1	驱动立柱2 Driving post 2	W8*13	3034	件	1	20	S355JR	热镀锌 平均锌层≥85μm	QZ2-3034	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	6	120	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	2	40	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
4	减速机底座 Slew drive base	80x183	244	件	2	40	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP064	铸造件
5	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	8	160	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
6	156轴承座上 Bearing housing cover	R116x73	288	件	8	160	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
7	重型牛腿 Bearing base	75x124	163	件	16	320	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
8	156阻尼支臂 156 Damper fixture	81.5x131	400	件	2	40	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
9	阻尼器套管 Round tube of damper	ф20x2.0	17	件	8	160	S250GD-ZM	锌铝镁 双面310g/m²	TG	
10	阻尼器下固定件-轻型 Damper lower fixture	102x130	220	件	4	80	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
11	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	48	960	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
12	主梁A5 Torque tube A5	S156x2.2	5990	件	2	40	S550GD-ZM	锌铝镁 双面310g/m²	L-5990-2.2	
13	主梁B6 Torque tube B6	S156x2.2	7045	件	4	80	S550GD-ZM	锌铝镁 双面310g/m²	L-7045-2.2	
14	主梁C5 Torque tube C5	S156x2.2	8893	件	2	40	S550GD-ZM	锌铝镁 双面310g/m²	L-8893-2.2	
15	檩条1 Purlin 1	J60x30x1.8	1450	件	44	880	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
16	156檩条垫块 156 Purlin pad block	42x43.5	28	件	44	880	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095	
17	156长垫块 156 Long purlin pad block	42x43.5	121	件	2	40	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095L	
18	加强垫板 Purlin reinforcement gasket	-3x22	35	件	88	1760	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
19	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	4	80	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
20	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	20	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
21	156主梁封盖 Torque tube cover	-1x25	158	件	2	40	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
22	减速机罩 Slew drive cover	-1x120x260	450	件	1	20	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
23	喉箍 Hoop	20	530	件	2	40	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
	螺栓配件									
23	外六角螺栓 Outer Hex Bolt	M18*80*50	8.8J	套	4	80	2大平1弹2母 大平垫尺寸：-3xф43xф19.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
24	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	16	320	1双叠自锁垫片 双叠垫片尺寸：-3.4xф30.7xф17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
25	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	8	160	1大平1双叠自锁垫片1母 大平垫尺寸：-3xф40xф17.5 双叠垫片尺寸：-3.4xф30.7xф17	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
26	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	16	320	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
27	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	16	320	1大平1双叠自锁垫片1母 大平垫尺寸：-3xф40xф17.5 双叠垫片尺寸：-3.4xф30.7xф17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
28	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	32	640	1大平1双叠自锁垫片1母 大平垫尺寸：-3xф37xф15.5 双叠垫片尺寸：-3.4xф30.7xф15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
29	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	8	160	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xф25.4xф13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
30	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	8	160	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
31	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	48	960	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xф16.6xф10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
32	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	4	80	2双叠自锁垫片2母 双叠垫片尺寸：-3.4xф25.4xф13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
33	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	44	880	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
34	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	168	3360	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径ф9	不锈钢		组件固定螺栓
35	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	60	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
36	薄螺母 thin nut	M12	C级	件	2	40	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
37	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	60	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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编码：XHY-FM-RD-29

产品BOM清单

0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（DESMO）	变更内容：	1版：
型 号：	1Px56-外围	订单套数：	28		2版：
组件功率W：	730	行数：	1	单套功率KW：40.88	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	56	总容量MW：1.14464	基础类型：	桩柱一体

序号	内容				数量		材质	防腐要求	零件号	备注
	名称	规格	长度（mm）	单位	单数	总数				
1	驱动立柱3 Driving post 3	W8*15	3024	件	1	28	S355JR	热镀锌 平均锌层≥85μm	QZ3-3024	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	6	168	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	非驱动立柱2 Non-driving post 2	W8*13	3096	件	2	56	S355JR	热镀锌 平均锌层≥85μm	FZ2-3096	
4	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	2	56	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
5	减速机底座 Slew drive base	80x183	244	件	2	56	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP064	铸造件
6	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	10	280	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
7	156轴承座上 Bearing housing cover	R116x73	288	件	10	280	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
8	重型牛腿 Bearing base	75x124	163	件	20	560	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
9	156阻尼支臂 156 Damper fixture	81.5x131	400	件	2	56	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
10	阻尼器套管 Round tube of damper	ϕ20x2.0	17	件	8	224	S250GD-ZM	锌铝镁 双面310g/m²	TG	
11	阻尼器下固定件-轻型 Damper lower fixture	102x130	220	件	4	112	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
12	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	64	1792	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
13	主梁A3 Torque tube A3	S156x2.8	6190	件	2	56	S550GD-ZM	锌铝镁 双面310g/m²	L-6190-2.8	
14	主梁B2 Torque tube B2	S156x2.8	7345	件	2	56	S550GD-ZM	锌铝镁 双面310g/m²	L-7345-2.8	
15	主梁B3 Torque tube B3	S156x2.8	7595	件	4	112	S550GD-ZM	锌铝镁 双面310g/m²	L-7595-2.8	
16	主梁C3 Torque tube C3	S156x2.8	9815	件	2	56	S550GD-ZM	锌铝镁 双面310g/m²	L-9815-2.8	
17	檩条1 Purlin 1	J60x30x1.8	1450	件	58	1624	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
18	156檩条垫块 156 Purlin pad block	42x43.5	28	件	58	1624	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095	
19	156长垫块 156 Long purlin pad block	42x43.5	121	件	2	56	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095L	
20	加强垫板 Purlin reinforcement gasket	-3x22	35	件	116	3248	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
21	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	4	112	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
22	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	28	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
23	156主梁封盖 Torque tube cover	-1x25	158	件	2	56	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
24	减速机罩 Slew drive cover	-1x120x260	450	件	1	28	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
25	喉箍 Hoop	20	530	件	2	56	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
螺栓配件										
25	外六角螺栓 Outer Hex Bolt	M18*80*50	8.8J	套	4	112	2大平1弹2母 大平垫尺寸：-3x ϕ 43x ϕ 19.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
26	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	16	448	1双叠自锁垫片 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
27	外六角螺栓 Outer Hex Bolt	M16*50*30	8.8J	套	8	224	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ40xϕ17.5 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
28	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	20	560	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
29	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	20	560	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ40xϕ17.5 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
30	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	40	1120	1大平1双叠自锁垫片1母 大平垫尺寸：-3x ϕ 37x ϕ 15.5 双叠垫片尺寸：-3.4x ϕ 30.7x ϕ 15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
31	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	8	224	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
32	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	8	224	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
33	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	64	1792	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xϕ16.6xϕ10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
34	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	4	112	2双叠自锁垫片2母 双叠垫片尺寸：-3.4x ϕ 25.4x ϕ 13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
35	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	58	1624	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
36	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	224	6272	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径 ϕ 9	不锈钢		组件固定螺栓
37	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	84	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
38	薄螺母 thin nut	M12	C级	件	2	56	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
39	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	84	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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编码：XHY-FM-RD-29　　产品BOM清单　　0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（DESMO）	变更内容：	1版：
型 号：	1Px70-外围	订单套数：	22		2版：
组件功率W：	730	行数：	1	单套功率KW：51.1	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	70	总容量MW：1.1242	基础类型：	桩柱一体

序号	内容 名称	规格	长度（mm）	单位	数量 单数	总数	材质	防腐要求	零件号	备注
1	驱动立柱4 Driving post 4	W8*13	3050	件	2	44	S355JR	热镀锌 平均锌层≥85μm	QZ4-3050	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	9	198	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	3	66	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
4	150底座 Slew drive base-150	77x206	177	件	4	88	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP088	铸造件
5	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	12	264	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
6	156轴承座上 Bearing housing cover	R116x73	288	件	12	264	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
7	重型牛腿 Bearing base	75x124	163	件	24	528	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
8	156阻尼支臂 156 Damper fixture	81.5x131	400	件	3	66	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
9	阻尼器套管 Round tube of damper	ϕ20x2.0	17	件	12	264	S250GD-ZM	锌铝镁 双面310g/m²	TG	
10	阻尼器下固定件-轻型 Damper lower fixture	102x130	220	件	6	132	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
11	156菱形主梁抱箍 Torque tube hoop	-4.5xS156	490	件	1	22	S355JR	热镀锌 平均锌层≥65μm	BG1	
12	主梁抱箍定制垫片 Torque tube hoop custom gasket	-3x37x30	69	件	4	88	S350GD-ZM	锌铝镁 双面310g/m²	DP	
13	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	72	1584	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
14	主梁A2 Torque tube A2	S156x2.8	6590	件	2	44	S550GD-ZM	锌铝镁 双面310g/m²	L-6590-2.8	
15	主梁A4 Torque tube A4	S156x2.8	7240	件	2	44	S550GD-ZM	锌铝镁 双面310g/m²	L-7240-2.8	
16	主梁B1 Torque tube B1	S156x2.8	6545	件	2	44	S550GD-ZM	锌铝镁 双面310g/m²	L-6845-2.8	
17	主梁B4 Torque tube B4	S156x2.8	7495	件	2	44	S550GD-ZM	锌铝镁 双面310g/m²	L-7495-2.8	
18	主梁B5 Torque tube B5	S156x2.8	7645	件	1	22	S550GD-ZM	锌铝镁 双面310g/m²	L-7645-2.8	
19	主梁C4 Torque tube C4	S156x2.8	8613	件	2	44	S550GD-ZM	锌铝镁 双面310g/m²	L-8613-2.8	
20	主梁D1 Torque tube D1	S156x2.8	5730	件	1	22	S550GD-ZM	锌铝镁 双面310g/m²	L-5730-2.8	
21	主梁D2 Torque tube D2	S156x2.8	9530	件	1	22	S550GD-ZM	锌铝镁 双面310g/m²	L-9530-2.8	
22	檩条1 Purlin 1	J60x30x1.8	1450	件	73	1606	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
23	156檩条垫块 156 Purlin pad block	42x43.5	28	件	73	1606	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/锌铝镁 双面310g/m²	BP095	
24	156长垫块 156 Long purlin pad block	42x43.5	121	件	3	66	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/锌铝镁 双面310g/m²	BP095L	
25	加强垫板 Purlin reinforcement gasket	-3x22	35	件	146	3212	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
26	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	8	176	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
27	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	22	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
28	156主梁封盖 Torque tube cover	-1x25	158	件	2	44	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
29	减速机罩 Slew drive cover	-1x120x260	450	件	2	44	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
30	喉箍 Hoop	20	530	件	3	66	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
	螺栓配件									
30	外六角螺栓 Outer Hex Bolt	M16*75*50	8.8J	套	8	176	2大平1弹2母 大平垫尺寸：-3xϕ40xϕ17.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
31	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	32	704	1双叠自锁垫片 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
32	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	16	352	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ37xϕ15.5 双叠垫片尺寸：-3.4xϕ30.7xϕ15.2	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
33	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	24	528	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
34	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	24	528	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ40xϕ17.5 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
35	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	48	1056	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ37xϕ15.5 双叠垫片尺寸：-3.4xϕ30.7xϕ15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
36	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	12	264	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
37	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	12	264	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
38	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	72	1584	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xϕ16.6xϕ10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
39	外六角螺栓 Outer Hex Bolt	M12*50*40	8.8J	套	5	110	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		主梁抱箍连接螺栓
40	马车螺栓 MC12 Carriage bolts	MC12*35*30	8.8J	套	4	88	1双叠自锁垫片1母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		主梁抱箍连接螺栓
41	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	6	132	2双叠自锁垫片2母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
42	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	73	1606	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
43	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	280	6160	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径ϕ9	不锈钢		组件固定螺栓
44	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	4	88	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
45	薄螺母 thin nut	M12	C级	件	2	44	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
46	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	66	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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编码：XHY-FM-RD-29

产品BOM清单

0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（GIULIETTA）	变更内容：	1版：
型 号：	1Px28-外围	订单套数：	66		2版：
组件功率W：	730	行数：	1	单套功率KW：20.44	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	28	总容量MW：1.34904	基础类型：	桩柱一体

序号	内容 名称	规格	长度(mm)	单位	数量 单数	数量 总数	材质	防腐要求	零件号	备注
1	驱动立柱1 Driving post 1	W8*10	3050	件	1	66	S355JR	热镀锌 平均锌层≥85μm	QZ1-3050	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	3	198	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	2	132	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
4	150底座 Slew drive base-150	77x206	177	件	2	132	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP088	铸造件
5	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	5	330	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
6	156轴承座上 Bearing housing cover	R116x73	288	件	5	330	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
7	重型牛腿 Bearing base	75x124	163	件	10	660	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
8	156阻尼支臂 156 Damper fixture	81.5x131	400	件	2	132	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
9	阻尼器套管 Round tube of damper	ϕ20x2.0	17	件	8	528	S250GD-ZM	锌铝镁 双面310g/m²	TG	
10	阻尼器下固定件-轻型 Damper lower fixture	102x130	220	件	4	264	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
11	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	24	1584	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
12	主梁A1 Torque tube A1	S156x2.8	7290	件	1	66	S550GD-ZM	锌铝镁 双面310g/m²	L-7290-2.8	
13	主梁A2 Torque tube A2	S156x2.8	6590	件	1	66	S550GD-ZM	锌铝镁 双面310g/m²	L-6590-2.8	
14	主梁B1 Torque tube B1	S156x2.8	6845	件	1	66	S550GD-ZM	锌铝镁 双面310g/m²	L-6845-2.8	
15	主梁C1 Torque tube C1	S156x2.8	8603	件	1	66	S550GD-ZM	锌铝镁 双面310g/m²	L-8603-2.8	
16	主梁C2 Torque tube C2	S156x2.8	9131	件	1	66	S550GD-ZM	锌铝镁 双面310g/m²	L-9131-2.8	
17	檩条1 Purlin 1	J60x30x1.8	1450	件	30	1980	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
18	156檩条垫块 156 Purlin pad block	42x43.5	28	件	30	1980	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/锌铝镁 双面310g/m²	BP095	
19	156长垫块 156 Long purlin pad block	42x43.5	121	件	2	132	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/锌铝镁 双面310g/m²	BP095L	
20	加强垫板 Purlin reinforcement gasket	-3x22	35	件	60	3960	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
21	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	4	264	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
22	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	66	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
23	156主梁封盖 Torque tube cover	-1x25	158	件	2	132	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
24	减速机罩 Slew drive cover	-1x120x260	450	件	1	66	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
25	喉箍 Hoop	20	530	件	2	132	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
螺栓配件										
25	外六角螺栓 Outer Hex Bolt	M16*75*50	8.8J	套	4	264	2大平1弹2母 大平垫尺寸：-3xϕ40xϕ17.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
26	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	16	1056	1双叠自锁垫片 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
27	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	8	528	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ37xϕ15.5 双叠垫片尺寸：-3.4xϕ30.7xϕ15.2	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
28	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	10	660	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
29	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	10	660	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ40xϕ17.5 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
30	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	20	1320	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ37xϕ15.5 双叠垫片尺寸：-3.4xϕ30.7xϕ15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
31	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	8	528	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
32	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	8	528	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
33	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	24	1584	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xϕ16.6xϕ10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
34	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	4	264	2双叠自锁垫片2母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
35	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	30	1980	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
36	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	112	7392	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径ϕ9	不锈钢		组件固定螺栓
37	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	198	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
38	薄螺母 thin nut	M12	C级	件	2	132	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
39	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	198	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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编码：XHY-FM-RD-29

产品BOM清单

0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（GIULIETTA）	变更内容：	1版：
型 号：	1Px42-外围	订单套数：	51		2版：
组件功率W：	730	行数：	1	单套功率KW：30.66	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	42	总容量MW：1.56366	基础类型：	桩柱一体

序号	内容				数量		材质	防腐要求	零件号	备注
	名称	规格	长度(mm)	单位	单数	总数				
1	驱动立柱2 Driving post 2	W8*13	3034	件	1	51	S355JR	热镀锌 平均锌层≥85μm	QZ2-3034	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	6	306	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	2	102	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
4	减速机底座 Slew drive base	80x183	244	件	2	102	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP064	铸造件
5	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	8	408	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
6	156轴承座上 Bearing housing cover	R116x73	288	件	8	408	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
7	重型牛腿 Bearing base	75x124	163	件	16	816	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
8	156阻尼支臂 156 Damper fixture	81.5x131	400	件	2	102	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
9	阻尼器套管 Round tube of damper	ф20x2.0	17	件	8	408	S250GD-ZM	锌铝镁 双面310g/m²	TG	
10	阻尼器下固定件-轻型 Damper lower fixture	102x130	220	件	4	204	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
11	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	48	2448	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
12	主梁A5 Torque tube A5	S156x2.2	5990	件	2	102	S550GD-ZM	锌铝镁 双面310g/m²	L-5990-2.2	
13	主梁B6 Torque tube B6	S156x2.2	7045	件	4	204	S550GD-ZM	锌铝镁 双面310g/m²	L-7045-2.2	
14	主梁C5 Torque tube C5	S156x2.2	8893	件	2	102	S550GD-ZM	锌铝镁 双面310g/m²	L-8893-2.2	
15	檩条1 Purlin 1	J60x30x1.8	1450	件	44	2244	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
16	156檩条垫块 156 Purlin pad block	42x43.5	28	件	44	2244	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095	
17	156长垫块 156 Long purlin pad block	42x43.5	121	件	2	102	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095L	
18	加强垫板 Purlin reinforcement gasket	-3x22	35	件	88	4488	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
19	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	4	204	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
20	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	51	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
21	156主梁封盖 Torque tube cover	-1x25	158	件	2	102	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
22	减速机罩 Slew drive cover	-1x120x260	450	件	1	51	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
23	喉箍 Hoop	20	530	件	2	102	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
螺栓配件										
23	外六角螺栓 Outer Hex Bolt	M18*80*50	8.8J	套	4	204	2大平1弹2母 大平垫尺寸：-3x φ 43x φ 19.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
24	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	16	816	1双叠自锁垫片 双叠垫片尺寸：-3.4xφ30.7xφ17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
25	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	8	408	1大平1双叠自锁垫片1母 大平垫尺寸：-3xφ40xφ17.5 双叠垫片尺寸：-3.4xφ30.7xφ17	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
26	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	16	816	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
27	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	16	816	1大平1双叠自锁垫片1母 大平垫尺寸：-3xφ40xφ17.5 双叠垫片尺寸：-3.4xφ30.7xφ17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
28	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	32	1632	1大平1双叠自锁垫片1母 大平垫尺寸：-3x φ 37x φ 15.5 双叠垫片尺寸：-3.4x φ 30.7x φ 15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
29	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	8	408	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xφ25.4xφ13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
30	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	8	408	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
31	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	48	2448	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xφ16.6xφ10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
32	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	4	204	2双叠自锁垫片2母 双叠垫片尺寸：-3.4x φ 25.4x φ 13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
33	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	44	2244	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
34	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	168	8568	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径 φ 9	不锈钢		组件固定螺栓
35	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	153	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
36	薄螺母 thin nut	M12	C级	件	2	102	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
37	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	153	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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江苏新恒源能源技术有限公司

编码：XHY-FM-RD-29

产品BOM清单

0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（GIULIETTA）	变更内容：	1版：
型 号：	1Px56-外围	订单套数：	61		2版：
组件功率W：	730	行数：	1	单套功率KW：40.88	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	56	总容量MW：2.49368	基础类型：	桩柱一体

序号	内容				数量		材质	防腐要求	零件号	备注
	名称	规格	长度（mm）	单位	单数	总数				
1	驱动立柱3 Driving post 3	W8*15	3024	件	1	61	S355JR	热镀锌 平均锌层≥85μm	QZ3-3024	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	6	366	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	非驱动立柱2 Non-driving post 2	W8*13	3096	件	2	122	S355JR	热镀锌 平均锌层≥85μm	FZ2-3096	
4	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	2	122	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
5	减速机底座 Slew drive base	80x183	244	件	2	122	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP064	铸造件
6	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	10	610	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
7	156轴承座上 Bearing housing cover	R116x73	288	件	10	610	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
8	重型牛腿 Bearing base	75x124	163	件	20	1220	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
9	156阻尼支臂 156 Damper fixture	81.5x131	400	件	2	122	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
10	阻尼器套管 Round tube of damper	ϕ20x2.0	17	件	8	488	S250GD-ZM	锌铝镁 双面310g/m²	TG	
11	阻尼器下固定件-轻型 Damper lower fixture	102x130	220	件	4	244	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
12	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	64	3904	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
13	主梁A3 Torque tube A3	S156x2.8	6190	件	2	122	S550GD-ZM	锌铝镁 双面310g/m²	L-6190-2.8	
14	主梁B2 Torque tube B2	S156x2.8	7345	件	2	122	S550GD-ZM	锌铝镁 双面310g/m²	L-7345-2.8	
15	主梁B3 Torque tube B3	S156x2.8	7595	件	4	244	S550GD-ZM	锌铝镁 双面310g/m²	L-7595-2.8	
16	主梁C3 Torque tube C3	S156x2.8	9815	件	2	122	S550GD-ZM	锌铝镁 双面310g/m²	L-9815-2.8	
17	檩条1 Purlin 1	J60x30x1.8	1450	件	58	3538	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
18	156檩条垫块 156 Purlin pad block	42x43.5	28	件	58	3538	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095	
19	156长垫块 156 Long purlin pad block	42x43.5	121	件	2	122	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/ 锌铝镁 双面310g/m²	BP095L	
20	加强垫板 Purlin reinforcement gasket	-3x22	35	件	116	7076	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
21	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	4	244	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
22	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	61	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
23	156主梁封盖 Torque tube cover	-1x25	158	件	2	122	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
24	减速机罩 Slew drive cover	-1x120x260	450	件	1	61	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
25	喉箍 Hoop	20	530	件	2	122	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
螺栓配件										
25	外六角螺栓 Outer Hex Bolt	M18*80*50	8.8J	套	4	244	2大平1弹2母 大平垫尺寸：-3xϕ43xϕ19.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
26	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	16	976	1双叠自锁垫片 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
27	外六角螺栓 Outer Hex Bolt	M16*50*30	8.8J	套	8	488	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ40xϕ17.5 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
28	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	20	1220	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
29	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	20	1220	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ40xϕ17.5 双叠垫片尺寸：-3.4xϕ30.7xϕ17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
30	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	40	2440	1大平1双叠自锁垫片1母 大平垫尺寸：-3xϕ37xϕ15.5 双叠垫片尺寸：-3.4xϕ30.7xϕ15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
31	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	8	488	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
32	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	8	488	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
33	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	64	3904	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xϕ16.6xϕ10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
34	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	4	244	2双叠自锁垫片2母 双叠垫片尺寸：-3.4xϕ25.4xϕ13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
35	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	58	3538	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
36	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	224	13664	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径ϕ9	不锈钢		组件固定螺栓
37	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	183	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
38	薄螺母 thin nut	M12	C级	件	2	122	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
39	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	183	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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产品BOM清单

编码：XHY-FM-RD-29　　0 版

订单号：	SP26033A	工程名称：	意大利跟踪支架（GIULIETTA）	变更内容：	1版：
型 号：	1Px70-外围	订单套数：	23		2版：
组件功率W：	730	行数：	1	单套功率KW：51.1	驱动类型：	回转减速机
组件尺寸：	1303x2384x33	列数：	70	总容量MW：1.1753	基础类型：	桩柱一体

序号	内容				数量		材质	防腐要求	零件号	备注
	名称	规格	长度(mm)	单位	单数	总数				
1	驱动立柱4 Driving post 4	W8*13	3050	件	2	46	S355JR	热镀锌 平均锌层≥85μm	QZ4-3050	
2	非驱动立柱1 Non-driving post 1	W8*10	3096	件	9	207	S355JR	热镀锌 平均锌层≥85μm	FZ1-3096	
3	带阻尼器非驱动立柱1 Damper post 1	W8*10	3096	件	3	69	S355JR	热镀锌 平均锌层≥85μm	NZ1-3096	
4	150底座 Slew drive base-150	77x206	177	件	4	92	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP088	铸造件
5	156轴承座下-H200 Bearing housing base-H200	R116x73	370	件	12	276	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP094-200	铸造件
6	156轴承座上 Bearing housing cover	R116x73	288	件	12	276	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP090	铸造件
7	重型牛腿 Bearing base	75x124	163	件	24	552	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP036H	铸造件
8	156阻尼支臂 156 Damper fixture	81.5x131	400	件	3	69	S350GD-ZM	锌铝镁 双面310g/m²	BP168	
9	阻尼器套管 Round tube of damper	ф20x2.0	17	件	12	276	S250GD-ZM	锌铝镁 双面310g/m²	TG	
10	阻尼器下固定件-轻型 Damper lower fixture	102x130	220	件	6	138	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP093	铸造件
11	156菱形主梁抱箍 Torque tube hoop	-4.5xS156	490	件	1	23	S355JR	热镀锌 平均锌层≥65μm	BG1	
12	主梁抱箍定制垫片 Torque tube hoop custom gasket	-3x37x30	69	件	4	92	S350GD-ZM	锌铝镁 双面310g/m²	DP	
13	抗剪导向垫片 Shear-resistant guide washer	M10	/	件	72	1656	EN-GJS-450-10	热镀锌 平均锌层≥65μm	BP086	铸造件
14	主梁A2 Torque tube A2	S156x2.8	6590	件	2	46	S550GD-ZM	锌铝镁 双面310g/m²	L-6590-2.8	
15	主梁A4 Torque tube A4	S156x2.8	7240	件	2	46	S550GD-ZM	锌铝镁 双面310g/m²	L-7240-2.8	
16	主梁B1 Torque tube B1	S156x2.8	6545	件	2	46	S550GD-ZM	锌铝镁 双面310g/m²	L-6845-2.8	
17	主梁B4 Torque tube B4	S156x2.8	7495	件	2	46	S550GD-ZM	锌铝镁 双面310g/m²	L-7495-2.8	
18	主梁B5 Torque tube B5	S156x2.8	7645	件	1	23	S550GD-ZM	锌铝镁 双面310g/m²	L-7645-2.8	
19	主梁C4 Torque tube C4	S156x2.8	8613	件	2	46	S550GD-ZM	锌铝镁 双面310g/m²	L-8613-2.8	
20	主梁D1 Torque tube D1	S156x2.8	5730	件	1	23	S550GD-ZM	锌铝镁 双面310g/m²	L-5730-2.8	
21	主梁D2 Torque tube D2	S156x2.8	9530	件	1	23	S550GD-ZM	锌铝镁 双面310g/m²	L-9530-2.8	
22	檩条1 Purlin 1	J60x30x1.8	1450	件	73	1679	S450GD-ZM	锌铝镁 双面310g/m²	LT1-1450	
23	156檩条垫块 156 Purlin pad block	42x43.5	28	件	73	1679	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/锌铝镁 双面310g/m²	BP095	
24	156长垫块 156 Long purlin pad block	42x43.5	121	件	3	69	S235JR/S250GD-ZM	热镀锌 平均锌层≥65μm/锌铝镁 双面310g/m²	BP095L	
25	加强垫板 Purlin reinforcement gasket	-3x22	35	件	146	3358	S350GD-ZM	锌铝镁 双面310g/m²	BP069	
26	156主梁加强件 156 Torque tube reinforcing plate	-3x50x145	145	件	8	184	S350GD-ZM	锌铝镁 双面310g/m²	BP099	
27	无线支架2 Wireless receiver bracket 2	-2x55x90	300	件	1	23	S250GD-ZM	锌铝镁 双面310g/m²	ZJ2	
28	156主梁封盖 Torque tube cover	-1x25	158	件	2	46	S250GD-ZM	锌铝镁 双面310g/m²	BP098	
29	减速机罩 Slew drive cover	-1x120x260	450	件	2	46	S250GD-ZM	锌铝镁 双面310g/m²	JSJZ	
30	喉箍 Hoop	20	530	件	3	69	SUS304	不锈钢	HG-530	配螺栓，螺栓热镀锌平均≥50μm
螺栓配件										
30	外六角螺栓 Outer Hex Bolt	M16*75*50	8.8J	套	8	184	2大平1弹2母 大平垫尺寸：-3xф40xф17.5	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与减速机连接螺栓
31	外六角螺栓 Outer Hex Bolt	M16*30*30	8.8J	套	32	736	1双叠自锁垫片 双叠垫片尺寸：-3.4xф30.7xф17	热镀锌平均≥50μm，局部最小≥40μm		减速机与主梁连接螺栓
32	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	16	368	1大平1双叠自锁垫片1母 大平垫尺寸：-3xф37xф15.5 双叠垫片尺寸：-3.4xф30.7xф15.2	热镀锌平均≥50μm，局部最小≥40μm		减速机底座与立柱连接螺栓
33	六角头带铰制孔螺栓 M10 bolt with reamed hole	M10*90*26	8.8J	套	24	552	1母1开口销（2.5x18） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		上下轴承座连接螺栓
34	外六角螺栓 Outer Hex Bolt	M16*45*30	8.8J	套	24	552	1大平1双叠自锁垫片1母 大平垫尺寸：-3xф40xф17.5 双叠垫片尺寸：-3.4xф30.7xф17	热镀锌平均≥50μm，局部最小≥40μm		轴承座与牛腿连接螺栓
35	外六角螺栓 Outer Hex Bolt	M14*40*30	8.8J	套	48	1104	1大平1双叠自锁垫片1母 大平垫尺寸：-3xф37xф15.5 双叠垫片尺寸：-3.4xф30.7xф15.2	热镀锌平均≥50μm，局部最小≥40μm		牛腿与立柱连接螺栓
36	外六角螺栓 Outer Hex Bolt	M12*40*25	8.8J	套	12	276	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xф25.4xф13.2	热镀锌平均≥50μm，局部最小≥40μm		阻尼器下固定件与立柱连接螺栓
37	外六角螺栓 Outer Hex Bolt	M14*80*25	8.8J	套	12	276	1开槽螺母1开口销（4x32） 开口销材质SUS304	热镀锌平均≥50μm，局部最小≥40μm		阻尼器固定螺栓
38	马车螺栓 MC10 Carriage bolts	MC10*31*29	8.8J	套	72	1656	1双叠自锁垫片1母 双叠垫片尺寸：-2.5xф16.6xф10.7	热镀锌平均≥50μm，局部最小≥40μm		缩径主梁连接螺栓
39	外六角螺栓 Outer Hex Bolt	M12*50*40	8.8J	套	5	115	1平1双叠自锁垫片1母 双叠垫片尺寸：-3.4xф25.4xф13.2	热镀锌平均≥50μm，局部最小≥40μm		主梁抱箍连接螺栓
40	马车螺栓 MC12 Carriage bolts	MC12*35*30	8.8J	套	4	92	1双叠自锁垫片1母 双叠垫片尺寸：-3.4xф25.4xф13.2	热镀锌平均≥50μm，局部最小≥40μm		主梁抱箍连接螺栓
41	U型螺栓 U bolt	UM12*168*195*35	6.8J	套	6	138	2双叠自锁垫片2母 双叠垫片尺寸：-3.4xф25.4xф13.2	热镀锌平均≥50μm，局部最小≥40μm		156阻尼支臂固定螺栓
42	U型螺栓 U bolt	UM10*166*188*30	6.8J	套	73	1679	2法兰螺母	热镀锌平均≥50μm，局部最小≥40μm		檩条固定螺栓
43	外六角螺栓 Outer Hex Bolt	M8*16*16	SUS304	套	280	6440	1矩形垫片1大六角法兰面螺母 矩形垫片：-1.8x25x15，内径ф9	不锈钢		组件固定螺栓
44	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	4	92	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓
45	薄螺母 thin nut	M12	C级	件	2	46	/	热镀锌平均≥50μm，局部最小≥40μm		接地用
46	外六角螺栓 Outer Hex Bolt	M10*25*25	6.8J	套	3	69	1平1母	热镀锌平均≥50μm，局部最小≥40μm		接地螺栓

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